

**APPLIED STATISTICS**  
( for AIML Students)  
**B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch**  
**Syllabus**

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**UNIT-1**

**The Role of Statistics in Engineering:**

The Engineering Method and Statistical Thinking – Data Collection

**Engineering Data and Basic Principles:**

Retrospective Studies – Observational Studies – Designed Experiments – Observing Processes Over Time – Mechanistic and Empirical Models.

**Data Description and Representation:**

Collection of Data – Classification and Tabulation of Data – Stem-and-Leaf Diagrams – Frequency Distributions and Histograms – Box Plots – Time Sequence Plots – Probability Plots.

**UNIT-2**

**Descriptive Statistics:**

Measures of Central Tendency – Measures of Dispersion – Skewness and Kurtosis.

**Correlation and Regression:**

Scatter Diagram – Types of Correlation – Karl Pearson's Coefficient of Correlation – Spearman's Rank Correlation – Method of Least Squares – Linear

**UNIT-3**

**Sampling:**

Types of Sampling – Sampling Distributions – Sampling Distribution of the Mean.

**Point Estimation of Parameters:**

General Concepts of Point Estimation – Unbiased Estimators – Variance of a Point Estimator – Standard Error – Methods of Point Estimation (Method of Moments, Method of Maximum Likelihood).

### **Statistical Intervals for a Single Sample:**

- (i) Confidence Interval for the Mean of a Normal Distribution (Variance Known)
- (ii) Confidence Interval for the Mean of a Normal Distribution (Variance Unknown)
- (iii) Confidence Interval for the Variance and Standard Deviation of a Normal Distribution
- (iv) Large-Sample Confidence Interval for a Population Proportion.

## **UNIT-4**

### **Tests of Hypotheses for a Single Sample:**

Tests of Statistical Hypotheses – General Procedure for Hypothesis Testing

- (i) Test on the Mean of a Normal Distribution (Variance Known)
- (ii) Test on the Mean of a Normal Distribution (Variance Unknown)
- (iii) Test on the Variance and Standard Deviation of a Normal Distribution.

## **UNIT-5**

### **Statistical Inference for Two Samples:**

- (i) Inference for the Difference in Means of Two Normal Distributions (Variances Known)
- (ii) Inference for the Difference in Means of Two Normal Distributions (Variances Unknown)
- (iii) Inference on the Variances of Two Normal Distributions
- (iv) Inference on Two Population Proportions.

### **Reference Books:**

01. Douglas C. Montgomery and George C. Runger. Applied Statistics and Probability for Engineers, (3rdEdn), John Wiley and Sons, Inc., New York, 2003.
02. Robert H. Carver and Jane Gradwohl Nash. Doing Data Analysis with SPSS Version 18.0, (Indian Edition) Cengage Learning, New Delhi, 2012.
03. Richard A. Johnson and C.B. Gupta, Probability and Statistics for Engineers, (7thEdn.), Pearson Education, Indian Impression 2006.

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Measures of Central Tendency

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Measures of Central Tendency

In statistics, a **measure of central tendency** is a numerical value that represents the central or typical value of a dataset. It provides a single value around which the observations are concentrated and summarizes the entire dataset in a simple and meaningful way. Measures of central tendency are also known as **averages**.

The three most commonly used measures of central tendency are:

1. Arithmetic Mean (Mean)
2. Median
3. Mode

Other important measures include:

- Geometric Mean (GM)
- Harmonic Mean (HM)
- Weighted Mean

### Characteristics of a Good Average

An ideal measure of central tendency should possess the following characteristics:

1. It should be **rigidly defined**.
2. It should be **easy to understand and compute**.
3. It should be **based on all observations**.
4. It should **not be unduly affected by extreme values (outliers)**.
5. It should be **capable of further mathematical treatment**.
6. It should be **stable** and show minimum variation from sample to sample.
7. It should provide a **true representation of the dataset**.

## Important Points

- A measure of central tendency summarizes a dataset using a single value.
- It helps describe the center of the data.
- The most commonly used measures are Mean, Median, and Mode.
- Different measures are suitable for different types of data.
- Measures of central tendency are extensively used in engineering, business, medicine, education, and scientific research.

## Arithmetic Mean (AM)

**Arithmetic Mean** is defined as the sum of all observations divided by the total number of observations.

In simple words,

$$\text{Arithmetic Mean} = \text{Total of all observations} \div \text{Number of observations}$$

### Individual Observations:

Then the arithmetic mean is

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where

$\bar{x}$  Arithmetic Mean

$x_i$   $i^{\text{th}}$  observation

$n$  Number of observations

$\sum x_i$  Sum of all observations

### Example

Find the arithmetic mean of

12, 15, 18, 20, 25

### Solution

$$n = \text{Number of observations} = 5$$

$$\sum x = \text{Sum of observations} = 12 + 15 + 18 + 20 + 25 = 90$$

$$\text{Arithmetic Mean} = \bar{x} = \frac{\sum x}{n} = \frac{90}{5} = 18$$

$$\boxed{\bar{x} = 18}$$

### Discrete Frequency Distribution:

For a discrete frequency distribution, the arithmetic mean is calculated by taking into account both the observations and their corresponding frequencies.

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Here,

- $x$  = Observation (or value of the variable)
- $f$  = Frequency of the observation
- $N = \sum f$  = Total frequency

### Example

Find the arithmetic mean of the following distribution.

| $x$ | $f$ |
|-----|-----|
| 5   | 2   |
| 7   | 3   |
| 9   | 5   |
| 11  | 4   |
| 13  | 1   |

### Solution

| $x$   | $f$           | $fx$            |
|-------|---------------|-----------------|
| 5     | 2             | 10              |
| 7     | 3             | 21              |
| 9     | 5             | 45              |
| 11    | 4             | 44              |
| 13    | 1             | 13              |
| Total | $\sum f = 15$ | $\sum fx = 133$ |

Therefore,

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{133}{15} = 8.87$$

$$\boxed{\bar{x} = 8.87}$$

### Example

The marks obtained by students are given below.

| Marks ( $x$ ) | Number of Students ( $f$ ) |
|---------------|----------------------------|
| 10            | 4                          |
| 20            | 6                          |
| 30            | 8                          |
| 40            | 5                          |
| 50            | 2                          |

Find the arithmetic mean.

### Importants:

The arithmetic mean possesses the following important properties.

1. It is based on all observations.
2. It is rigidly defined.
3. It is easy to calculate.
4. It is unique for a given dataset.
5. It can be used for further mathematical analysis.
6. It is sensitive to extreme observations (outliers).

### Continuous Frequency Distribution: Direct Method

For a **continuous frequency distribution**, the arithmetic mean is calculated using the **class mid-values** and their corresponding frequencies. The mid-value of each class is taken as the representative value of that class.

Then the arithmetic mean is

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Here,

- $x$  = Mid value of the class
- $f$  = Frequency
- $N = \sum f$  = Total frequency

$$x = \text{Mid-Value} = \frac{\text{Lower Class Limit} + \text{Upper Class Limit}}{2}$$

### Example:

Calculate the arithmetic mean of the following distribution.

| Class Interval | Frequency ( $f$ ) |
|----------------|-------------------|
| 0–10           | 4                 |
| 10–20          | 6                 |
| 20–30          | 10                |
| 30–40          | 8                 |
| 40–50          | 2                 |

**Solution:**

| Class Interval | $x$ | $f$           | $fx$            |
|----------------|-----|---------------|-----------------|
| 0–10           | 5   | 4             | 20              |
| 10–20          | 15  | 6             | 90              |
| 20–30          | 25  | 10            | 250             |
| 30–40          | 35  | 8             | 280             |
| 40–50          | 45  | 2             | 90              |
| Total          |     | $\sum f = 30$ | $\sum fx = 730$ |

Therefore,

$$\bar{x} = \frac{730}{30} = 24.33$$

$$\boxed{\bar{x} = 24.33}$$

### Importants

- The actual observations are not available.
- Each class interval is represented by its class mark.
- The direct method gives an exact mean when the class mark adequately represents the observations in each class.
- This method is simple and is suitable when the class marks are small.

### Continuous Frequency Distribution: Step-Deviation Method

When class intervals are of equal width and class mid-values are large, the **Step-Deviation Method** (or **Coding Method**) is preferred over the Direct Method. It simplifies calculations by dividing the deviations from an assumed mean by the common class width.

Then the arithmetic mean is given by

$$\boxed{\bar{x} = A + \left( \frac{\sum fu}{\sum f} \right) h}$$

where

$$u = \frac{x - A}{h}.$$

and

- $A$  = Assumed Mean
- $h$  = Common class width
- $x$  = class mid-value
- $u = \frac{x - A}{h}$

- $f$  = Frequency
- $N = \sum f$

### Example

The weekly wages (in hundreds of rupees) of workers are given below.

| Weekly Wages | Frequency |
|--------------|-----------|
| 20–30        | 5         |
| 30–40        | 9         |
| 40–50        | 15        |
| 50–60        | 12        |
| 60–70        | 9         |

Find the arithmetic mean using the Step-Deviation Method.

### Solution

Take

$$A = 45, \quad h = 10.$$

| Class | $x$ | $f$           | $u$ | $fu$           |
|-------|-----|---------------|-----|----------------|
| 20–30 | 25  | 5             | –2  | –10            |
| 30–40 | 35  | 9             | –1  | –9             |
| 40–50 | 45  | 15            | 0   | 0              |
| 50–60 | 55  | 12            | 1   | 12             |
| 60–70 | 65  | 9             | 2   | 18             |
| Total |     | $\sum f = 50$ |     | $\sum fu = 11$ |

Therefore,

$$\bar{x} = 45 + \left(\frac{11}{50}\right)(10) = 47.2$$

$$\boxed{\bar{x} = 47.2}$$

### Importants

- Simplifies lengthy calculations.
- Uses smaller numbers, reducing computational effort.
- Particularly useful for grouped data with equal class widths.
- Produces the same result as the Direct Method.
- Reduces the chances of arithmetic errors.

## Combined Mean

The **Combined Mean** is the arithmetic mean of two or more groups after combining all the observations into a single group.

Suppose there are two groups.

Group 1: Number of observations =  $n_1$ , Mean =  $\bar{x}_1$

Group 2: Number of observations =  $n_2$ , Mean =  $\bar{x}_2$

Then the combined mean is

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

For  $k$  groups,

$$\bar{x} = \frac{\sum n_i\bar{x}_i}{\sum n_i}$$

### Example 1

The average marks of two classes are given below.

| Class | Number of Students | Average Marks |
|-------|--------------------|---------------|
| A     | 40                 | 65            |
| B     | 60                 | 72            |

Find the combined average marks.

### Solution

Given,

$$n_1 = 40, \quad \bar{x}_1 = 65$$

$$n_2 = 60, \quad \bar{x}_2 = 72$$

Using

$$\begin{aligned}\bar{x} &= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} \\ &= \frac{40(65) + 60(72)}{40 + 60} \\ &= \frac{2600 + 4320}{100} = 69.2\end{aligned}$$

$$\boxed{\bar{x} = 69.2}$$

### Example 2

Department A has 80 employees with an average salary of Rs. 45,000. Department B has 120 employees with an average salary of Rs. 52,000.

Find the combined average salary.

### Solution

$$n_1 = 80, \quad \bar{x}_1 = 45000$$

$$n_2 = 120, \quad \bar{x}_2 = 52000$$

$$\bar{x} = \frac{80(45000) + 120(52000)}{200}$$

$$= \frac{3600000 + 6240000}{200}$$

$$= 49200$$

$$\boxed{\bar{x} = \text{Rs. } 49,200}$$

### Weighted Mean

The **Weighted Mean** is the average obtained by assigning different weights to different observations according to their importance.

It is used whenever some observations contribute more than others.

Then

$$\boxed{\text{weighted mean} = \bar{x}_w = \frac{\sum wx}{\sum w}}$$

Here,

- $x$  = observation
- $w$  = corresponding weight

### Example 4

A student obtains the following grades.

| Course      | Grade Point | Credits |
|-------------|-------------|---------|
| Mathematics | 9           | 4       |
| Physics     | 8           | 3       |
| Chemistry   | 7           | 3       |
| Programming | 10          | 4       |

Find the GPA.

### Solution

| Course      | Grade Point | Credits       | Credit $\times$ Grade |
|-------------|-------------|---------------|-----------------------|
| Mathematics | 9           | 4             | 36                    |
| Physics     | 8           | 3             | 24                    |
| Chemistry   | 7           | 3             | 21                    |
| Programming | 10          | 4             | 40                    |
| Total       |             | $\sum w = 14$ | $\sum wx = 121$       |

Hence,

$$\text{GPA} = \frac{121}{14} = 8.64$$

GPA = 8.64

### Importants

- Combined Mean is used when two or more groups are merged.
- Weighted Mean is used when observations have different levels of importance.

### Advantages and Disadvantages of Arithmetic Mean

| Advantages (Merits)                    | Disadvantages (Demerits)             |
|----------------------------------------|--------------------------------------|
| 1. Simple to understand                | 1. Affected by extreme values        |
| 2. Easy to compute                     | 2. Not suitable for skewed data      |
| 3. Based on all observations           | 3. Not suitable for qualitative data |
| 4. Rigidly defined                     | 4. May not be an actual observation  |
| 5. Suitable for mathematical treatment | 5. Cannot be obtained by inspection  |
| 6. Representative value                |                                      |
| 7. Stable from sample to sample        |                                      |
| 8. Useful for comparison               |                                      |

### Advantages and Disadvantages of Arithmetic Mean

The Arithmetic Mean (AM) is the most commonly used measure of central tendency because of its simplicity and mathematical properties. However, it also has certain limitations. The merits and demerits of Arithmetic Mean are explained below.

#### Advantages (Merits)

##### 1. Simple to Understand

The concept of arithmetic mean is very easy to understand. It represents the average value obtained by dividing the sum of all observations by the total number

of observations. Even people with little statistical knowledge can easily interpret it.

**Example:** The average marks of students in a class are often expressed using the arithmetic mean.

## 2. Easy to Compute

Arithmetic mean can be calculated using a simple mathematical formula. It requires only the sum of observations and the number of observations. It can also be computed easily for individual, discrete, and continuous data.

**Example:** For the data 5, 8, 10, the mean is

$$\bar{x} = \frac{5 + 8 + 10}{3} = \frac{23}{3} = 7.67.$$

## 3. Based on All Observations

Every observation in the dataset contributes to the calculation of the arithmetic mean. Hence, it provides a comprehensive measure of the entire dataset rather than depending on only a few values.

**Example:** If one observation changes, the mean also changes because every value is considered.

## 4. Rigidly Defined

The arithmetic mean has a unique value for a given dataset. There is no ambiguity in its computation, and different people using the same data will obtain the same result.

**Example:** The mean of the data 2, 4, 6, 8 is always 5.

## 5. Suitable for Mathematical Treatment

Arithmetic mean possesses many algebraic properties, making it useful in advanced statistical analysis. It is widely used in calculating variance, standard deviation, correlation, regression, hypothesis testing, and many other statistical methods.

**Example:** The formulas for variance and standard deviation are based on the arithmetic mean.

## 6. Representative Value

Since it considers every observation, the arithmetic mean generally represents the central value of the data effectively, especially when the data are fairly symmetrical.

**Example:** The average monthly rainfall provides a good representation of rainfall over a year.

## 7. Stable from Sample to Sample

The arithmetic mean shows relatively small variation when different random samples are drawn from the same population. This property makes it a reliable estimator of the population mean.

**Example:** If several random samples are selected from a population, their sample means are usually close to one another.

## 8. Useful for Comparison

Arithmetic mean provides a single numerical value that makes it easy to compare two or more datasets.

**Example:** The average salary of employees in two companies can be compared using their arithmetic means.

### Disadvantages (Demerits)

#### 1. Affected by Extreme Values

Arithmetic mean is highly sensitive to very large or very small observations (outliers). A single extreme value can significantly alter the mean and give a misleading picture of the data.

**Example:**

Data: 10, 12, 13, 15, 100

Mean:

$$\bar{x} = \frac{10 + 12 + 13 + 15 + 100}{5} = 30,$$

which is much higher than most observations.

#### 2. Not Suitable for Skewed Data

In highly skewed distributions, the arithmetic mean is pulled toward the long tail and may not represent the typical value of the dataset accurately.

**Example:** Income distributions are usually positively skewed because a few individuals earn extremely high incomes.

#### 3. Not Suitable for Qualitative Data

Arithmetic mean can only be calculated for numerical data. It cannot be applied to qualitative or categorical data such as gender, blood group, religion, or eye color.

**Example:** It is meaningless to calculate the average blood group of a group of people.

#### 4. May Not Be an Actual Observation

The arithmetic mean may produce a value that does not exist in the original dataset.

**Example:**

Data: 2, 3, 5

Mean:

$$\bar{x} = \frac{2 + 3 + 5}{3} = 3.33,$$

but 3.33 is not one of the observations.

#### 5. Cannot Be Obtained by Inspection

Unlike the mode or, in some cases, the median, the arithmetic mean cannot usually be identified simply by looking at the data. It requires actual calculation.

**Example:** For the data

12, 18, 25, 31, 40,

the mean cannot be determined without performing the calculation.

## Summary

- Arithmetic Mean is the most widely used measure of central tendency because it is simple, objective, and mathematically tractable.
- It uses every observation in the dataset and serves as the foundation for many statistical techniques.
- However, it is sensitive to extreme values and is not suitable for skewed or qualitative data.
- Therefore, the arithmetic mean should be used carefully, considering the nature and distribution of the data.

## Properties of Arithmetic Mean

The arithmetic mean possesses several important mathematical properties.

### Property 1

#### Statement:

The algebraic sum of the deviations of the observations from their arithmetic mean is zero.

$$\sum (x - \bar{x}) = 0$$

#### Proof:

Let the observations be  $x_1, x_2, \dots, x_n$ . Then, A.M =  $\bar{x} = \frac{\sum x}{n}$ .

Now,

$$\begin{aligned}\sum (x - \bar{x}) &= \sum x - \sum \bar{x} \\ &= \sum x - n\bar{x} \\ &= \sum x - n \left( \frac{\sum x}{n} \right) = \sum x - \sum x = 0.\end{aligned}$$

Hence,

$$\sum (x - \bar{x}) = 0.$$

### Property 2

#### Statement:

The arithmetic mean minimizes the sum of squared deviations.

$$\sum (x - \bar{x})^2 < \sum (x - a)^2,$$

where  $a$  is any value other than  $\bar{x}$ .

**Proof:**

Let the observations be  $x_1, x_2, \dots, x_n$ , and let  $a$  be any arbitrary constant.

Write

$$x - a = (x - \bar{x}) + (\bar{x} - a).$$

Squaring both sides,

$$(x - a)^2 = [(x - \bar{x}) + (\bar{x} - a)]^2.$$

$$(x - a)^2 = (x - \bar{x})^2 + 2(x - \bar{x})(\bar{x} - a) + (\bar{x} - a)^2.$$

Summing over all observations,

$$\sum (x - a)^2 = \sum (x - \bar{x})^2 + 2(\bar{x} - a) \sum (x - \bar{x}) + \sum (\bar{x} - a)^2.$$

Since

$$\sum (x - \bar{x}) = 0,$$

(by Property 1), the middle term vanishes.

Hence,

$$\boxed{\sum (x - a)^2 = \sum (x - \bar{x})^2 + n(\bar{x} - a)^2.}$$

Since

$$n(\bar{x} - a)^2 \geq 0,$$

we obtain

$$\sum (x - a)^2 \geq \sum (x - \bar{x})^2.$$

The equality holds only when

$$(\bar{x} - a)^2 = 0,$$

that is,

$$a = \bar{x}.$$

Therefore, for every value  $a \neq \bar{x}$ ,

$$\boxed{\sum (x - \bar{x})^2 < \sum (x - a)^2.}$$

Hence proved. ■

### Property 3

**Statement:** Adding or subtracting a constant from every observation changes the arithmetic mean by the same constant.

$$\text{If } y = x + c, \quad \text{then } \boxed{\bar{y} = \bar{x} + c}$$

#### Proof:

Let the observations be

$$x_1, x_2, \dots, x_n.$$

Suppose a constant  $c$  is added to each observation. Then the new observations are

$$y_1 = x_1 + c, y_2 = x_2 + c, \dots, y_n = x_n + c.$$

The arithmetic mean of the new observations is

$$\begin{aligned} \bar{y} &= \frac{y_1 + y_2 + \dots + y_n}{n} \\ &= \frac{(x_1 + c) + (x_2 + c) + \dots + (x_n + c)}{n} \\ &= \frac{(x_1 + x_2 + \dots + x_n) + nc}{n} \\ &= \frac{\sum x}{n} + \frac{nc}{n} \\ &= \bar{x} + c. \end{aligned}$$

Hence,

$$\boxed{\bar{y} = \bar{x} + c.}$$

Similarly, if a constant  $c$  is subtracted from every observation, then

$$\boxed{\bar{y} = \bar{x} - c.}$$

Hence proved. ■

### Property 4

#### Statement:

Multiplying every observation by a constant multiplies the arithmetic mean by the same constant.

$$\text{If } y = cx, \quad \text{then } \boxed{\bar{y} = c\bar{x}}$$

**Proof:**

Let the observations be

$$x_1, x_2, \dots, x_n.$$

Suppose each observation is multiplied by a constant  $c$ . Then the new observations are

$$y_1 = cx_1, y_2 = cx_2, \dots, y_n = cx_n.$$

The arithmetic mean of the new observations is

$$\begin{aligned}\bar{y} &= \frac{y_1 + y_2 + \dots + y_n}{n} \\ &= \frac{cx_1 + cx_2 + \dots + cx_n}{n} \\ &= \frac{c(x_1 + x_2 + \dots + x_n)}{n} \\ &= c \left( \frac{\sum x}{n} \right) \\ &= c\bar{x}.\end{aligned}$$

Hence,

$$\boxed{\bar{y} = c\bar{x}.$$

Similarly, if every observation is divided by a non-zero constant  $c$ , then

$$\boxed{\bar{y} = \frac{\bar{x}}{c}.$$

Hence proved. ■

**Property 5:****Statement:**

If two groups have sizes  $n_1$  and  $n_2$  with arithmetic means  $\bar{x}_1$  and  $\bar{x}_2$ , respectively, then the arithmetic mean of the combined group is

$$\boxed{\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}}$$

**Proof:**

Let the first group contain  $n_1$  observations with mean  $\bar{x}_1$ .

Then,

$$\bar{x}_1 = \frac{\sum x_1}{n_1}, \quad \implies \sum x_1 = n_1\bar{x}_1.$$

Similarly, for the second group,

$$\bar{x}_2 = \frac{\sum x_2}{n_2}, \quad \Rightarrow \quad \sum x_2 = n_2 \bar{x}_2.$$

The total number of observations in the combined group is

$$n = n_1 + n_2.$$

The total sum of all observations is

$$\sum x = \sum x_1 + \sum x_2.$$

Substituting the above results,

$$\sum x = n_1 \bar{x}_1 + n_2 \bar{x}_2.$$

Therefore, the arithmetic mean of the combined group is

$$\begin{aligned} \bar{x} &= \frac{\sum x}{n} \\ &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}. \end{aligned}$$

Hence,

$$\boxed{\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}.$$

Hence proved. ■

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Measures of Central Tendency

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Median

Median is the middle value of an ordered dataset that divides the observations into two equal halves.

If the observations are arranged in ascending order,

$$50\% \text{ observations} < \text{Median} < 50\% \text{ observations}$$

### Median for Ungrouped Data

#### Case I: Odd Number of Observations

If the number of observations is odd,

$$n = 2k + 1$$

then

$$\text{Median} = \left( \frac{n+1}{2} \right)^{\text{th}} \text{ observation}$$

#### Case II: Even Number of Observations

If the number of observations is even,

$$n = 2k$$

there is no single middle observation.

The median is the average of the two middle observations.

$$\text{Median} = \frac{\left( \frac{n}{2} \right)^{\text{th}} + \left( \frac{n}{2} + 1 \right)^{\text{th}}}{2}$$

### Examples:

Find the median for following data.

1. 18, 12, 25, 15, 20
2. 30, 28, 34, 32, 25, 31, 27
3. 22, 15, 18, 24, 20, 17
4. 45, 52, 48, 50, 55, 47, 53, 51

5. The heights (in cm) of nine students are: 152, 158, 150, 160, 154, 149, 156, 155, 153
6. The weights (kg) of ten athletes are : 60, 58, 65, 62, 59, 64, 61, 63, 66, 67

**Answers:**

The median of given data are:

1. 18
2. 30
3. 19
4. 50.5
5. 154 cm
6. 62.5 kg

**Median for Discrete Frequency Distribution**

Let's consider a discrete frequency distribution is provided. Then, the median can be find as follows:

1. Calculate the cumulative frequencies.
2. Compute  $\frac{N}{2}$ . Here,  $N = \sum f$ .
3. Identify the first cumulative frequency that is greater than or equal to  $\frac{N}{2}$ .
4. The corresponding observation is the median.

**Examples:**

Find the median for following data.

1. Find the median of the following distribution.

| $x$ | $f$ |
|-----|-----|
| 10  | 3   |
| 20  | 5   |
| 30  | 8   |
| 40  | 6   |
| 50  | 3   |

2. Find the median of the following distribution.

| Marks ( $x$ ) | Frequency ( $f$ ) |
|---------------|-------------------|
| 15            | 4                 |
| 20            | 6                 |
| 25            | 10                |
| 30            | 8                 |
| 35            | 2                 |

3. Find the median

| Number of Books | Frequency |
|-----------------|-----------|
| 1               | 5         |
| 2               | 8         |
| 3               | 12        |
| 4               | 10        |
| 5               | 5         |

4. Find the median

| Age (Years) | Frequency |
|-------------|-----------|
| 18          | 4         |
| 19          | 6         |
| 20          | 9         |
| 21          | 5         |
| 22          | 2         |

5. Find the median

| Defective Items | Frequency |
|-----------------|-----------|
| 0               | 3         |
| 1               | 7         |
| 2               | 12        |
| 3               | 8         |
| 4               | 5         |

**Answers:**

1. Median = 30
2. Median = 25
3. Median = 3
4. Median = 20
5. Median = 2

### Median for Continuous Frequency Distribution

The median of a continuous frequency distribution is given by

$$\text{Median} = L + \left( \frac{\frac{N}{2} - C}{f} \right) h$$

where

| Symbol | Meaning                                             |
|--------|-----------------------------------------------------|
| $L$ :  | Lower boundary (or lower limit) of the median class |
| $N$ :  | Total frequency ( $\sum f$ )                        |
| $C$ :  | Cumulative frequency preceding the median class     |
| $f$ :  | Frequency of the median class                       |
| $h$ :  | Width of the median class                           |

The **Median Class** is the class interval whose cumulative frequency is equal to or immediately greater than

$$\boxed{\frac{N}{2}} \quad \text{where} \quad N = \sum f.$$

### Example 1

Find the median of the following distribution.

| Class Interval | Frequency ( $f$ ) | Cumulative Frequency |
|----------------|-------------------|----------------------|
| 0–10           | 5                 | 5                    |
| 10–20          | 8                 | 13                   |
| 20–30          | 12                | 25                   |
| 30–40          | 10                | 35                   |
| 40–50          | 5                 | 40                   |

### Solution

Total frequency

$$N = 40$$

Therefore,

$$\frac{N}{2} = 20$$

The first cumulative frequency greater than or equal to 20 is

$$25.$$

Hence, the median class is

$$20 - 30.$$

Thus,

$$L = 20, \quad C = 13, \quad f = 12, \quad h = 10.$$

Using the formula,

$$\begin{aligned} \text{Median} &= 20 + \left( \frac{20 - 13}{12} \right) (10) \\ &= 20 + \frac{70}{12} = 25.83 \end{aligned}$$

$$\boxed{\text{Median} = 25.83}$$

**Example 2**

Calculate the median.

| Marks | Frequency | CF |
|-------|-----------|----|
| 10–20 | 4         | 4  |
| 20–30 | 6         | 10 |
| 30–40 | 15        | 25 |
| 40–50 | 10        | 35 |
| 50–60 | 5         | 40 |

**Solution**

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

$$30 - 40$$

Therefore,

$$L = 30, \quad C = 10, \quad f = 15, \quad h = 10$$

$$\begin{aligned} \text{Median} &= 30 + \left( \frac{20 - 10}{15} \right) (10) \\ &= 30 + 6.67 = 36.67 \end{aligned}$$

|                |
|----------------|
| Median = 36.67 |
|----------------|

**Example 3**

Find the median.

| Income    | Frequency | CF |
|-----------|-----------|----|
| 1000–2000 | 5         | 5  |
| 2000–3000 | 9         | 14 |
| 3000–4000 | 16        | 30 |
| 4000–5000 | 10        | 40 |
| 5000–6000 | 10        | 50 |

**Solution**

$$N = 50, \quad \frac{N}{2} = 25$$

Median class

$$3000 - 4000$$

Hence,

$$L = 3000, \quad C = 14, \quad f = 16, \quad h = 1000$$

$$\begin{aligned}\text{Median} &= 3000 + \left( \frac{25 - 14}{16} \right) 1000 \\ &= 3000 + 687.5 = 3687.5\end{aligned}$$

$$\boxed{\text{Median} = 3687.5}$$

#### Example 4

The following table shows the ages of employees.

| Age   | Frequency | CF |
|-------|-----------|----|
| 20–25 | 6         | 6  |
| 25–30 | 8         | 14 |
| 30–35 | 12        | 26 |
| 35–40 | 9         | 35 |
| 40–45 | 5         | 40 |

Find the median age.

#### Solution

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

30 – 35

Therefore,

$$L = 30, \quad C = 14, \quad f = 12, \quad h = 5$$

$$\text{Median} = 30 + \left( \frac{20 - 14}{12} \right) 5 = 32.5$$

$$\boxed{\text{Median} = 32.5 \text{ years}}$$

#### Example 5

Find the median of the following grouped data.

| Class Interval | Frequency | CF |
|----------------|-----------|----|
| 50–60          | 3         | 3  |
| 60–70          | 7         | 10 |
| 70–80          | 12        | 22 |
| 80–90          | 10        | 32 |
| 90–100         | 8         | 40 |

### Solution

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

$$70 - 80$$

Hence,

$$L = 70, \quad C = 10, \quad f = 12, \quad h = 10$$

$$\text{Median} = 70 + \left( \frac{20 - 10}{12} \right) 10 = 78.33$$

$$\boxed{\text{Median} = 78.33}$$

### Partition Value

Measures such as the **median** divide a dataset into two equal parts. Sometimes, it is useful to divide the data into more than two equal parts to understand the distribution in greater detail. These values are known as **Partition Values**.

**Partition values** are positional measures that divide an ordered dataset into a specified number of equal parts.

| Partition Value | Number of Equal Parts | Number of Partition Points |
|-----------------|-----------------------|----------------------------|
| Median          | 2                     | 1                          |
| Quartiles       | 4                     | 3                          |
| Deciles         | 10                    | 9                          |
| Percentiles     | 100                   | 99                         |

### Quartiles

**Quartiles** divide an ordered dataset into four equal parts.

There are three quartiles.

- $Q_1$  : First Quartile (25%)
- $Q_2$  : Second Quartile (Median, 50%)
- $Q_3$  : Third Quartile (75%)

To find the quartiles, arrange the observations in ascending order.

If there are  $n$  observations,

$$\boxed{Q_1 = \left( \frac{n+1}{4} \right)^{th} \text{ observation}}$$

$$Q_2 = \left( \frac{n+1}{2} \right)^{th} \text{ observation}$$

$$Q_3 = \left( \frac{3(n+1)}{4} \right)^{th} \text{ observation}$$

### Example

Find the quartiles of

18, 20, 12, 15, 22, 27, 25, 30, 35

### Solution

The ordered data is

12, 15, 18, 20, 22, 25, 27, 30, 35

$$n = 9$$

First Quartile

$$Q_1 = \frac{9+1}{4} = 2.5^{th} \text{ observation}$$

$$Q_1 = \frac{15+18}{2} = 16.5$$

Second Quartile

$$Q_2 = \frac{10}{2} = 5^{th} \text{ observation} = 22$$

Third Quartile

$$Q_3 = \frac{3(10)}{4} = 7.5^{th} \text{ observation}$$

$$Q_3 = \frac{27+30}{2} = 28.5$$

Hence,

$$Q_1 = 16.5, \quad Q_2 = 22, \quad Q_3 = 28.5$$

### Quartiles for Discrete Frequency Distribution

For a discrete frequency distribution,

$$N = \sum f$$

The quartile positions are

$$Q_1 = \frac{N}{4}$$

$$Q_2 = \frac{N}{2}$$

$$Q_3 = \frac{3N}{4}$$

Locate these positions using the cumulative frequencies.

| Measure                   | Position       | Required Value                                              |
|---------------------------|----------------|-------------------------------------------------------------|
| First Quartile ( $Q_1$ )  | $\frac{N}{4}$  | Observation whose CF is the first to be $\geq \frac{N}{4}$  |
| Second Quartile ( $Q_2$ ) | $\frac{N}{2}$  | Median                                                      |
| Third Quartile ( $Q_3$ )  | $\frac{3N}{4}$ | Observation whose CF is the first to be $\geq \frac{3N}{4}$ |

### Example

Find the quartiles.

| $x$ | $f$ | CF |
|-----|-----|----|
| 10  | 4   | 4  |
| 20  | 6   | 10 |
| 30  | 8   | 18 |
| 40  | 5   | 23 |
| 50  | 2   | 25 |

### Solution

$$N = 25$$

$$Q_1 = \frac{25}{4} = 6.25$$

The first CF greater than 6.25 is

$$10$$

Therefore,

$$Q_1 = 20$$

Similarly,

$$Q_2 = \frac{25}{2} = 12.5$$

First CF greater than 12.5 is

$$18$$

Hence,

$$Q_2 = 30$$

Finally,

$$Q_3 = \frac{75}{4} = 18.75$$

First CF greater than 18.75 is

$$23$$

Therefore,

$$Q_3 = 40$$

Hence,

$$Q_1 = 20, \quad Q_2 = 30, \quad Q_3 = 40$$

### Deciles

Deciles divide an ordered dataset into **ten equal parts**.

### Ungrouped Data

$$D_k = \left( \frac{k(n+1)}{10} \right)^{th} \text{ observation}$$

where

$$k = 1, 2, \dots, 9.$$

### Example

Find the fifth decile of

$$8, 10, 12, 14, 16, 18, 20, 22, 24$$

### Solution

$$n = 9$$

$$D_5 = \frac{5(10)}{10} = 5^{th} \text{ observation}$$

The fifth observation is

$$16$$

Therefore,

$$D_5 = 16$$

### Discrete Frequency Distribution

For discrete data,

$$D_k = \frac{kN}{10}$$

Locate the corresponding cumulative frequency.

| Measure                          | Position        | Required Value                                               |
|----------------------------------|-----------------|--------------------------------------------------------------|
| $k^{\text{th}}$ Decile ( $D_k$ ) | $\frac{kN}{10}$ | Observation whose CF is the first to be $\geq \frac{kN}{10}$ |

### Example

Find  $D_7$  from the following data.

| $x$ | $f$ | CF |
|-----|-----|----|
| 5   | 3   | 3  |
| 10  | 5   | 8  |
| 15  | 6   | 14 |
| 20  | 4   | 18 |
| 25  | 2   | 20 |

### Solution

$$N = 20$$

$$D_7 = \frac{7(20)}{10} = 14$$

The first cumulative frequency greater than or equal to 14 is

$$14$$

Therefore,

$$D_7 = 15$$

### Percentiles

Percentiles divide a dataset into **100 equal parts**.

### Ungrouped Data

$$P_k = \left( \frac{k(n+1)}{100} \right)^{th} \text{ observation}$$

where

$$k = 1, 2, \dots, 99.$$

### Example

Find

$$P_{60}$$

of the following observations.

$$5, 8, 10, 12, 15, 18, 20, 24, 28, 30$$

### Solution

$$n = 10$$

Position

$$P_{60} = \frac{60(11)}{100} = 6.6^{th} \text{ observation}$$

The sixth and seventh observations are

$$18, 20.$$

Hence,

$$P_{60} = 18 + 0.6(20 - 18) = 18 + 1.2 = 19.2$$

Therefore,

$$P_{60} = 19.2$$

### Discrete Frequency Distribution

$$P_k = \frac{kN}{100}$$

Locate the corresponding cumulative frequency.

| Measure                       | Position         | Required Value                                                |
|-------------------------------|------------------|---------------------------------------------------------------|
| $k^{th}$ Percentile ( $P_k$ ) | $\frac{kN}{100}$ | Observation whose CF is the first to be $\geq \frac{kN}{100}$ |

### Example

Find  $D_5$ ,  $D_7$ ,  $D_9$ .

|     |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|
| $x$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 |
| $f$ | 2  | 3  | 5  | 4  | 3  | 2  | 1  |

### Solution

| $x$   | $f$      | CF |
|-------|----------|----|
| 10    | 2        | 2  |
| 20    | 3        | 5  |
| 30    | 5        | 10 |
| 40    | 4        | 14 |
| 50    | 3        | 17 |
| 60    | 2        | 19 |
| 70    | 1        | 20 |
| Total | $N = 20$ |    |

| Measure                          | Position Formula | Position                      | Required Value |
|----------------------------------|------------------|-------------------------------|----------------|
| 5 <sup>th</sup> Decile ( $D_5$ ) | $\frac{5N}{10}$  | $\frac{5 \times 20}{10} = 10$ | 30             |
| 7 <sup>th</sup> Decile ( $D_7$ ) | $\frac{7N}{10}$  | $\frac{7 \times 20}{10} = 14$ | 40             |
| 9 <sup>th</sup> Decile ( $D_9$ ) | $\frac{9N}{10}$  | $\frac{9 \times 20}{10} = 18$ | 60             |

## Continuous Frequency Distribution

### Median

The median of a continuous frequency distribution is given by

$$\text{Median} = L + \left( \frac{\frac{N}{2} - C}{f} \right) h$$

where

| Symbol | Meaning                                             |
|--------|-----------------------------------------------------|
| $L$    | Lower boundary (or lower limit) of the median class |
| $N$    | Total frequency ( $\sum f$ )                        |
| $C$    | Cumulative frequency preceding the median class     |
| $f$    | Frequency of the median class                       |
| $h$    | Width of the median class                           |

### Example 1

Find the median of the following distribution.

| Class Interval | Frequency ( $f$ ) | Cumulative Frequency |
|----------------|-------------------|----------------------|
| 0–10           | 5                 | 5                    |
| 10–20          | 8                 | 13                   |
| 20–30          | 12                | 25                   |
| 30–40          | 10                | 35                   |
| 40–50          | 5                 | 40                   |

### Solution

Total frequency

$$N = 40$$

Therefore,

$$\frac{N}{2} = 20$$

The first cumulative frequency greater than or equal to 20 is

25.

Hence, the median class is

20 – 30.

Thus,

$$L = 20, \quad C = 13, \quad f = 12, \quad h = 10.$$

Using the formula,

$$\begin{aligned}\text{Median} &= 20 + \left( \frac{20 - 13}{12} \right) (10) \\ &= 20 + \frac{70}{12} = 25.83\end{aligned}$$

|                |
|----------------|
| Median = 25.83 |
|----------------|

### Example 2

Calculate the median.

| Marks | Frequency | CF |
|-------|-----------|----|
| 10-20 | 4         | 4  |
| 20-30 | 6         | 10 |
| 30-40 | 15        | 25 |
| 40-50 | 10        | 35 |
| 50-60 | 5         | 40 |

### Solution

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

$$30 - 40$$

Therefore,

$$L = 30, \quad C = 10, \quad f = 15, \quad h = 10$$

$$\begin{aligned}\text{Median} &= 30 + \left( \frac{20 - 10}{15} \right) (10) \\ &= 30 + 6.67 = 36.67\end{aligned}$$

|                |
|----------------|
| Median = 36.67 |
|----------------|

**Example 3**

Find the median.

| Income    | Frequency | CF |
|-----------|-----------|----|
| 1000–2000 | 5         | 5  |
| 2000–3000 | 9         | 14 |
| 3000–4000 | 16        | 30 |
| 4000–5000 | 10        | 40 |
| 5000–6000 | 10        | 50 |

**Solution**

$$N = 50, \quad \frac{N}{2} = 25$$

Median class

$$3000 - 4000$$

Hence,

$$L = 3000, \quad C = 14, \quad f = 16, \quad h = 1000$$

$$\begin{aligned} \text{Median} &= 3000 + \left( \frac{25 - 14}{16} \right) 1000 \\ &= 3000 + 687.5 = 3687.5 \end{aligned}$$

$$\boxed{\text{Median} = 3687.5}$$

**Example 4**

The following table shows the ages of employees.

| Age   | Frequency | CF |
|-------|-----------|----|
| 20–25 | 6         | 6  |
| 25–30 | 8         | 14 |
| 30–35 | 12        | 26 |
| 35–40 | 9         | 35 |
| 40–45 | 5         | 40 |

Find the median age.

**Solution**

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

$$30 - 35$$

Therefore,

$$L = 30, \quad C = 14, \quad f = 12, \quad h = 5$$

$$\text{Median} = 30 + \left( \frac{20 - 14}{12} \right) 5 = 32.5$$

|                     |
|---------------------|
| Median = 32.5 years |
|---------------------|

**Example 5**

Find the median of the following grouped data.

| Class Interval | Frequency | CF |
|----------------|-----------|----|
| 50-60          | 3         | 3  |
| 60-70          | 7         | 10 |
| 70-80          | 12        | 22 |
| 80-90          | 10        | 32 |
| 90-100         | 8         | 40 |

**Solution**

$$N = 40, \quad \frac{N}{2} = 20$$

Median class

$$70 - 80$$

Hence,

$$L = 70, \quad C = 10, \quad f = 12, \quad h = 10$$

$$\text{Median} = 70 + \left( \frac{20 - 10}{12} \right) 10 = 78.33$$

|                |
|----------------|
| Median = 78.33 |
|----------------|

## Quartile

$$Q_1 = L + \left( \frac{\frac{N}{4} - C}{f} \right) h \quad \text{where}$$

$$Q_2 = L + \left( \frac{\frac{N}{2} - C}{f} \right) h$$

$$Q_3 = L + \left( \frac{\frac{3N}{4} - C}{f} \right) h$$

| Symbol | Meaning                                               |
|--------|-------------------------------------------------------|
| $L$    | Lower boundary (or lower limit) of the required class |
| $C$    | Cumulative frequency preceding the required class     |
| $f$    | Frequency of the required class                       |
| $h$    | Width of the class interval                           |

### Example 1

Find the quartiles.

| Class | $f$ | CF |
|-------|-----|----|
| 0–10  | 5   | 5  |
| 10–20 | 8   | 13 |
| 20–30 | 12  | 25 |
| 30–40 | 10  | 35 |
| 40–50 | 5   | 40 |

### Solution

$$N = 40$$

### First Quartile

$$\frac{N}{4} = 10$$

The quartile class is

$$10 - 20$$

Hence,

$$L = 10, C = 5, f = 8, h = 10$$

$$Q_1 = 10 + \left( \frac{10 - 5}{8} \right) 10 = 16.25$$

### Second Quartile

$$\frac{N}{2} = 20$$

Quartile class

$$20 - 30$$

$$L = 20, C = 13, f = 12$$

$$Q_2 = 20 + \frac{20 - 13}{12} \times 10 = 25.83$$

### Third Quartile

$$\frac{3N}{4} = 30$$

Quartile class

$$30 - 40$$

$$L = 30, C = 25, f = 10$$

$$Q_3 = 30 + \frac{30 - 25}{10} \times 10 = 35$$

Therefore,

|                |                |            |
|----------------|----------------|------------|
| $Q_1 = 16.25,$ | $Q_2 = 25.83,$ | $Q_3 = 35$ |
|----------------|----------------|------------|

### Example 2

Find the third quartile.

| Class  | Frequency | CF |
|--------|-----------|----|
| 50–60  | 4         | 4  |
| 60–70  | 8         | 12 |
| 70–80  | 12        | 24 |
| 80–90  | 10        | 34 |
| 90–100 | 6         | 40 |

### Solution

$$N = 40$$

$$\frac{3N}{4} = 30$$

The quartile class is

$$80 - 90$$

Thus,

$$L = 80, C = 24, f = 10, h = 10$$

$$Q_3 = 80 + \frac{30 - 24}{10} \times 10 = 86$$

$$\boxed{Q_3 = 86}$$

### Decile

The  $k^{th}$  decile is

and

$$D_k = L + \left( \frac{\frac{kN}{10} - C}{f} \right) h$$

where

$$k = 1, 2, \dots, 9.$$

| Symbol | Meaning                                               |
|--------|-------------------------------------------------------|
| $L$    | Lower boundary (or lower limit) of the required class |
| $C$    | Cumulative frequency preceding the required class     |
| $f$    | Frequency of the required class                       |
| $h$    | Width of the class interval                           |

### Example 1

Find the sixth decile.

| Class | Frequency | CF |
|-------|-----------|----|
| 10–20 | 5         | 5  |
| 20–30 | 8         | 13 |
| 30–40 | 12        | 25 |
| 40–50 | 10        | 35 |
| 50–60 | 5         | 40 |

**Solution**

$$N = 40$$

$$D_6 = \frac{6N}{10} = 24$$

The decile class is

$$30 - 40$$

Therefore,

$$L = 30, C = 13, f = 12, h = 10$$

$$D_6 = 30 + \frac{24 - 13}{12} \times 10 = 39.17$$

|               |
|---------------|
| $D_6 = 39.17$ |
|---------------|

**Example 2**

Find the second decile.

| Class | Frequency | CF |
|-------|-----------|----|
| 0-10  | 3         | 3  |
| 10-20 | 7         | 10 |
| 20-30 | 12        | 22 |
| 30-40 | 10        | 32 |
| 40-50 | 8         | 40 |

**Solution**

$$N = 40$$

$$D_2 = \frac{2(40)}{10} = 8$$

The decile class is

$$10 - 20$$

$$L = 10, C = 3, f = 7, h = 10$$

$$D_2 = 10 + \frac{8 - 3}{7} \times 10 = 17.14$$

|               |
|---------------|
| $D_2 = 17.14$ |
|---------------|

## Percentile

The  $k^{th}$  percentile is

and

$$P_k = L + \left( \frac{\frac{kN}{100} - C}{f} \right) h$$

where

$$k = 1, 2, \dots, 99.$$

| Symbol | Meaning                                               |
|--------|-------------------------------------------------------|
| $L$    | Lower boundary (or lower limit) of the required class |
| $C$    | Cumulative frequency preceding the required class     |
| $f$    | Frequency of the required class                       |
| $h$    | Width of the class interval                           |

### Example 1

Find the 40th percentile.

| Class | Frequency | CF |
|-------|-----------|----|
| 20–30 | 6         | 6  |
| 30–40 | 8         | 14 |
| 40–50 | 12        | 26 |
| 50–60 | 9         | 35 |
| 60–70 | 5         | 40 |

### Solution

$$N = 40$$

$$P_{40} = \frac{40 \times 40}{100} = 16$$

The percentile class is

$$40 - 50$$

Hence,

$$L = 40, C = 14, f = 12, h = 10$$

$$P_{40} = 40 + \frac{16 - 14}{12} \times 10 = 41.67$$

$$P_{40} = 41.67$$

**Example 2**

Find the 75th percentile.

| Class   | Frequency | CF |
|---------|-----------|----|
| 100–120 | 5         | 5  |
| 120–140 | 10        | 15 |
| 140–160 | 15        | 30 |
| 160–180 | 12        | 42 |
| 180–200 | 8         | 50 |

**Solution**

$$N = 50$$

$$P_{75} = \frac{75 \times 50}{100} = 37.5$$

The percentile class is

$$160 - 180$$

Therefore,

$$L = 160, C = 30, f = 12, h = 20$$

$$P_{75} = 160 + \frac{37.5 - 30}{12} \times 20 = 172.5$$

$$\boxed{P_{75} = 172.5}$$

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Measures of Central Tendency

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Mode

The **Mode** is the value that occurs **most frequently** in a data set. It represents the observation with the highest frequency and is one of the three important measures of central tendency.

Mode = Value having the highest frequency

### Example:

Data: 2, 4, 5, 5, 6, 7, 5, 8

Since 5 occurs three times, which is more than any other value,

Mode = 5

## Mode for Ungrouped Data

Arrange the observations (if necessary) and identify the value that appears most frequently.

### Example

Find the mode of

8, 10, 12, 15, 12, 18, 20, 12, 25, 18

### Solution

Frequency of each observation:

|     |   |    |    |    |    |    |    |
|-----|---|----|----|----|----|----|----|
| $x$ | 8 | 10 | 12 | 15 | 18 | 20 | 25 |
| $f$ | 1 | 1  | 3  | 1  | 2  | 1  | 1  |

Since 12 has the highest frequency,

Mode = 12

## Mode for Discrete Frequency Distribution

For a discrete frequency distribution, the observation having the highest frequency is the mode.

### Example

| $x$ | Frequency ( $f$ ) |
|-----|-------------------|
| 10  | 4                 |
| 20  | 8                 |
| 30  | 15                |
| 40  | 10                |
| 50  | 3                 |

The highest frequency is 15, corresponding to  $x = 30$ .

$$\text{Mode} = 30$$

### Mode for Continuous Frequency Distribution

For grouped continuous data, the exact modal value is estimated using

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where

$L$  Lower boundary of modal class

$f_1$  Frequency of modal class

$f_0$  Frequency of preceding class

$f_2$  Frequency of succeeding class

$h$  Class width

### Example

| Class | Frequency |
|-------|-----------|
| 0–10  | 5         |
| 10–20 | 8         |
| 20–30 | 14        |
| 30–40 | 10        |
| 40–50 | 6         |

The modal class is 20–30.

Hence,

$$L = 20, \quad f_1 = 14, \quad f_0 = 8, \quad f_2 = 10, \quad h = 10$$

$$\begin{aligned}
 \text{Mode} &= 20 + \frac{14 - 8}{2(14) - 8 - 10} \times 10 \\
 &= 20 + \frac{6}{10} \times 10 \\
 &= 20 + 6 \\
 &= 26
 \end{aligned}$$

|           |
|-----------|
| Mode = 26 |
|-----------|

Dr. Boina Anil Kumar

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## Geometric Mean (GM)

The **Geometric Mean (GM)** is the average obtained by multiplying all observations and taking the  $n^{\text{th}}$  root of the product.

It is particularly useful when dealing with:

- Growth rates
- Compound interest
- Population growth
- Financial returns
- Percentage changes

## Individual Observations

For observations

$$x_1, x_2, \dots, x_n,$$

the geometric mean is

$$GM = \sqrt[n]{x_1 x_2 \cdots x_n}$$

or equivalently,

$$\log(GM) = \frac{1}{n} \sum_{i=1}^n \log x_i$$

## Discrete Frequency Distribution

If observations have frequencies  $f_i$ ,

$$GM = \text{Antilog} \left( \frac{\sum f \log x}{\sum f} \right)$$

## Example

Find the geometric mean of

2, 4, 8.

**Solution**

$$GM = \sqrt[3]{2 \times 4 \times 8} = \sqrt[3]{64} = 4$$

$$\boxed{GM = 4}$$

**Example**

Find the geometric mean of the numbers

$$4, 16, 64.$$

**Solution**

$$GM = \sqrt[3]{4 \times 16 \times 64} = \sqrt[3]{4096} = 16.$$

$$\boxed{GM = 16}$$

**Example**

Find the geometric mean of

$$3, 12, 48.$$

**Solution**

$$GM = \sqrt[3]{3 \times 12 \times 48} = \sqrt[3]{1728} = 12.$$

$$\boxed{GM = 12}$$

**Example**

Find the geometric mean of the following data.

|     |   |   |   |    |
|-----|---|---|---|----|
| $x$ | 2 | 4 | 8 | 16 |
| $f$ | 1 | 2 | 3 | 2  |

**Solution**

$$N = \sum f = 8$$

| $x$ | $f$ | $f \log x$ |
|-----|-----|------------|
| 2   | 1   | 0.3010     |
| 4   | 2   | 1.2040     |
| 8   | 3   | 2.7093     |
| 16  | 2   | 2.4082     |
|     |     | 6.6225     |

$$\log(GM) = \frac{6.6225}{8} = 0.8278$$

$$GM = \text{Antilog}(0.8278) \approx 6.72$$

$$\boxed{GM \approx 6.72}$$

### Harmonic Mean (HM)

The **Harmonic Mean (HM)** is the reciprocal of the arithmetic mean of reciprocals of the observations.

It is useful when observations are expressed as rates.

Examples:

- Speed
- Cost per unit
- Price per kilogram
- Fuel efficiency

### Individual Observations

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

### Discrete Frequency Distribution

$$HM = \frac{\sum f}{\sum \frac{f}{x}}$$

### Example

A car travels

- 60 km at 40 km/h
- 60 km at 60 km/h

Find the average speed.

### Solution

Since equal distances are covered,

$$HM = \frac{2}{\frac{1}{40} + \frac{1}{60}} = 48 \text{ km/h}$$

$$\boxed{HM = 48 \text{ km/h}}$$

**Example**

Find the harmonic mean of

2, 3, 6.

**Solution**

$$HM = \frac{3}{\frac{1}{2} + \frac{1}{3} + \frac{1}{6}} = \frac{3}{1} = 3.$$

$$\boxed{HM = 3}$$

**Example**

A car travels

- 120 km at 60 km/h,
- 120 km at 40 km/h.

Find the average speed.

**Solution**

Since equal distances are covered,

$$HM = \frac{2}{\frac{1}{60} + \frac{1}{40}} = 48 \text{ km/h.}$$

$$\boxed{48 \text{ km/h}}$$

**Example 3**

Find the harmonic mean of the following data.

|     |   |   |   |    |
|-----|---|---|---|----|
| $x$ | 2 | 4 | 5 | 10 |
| $f$ | 3 | 2 | 4 | 1  |

**Solution**

$$N = \sum f = 10$$

| $x$ | $f$ | $\frac{f}{x}$ |
|-----|-----|---------------|
| 2   | 3   | 1.50          |
| 4   | 2   | 0.50          |
| 5   | 4   | 0.80          |
| 10  | 1   | 0.10          |
|     |     | 2.90          |

$$HM = \frac{10}{2.90} = 3.45.$$

$$HM \approx 3.45$$

### Important Points

- GM is suitable for multiplicative data.
- HM is suitable for rates.
- GM and HM require all observations to be positive.
- For positive data:

$$AM \geq GM \geq HM.$$

Dr. Boina Anil Kumar

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Measures of Central Tendency

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Empirical Relation between Mean, Median and Mode

The empirical relation is used to estimate one measure of central tendency when the other two are known. This relation is applicable for **moderately skewed distributions**.

## Empirical Formula

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

Equivalent forms of the relation are

$$\text{Median} = \frac{\text{Mode} + 2(\text{Mean})}{3}$$

and

$$\text{Mean} = \frac{3(\text{Median}) - \text{Mode}}{2}$$

## Note

The above relation is only an approximation and should be used only for moderately skewed distributions. It is not valid for highly skewed or multimodal distributions.

## Example 1

The mean of a distribution is 40 and the median is 36. Find the mode.

## Solution

Using the empirical relation,

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

$$= 3(36) - 2(40)$$

$$= 108 - 80$$

$$= 28.$$

$$\text{Mode} = 28$$

**Example 2**

The mean is 25 and the mode is 19. Find the median.

**Solution**

Using

$$\begin{aligned}\text{Median} &= \frac{\text{Mode} + 2(\text{Mean})}{3} \\ &= \frac{19 + 2(25)}{3} = \frac{69}{3} = 23.\end{aligned}$$

$$\boxed{\text{Median} = 23}$$

**Example 3**

The median of a distribution is 35 and the mode is 29. Find the mean.

**Solution**

Using

$$\begin{aligned}\text{Mean} &= \frac{3(\text{Median}) - \text{Mode}}{2} \\ &= \frac{3(35) - 29}{2} = \frac{105 - 29}{2} = \frac{76}{2} = 38.\end{aligned}$$

$$\boxed{\text{Mean} = 38}$$

**Example 4**

The mean is 52 and the mode is 46. Find the median.

**Solution**

$$\begin{aligned}\text{Median} &= \frac{\text{Mode} + 2(\text{Mean})}{3} \\ &= \frac{46 + 2(52)}{3} = \frac{150}{3} = 50.\end{aligned}$$

$$\boxed{\text{Median} = 50}$$

**Example 5**

The mean of a distribution is 32.5 and the median is 30. Find the mode.

### Solution

$$\begin{aligned}\text{Mode} &= 3(\text{Median}) - 2(\text{Mean}) \\ &= 3(30) - 2(32.5) = 90 - 65 = 25.\end{aligned}$$

$$\boxed{\text{Mode} = 25}$$

### Example 6

The median is 44 and the mode is 38. Find the mean.

### Solution

$$\begin{aligned}\text{Mean} &= \frac{3(\text{Median}) - \text{Mode}}{2} \\ &= \frac{3(44) - 38}{2} = \frac{132 - 38}{2} = \frac{94}{2} = 47.\end{aligned}$$

$$\boxed{\text{Mean} = 47}$$

### Practice Problems

Use the empirical relation to solve the following.

1. Mean = 48, Median = 45. Find the Mode.
2. Mean = 36, Mode = 30. Find the Median.
3. Median = 40, Mode = 34. Find the Mean.
4. Mean = 65, Median = 61. Find the Mode.
5. Mean = 27.5, Mode = 21.5. Find the Median.

### Answers

1. Mode = 39
2. Median = 34
3. Mean = 43
4. Mode = 53
5. Median = 25.5

### Example

Calculate the median of the following distribution.

| Annual Sales (Rs. '000) | Frequency | Annual Sales (Rs. '000) | Frequency |
|-------------------------|-----------|-------------------------|-----------|
| Less than 10            | 4         | Less than 40            | 55        |
| Less than 20            | 20        | Less than 50            | 62        |
| Less than 30            | 35        | Less than 60            | 67        |

Is it possible to calculate the arithmetic mean? If possible, calculate it. Hence obtain an estimate of the mode.

### Solution

The given data are in the form of a *less than cumulative frequency distribution*. First, convert the cumulative frequencies into an ordinary continuous frequency distribution.

| Annual Sales<br>( '000 Rs.) | Mid-value<br>( $X$ ) | Frequency<br>( $f$ ) | Less than<br>c.f. | $d = \frac{X - 35}{10}$ | $fd$            |
|-----------------------------|----------------------|----------------------|-------------------|-------------------------|-----------------|
| 0–10                        | 5                    | 4                    | 4                 | –3                      | –12             |
| 10–20                       | 15                   | $20 - 4 = 16$        | 20                | –2                      | –32             |
| 20–30                       | 25                   | $35 - 20 = 15$       | 35                | –1                      | –15             |
| 30–40                       | 35                   | $55 - 35 = 20$       | 55                | 0                       | 0               |
| 40–50                       | 45                   | $62 - 55 = 7$        | 62                | 1                       | 7               |
| 50–60                       | 55                   | $67 - 62 = 5$        | 67                | 2                       | 10              |
| <b>Total</b>                |                      | $\sum f = N = 67$    |                   |                         | $\sum fd = -42$ |

### Median

The total frequency is

$$N = 67.$$

Hence,

$$\frac{N}{2} = \frac{67}{2} = 33.5.$$

The cumulative frequency just greater than 33.5 is 35.

Therefore, the corresponding class

$$20-30$$

is the **median class**.

Using the median formula,

$$\text{Median} = L + \frac{\left(\frac{N}{2} - C\right)}{f} \times h,$$

where

$$L = 20, \quad C = 20, \quad f = 15, \quad h = 10.$$

Substituting these values,

$$\begin{aligned} \text{Median} &= 20 + \frac{(33.5 - 20)}{15} \times 10 \\ &= 20 + \frac{13.5 \times 10}{15} \\ &= 20 + 9 \\ &= 29. \end{aligned}$$

Hence,

$$\boxed{\text{Median Sales} = 29 \text{ ('000 Rs.)}}$$

or

$$\boxed{\text{Median Sales} = \text{Rs. } 29,000.}$$

### Arithmetic Mean

Since the grouped frequency distribution is available, the arithmetic mean can also be calculated by the **Step Deviation Method**.

Here,

$$A = 35, \quad h = 10, \quad N = 67, \quad \sum fd = -42.$$

Using the formula,

$$\bar{X} = A + \frac{h \sum fd}{N},$$

we get

$$\begin{aligned} \bar{X} &= 35 + \frac{10(-42)}{67} \\ &= 35 - \frac{420}{67} \\ &= 35 - 6.269 \\ &= 28.731. \end{aligned}$$

Hence,

$$\boxed{\text{Mean Sales} = 28.731 \text{ ('000 Rs.)}}$$

or

$$\text{Mean Sales} = \text{Rs. } 28,731.$$

### Mode

For a moderately skewed distribution, the empirical relation between the mean, median and mode is

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

From the previous calculations,

$$\text{Median} = 29 \text{ ('000 Rs.)},$$

and

$$\text{Mean} = 28.731 \text{ ('000 Rs.)}.$$

Substituting these values into the empirical formula,

$$\begin{aligned}\text{Mode} &= 3(29) - 2(28.731) \\ &= 87 - 57.462 \\ &= 29.538.\end{aligned}$$

Hence,

$$\text{Mode Sales} = 29.538 \text{ ('000 Rs.)}$$

or

$$\text{Mode Sales} = \text{Rs. } 29,538.$$

### Result

Thus, the three measures of central tendency are

$$\begin{aligned}\text{Median} &= \text{Rs. } 29,000, \\ \text{Mean} &= \text{Rs. } 28,731, \\ \text{Mode} &= \text{Rs. } 29,538.\end{aligned}$$

## Problem Discussion

### Question-1

The numbers 3.2, 5.8, 7.9 and 4.5 have frequencies  $x$ ,  $(x + 2)$ ,  $(x - 3)$  and  $(x + 6)$  respectively. If the arithmetic mean is 4.876, find the value of  $x$ .

### Solution

Let the observations and their corresponding frequencies be as follows.

| Number ( $X$ ) | Frequency ( $f$ ) | $fX$         |
|----------------|-------------------|--------------|
| 3.2            | $x$               | $3.2x$       |
| 5.8            | $x + 2$           | $5.8(x + 2)$ |
| 7.9            | $x - 3$           | $7.9(x - 3)$ |
| 4.5            | $x + 6$           | $4.5(x + 6)$ |

Hence,

$$\sum f = x + (x + 2) + (x - 3) + (x + 6) = 4x + 5.$$

Also,

$$\begin{aligned}\sum fX &= 3.2x + 5.8(x + 2) + 7.9(x - 3) + 4.5(x + 6) \\ &= (3.2 + 5.8 + 7.9 + 4.5)x + 11.6 - 23.7 + 27 \\ &= 21.4x + 14.9.\end{aligned}$$

Since,

$$\bar{X} = \frac{\sum fX}{\sum f},$$

and the arithmetic mean is given as 4.876,

$$\frac{21.4x + 14.9}{4x + 5} = 4.876.$$

Therefore,

$$21.4x + 14.9 = 4.876(4x + 5).$$

Expanding,

$$21.4x + 14.9 = 19.504x + 24.380.$$

Rearranging,

$$21.4x - 19.504x = 24.380 - 14.900,$$

$$1.896x = 9.480.$$

Hence,

$$x = \frac{9.480}{1.896} = 5.$$

$$\boxed{x = 5}$$

### Question-2

In the following grouped data,  $X$  are the mid-values of the class intervals and  $c$  is an arbitrary constant. If the arithmetic mean of the original distribution is 35.84, find the corresponding class intervals.

| $X - c$ | -21 | -14 | -7 | 0  | 7  | 14 | 21 | Total |
|---------|-----|-----|----|----|----|----|----|-------|
| $f$     | 2   | 12  | 19 | 29 | 20 | 13 | 5  | 100   |

### Solution

Here,  $X - c$  represents the deviation of the mid-value  $X$  from the arbitrary constant  $c$ . Thus,

$$d = X - c.$$

The arithmetic mean is given by

$$\bar{X} = c + \frac{\sum fd}{N} = c + \frac{\sum f(X - c)}{N},$$

where

$$N = \sum f.$$

| $X - c$      | $f$            | $f(X - c)$                             | $X$ | Class Interval |
|--------------|----------------|----------------------------------------|-----|----------------|
| -21          | 2              | -42                                    | 14  | 10.5-17.5      |
| -14          | 12             | -168                                   | 21  | 17.5-24.5      |
| -7           | 19             | -133                                   | 28  | 24.5-31.5      |
| 0            | 29             | 0                                      | 35  | 31.5-38.5      |
| 7            | 20             | 140                                    | 42  | 38.5-45.5      |
| 14           | 13             | 182                                    | 49  | 45.5-52.5      |
| 21           | 5              | 105                                    | 56  | 52.5-59.5      |
| <b>Total</b> | <b>N = 100</b> | <b><math>\sum f(X - c) = 84</math></b> |     |                |

Using the formula,

$$\bar{X} = c + \frac{\sum f(X - c)}{N},$$

and substituting the given values,

$$35.84 = c + \frac{84}{100}.$$

Hence,

$$c + \frac{84}{100} = 35.84,$$

$$c = 35.84 - 0.84 = 35.$$

Therefore,

$$X - c = 0 \implies X = c = 35.$$

Since the class width is

$$7,$$

the half-width is

$$\frac{7}{2} = 3.5.$$

Hence, the class corresponding to the midpoint 35 is

$$35 - 3.5 \text{ to } 35 + 3.5,$$

that is,

$$31.5-38.5.$$

Similarly, the remaining class intervals are obtained by adding and subtracting 3.5 from each midpoint.

Thus, the required class intervals are

|                                                                                 |
|---------------------------------------------------------------------------------|
| 10.5–17.5, 17.5–24.5, 24.5–31.5,<br>31.5–38.5, 38.5–45.5, 45.5–52.5, 52.5–59.5. |
|---------------------------------------------------------------------------------|

### Question-3

The following table gives the step deviations ( $d$ ) and frequencies of a grouped frequency distribution. The assumed mean is  $A = 35$ . If the arithmetic mean of the distribution is 33, determine the class intervals.

| Step Deviation ( $d$ ) | −3 | −2 | −1 | 0  | 1  | 2  |
|------------------------|----|----|----|----|----|----|
| Frequency ( $f$ )      | 5  | 10 | 25 | 30 | 20 | 10 |

### Solution

The given step deviations are

$$d = \frac{X - A}{h},$$

where

$$A = 35, \quad N = \sum f.$$

Hence,

$$\bar{X} = A + \frac{h \sum fd}{N}.$$

| Step Deviation ( $d$ ) | Frequency ( $f$ ) | $fd$                              | Mid-value ( $X$ ) | Class Interval |
|------------------------|-------------------|-----------------------------------|-------------------|----------------|
| -3                     | 5                 | -15                               | 5                 | 0-10           |
| -2                     | 10                | -20                               | 15                | 10-20          |
| -1                     | 25                | -25                               | 25                | 20-30          |
| 0                      | 30                | 0                                 | 35                | 30-40          |
| 1                      | 20                | 20                                | 45                | 40-50          |
| 2                      | 10                | 20                                | 55                | 50-60          |
| <b>Total</b>           | <b>N = 100</b>    | <b><math>\sum fd = -20</math></b> |                   |                |

Using

$$\bar{X} = A + \frac{h \sum fd}{N},$$

we obtain

$$33 = 35 + \frac{h(-20)}{100}.$$

Therefore,

$$33 = 35 - \frac{20h}{100},$$

or,

$$35 - 33 = \frac{20h}{100},$$

$$2 = 0.2h.$$

Hence,

$$\boxed{h = 10.}$$

Now,

$$d = \frac{X - A}{h},$$

therefore,

$$X = A + hd.$$

Substituting  $A = 35$  and  $h = 10$ ,

$$X = 35 + 10d.$$

Thus, the corresponding mid-values are

$$d = -3 \Rightarrow X = 5,$$

$$d = -2 \Rightarrow X = 15,$$

$$d = -1 \Rightarrow X = 25,$$

$$d = 0 \Rightarrow X = 35,$$

$$d = 1 \Rightarrow X = 45,$$

$$d = 2 \Rightarrow X = 55.$$

Since the class width is

$$h = 10,$$

each class interval extends

$$\frac{10}{2} = 5$$

units on either side of its midpoint.

Hence,

$$5 \Rightarrow 0-10,$$

$$15 \Rightarrow 10-20,$$

$$25 \Rightarrow 20-30,$$

$$35 \Rightarrow 30-40,$$

$$45 \Rightarrow 40-50,$$

$$55 \Rightarrow 50-60.$$

Therefore, the required class intervals are

$$\boxed{0-10, 10-20, 20-30, 30-40, 40-50, 50-60.}$$

#### Question 4

Find the missing frequencies. The mean of the following frequency distribution is 1.46.

| No. of Accidents        | 0  | 1 | 2 | 3  | 4  | 5 |
|-------------------------|----|---|---|----|----|---|
| Frequency (No. of Days) | 46 | ? | ? | 25 | 10 | 5 |

$$\text{Total Frequency} = 200$$

### Solution

Let the missing frequencies corresponding to 1 and 2 accidents be  $f_1$  and  $f_2$ , respectively.

Since the total frequency is 200,

$$46 + f_1 + f_2 + 25 + 10 + 5 = 200.$$

Hence,

$$f_1 + f_2 = 114.$$

The arithmetic mean is given as 1.46.

That is,

$$\bar{X} = \frac{\sum fX}{\sum f},$$

we have

$$1.46 = \frac{0(46) + 1(f_1) + 2(f_2) + 3(25) + 4(10) + 5(5)}{200}.$$

Simplifying,

$$1.46 = \frac{f_1 + 2f_2 + 75 + 40 + 25}{200},$$

$$1.46 = \frac{f_1 + 2f_2 + 140}{200}.$$

Multiplying both sides by 200,

$$292 = f_1 + 2f_2 + 140,$$

or,

$$f_1 + 2f_2 = 152.$$

Now solve the simultaneous equations

$$f_1 + f_2 = 114,$$

$$f_1 + 2f_2 = 152.$$

Subtracting the first equation from the second,

$$f_2 = 38.$$

Substituting into the first equation,

$$f_1 = 114 - 38 = 76.$$

Therefore,

$$\boxed{f_1 = 76, \quad f_2 = 38.}$$

### Question-5

From the following cumulative frequency distribution of income, calculate the arithmetic mean. It is given that

1. The total income of persons in the highest income group is Rs. 435.
2. None of the persons is earning less than Rs. 20.

| Income (Rs.) | No. of Persons | Income (Rs.) | No. of Persons |
|--------------|----------------|--------------|----------------|
| Below 30     | 16             | Below 70     | 87             |
| Below 40     | 36             | Below 80     | 95             |
| Below 50     | 61             | 80 and over  | 5              |
| Below 60     | 76             |              |                |

### Solution

Since none of the persons is earning less than Rs. 20, the open-end class “Below 30” is taken as the class interval

20–30.

The given cumulative frequency distribution is first converted into an ordinary frequency distribution.

| Income (Rs.) | Mid-value ( $X$ ) | No. of Persons ( $f$ ) | $fX$             |
|--------------|-------------------|------------------------|------------------|
| 20–30        | 25                | 16                     | 400              |
| 30–40        | 35                | $36 - 16 = 20$         | 700              |
| 40–50        | 45                | $61 - 36 = 25$         | 1125             |
| 50–60        | 55                | $76 - 61 = 15$         | 825              |
| 60–70        | 65                | $87 - 76 = 11$         | 715              |
| 70–80        | 75                | $95 - 87 = 8$          | 600              |
| 80 and over  | —                 | 5                      | 435*             |
| <b>Total</b> |                   | $\sum f = 100$         | $\sum fX = 4800$ |

\*It is given that the total income of persons in the highest income group is Rs. 435.

The arithmetic mean is

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{4800}{100} = 48.$$

Hence,

|                           |
|---------------------------|
| Arithmetic Mean = Rs. 48. |
|---------------------------|

### Remarks

- Since the data are in cumulative frequency form, the frequencies are obtained by taking successive differences.
- The first class is an open-end class. As it is given that no person earns less than Rs. 20, it is treated as the class interval 20–30.
- The last class is also open-ended. Therefore, its midpoint is not required because the total income of that class is directly given as Rs. 435.
- The arithmetic mean is finally obtained using the formula

$$\bar{X} = \frac{\sum fX}{\sum f}.$$

### Question 6

The arithmetic mean of the following frequency distribution is 50. Find the missing frequencies  $f_1$  and  $f_2$ .

| Class Interval | Frequency |
|----------------|-----------|
| 0–20           | 17        |
| 20–40          | $f_1$     |
| 40–60          | 32        |
| 60–80          | $f_2$     |
| 80–100         | 19        |
| Total          | 120       |

### Solution

Let the unknown frequencies be  $f_1$  and  $f_2$ .

Since the total frequency is 120,

$$17 + f_1 + 32 + f_2 + 19 = 120,$$

or,

$$f_1 + f_2 = 52.$$

The mid-values of the class intervals are

$$10, 30, 50, 70, 90.$$

Hence, we prepare the following computation table.

| Class Interval | Mid-value ( $X$ ) | Frequency ( $f$ ) | $fX$                   |
|----------------|-------------------|-------------------|------------------------|
| 0–20           | 10                | 17                | 170                    |
| 20–40          | 30                | $f_1$             | $30f_1$                |
| 40–60          | 50                | 32                | 1600                   |
| 60–80          | 70                | $f_2$             | $70f_2$                |
| 80–100         | 90                | 19                | 1710                   |
| Total          |                   | 120               | $3480 + 30f_1 + 70f_2$ |

Since

$$\bar{X} = \frac{\sum fX}{\sum f},$$

and the arithmetic mean is 50,

$$50 = \frac{3480 + 30f_1 + 70f_2}{120}.$$

Therefore,

$$6000 = 3480 + 30f_1 + 70f_2,$$

or,

$$30f_1 + 70f_2 = 2520.$$

Dividing by 10,

$$3f_1 + 7f_2 = 252.$$

Now,

$$f_1 + f_2 = 52.$$

Multiplying the second equation by 3,

$$3f_1 + 3f_2 = 156.$$

Subtracting,

$$(3f_1 + 7f_2) - (3f_1 + 3f_2) = 252 - 156,$$

$$4f_2 = 96,$$

$$\boxed{f_2 = 24.}$$

Hence,

$$f_1 = 52 - 24 = 28.$$

Therefore,

$$\boxed{f_1 = 28, \quad f_2 = 24.}$$

### Question-7

In a batch of 15 students, 5 students failed in a test. The marks of the 10 students who passed were

90, 60, 70, 80, 80, 90, 60, 50, 40, 70.

Find the median marks of all the 15 students.

### Solution:

Arrange the marks of the students who passed in ascending order:

40, 50, 60, 60, 70, 70, 80, 80, 90, 90.

Since the five students who failed must have scored less than 40 marks, the complete ordered series of all 15 students is

-, -, -, -, -, 40, 50, 60, 60, 70, 70, 80, 80, 90, 90.

Here,

$$n = 15.$$

Hence, the median is the

$$\frac{n+1}{2} = \frac{15+1}{2} = 8^{\text{th}}$$

observation.

The 8<sup>th</sup> observation is

60

Therefore,

Median = 60.

### Question-8

The following table shows the age distribution of persons in a particular region.

| Age (years) | No. of persons (in thousands) |
|-------------|-------------------------------|
| Below 10    | 2                             |
| Below 20    | 5                             |
| Below 30    | 9                             |
| Below 40    | 12                            |
| Below 50    | 14                            |
| Below 60    | 15                            |
| Below 70    | 15.5                          |
| 70 and over | 15.6                          |

Find

1. Median age.
2. Why is the median a more suitable measure of central tendency than the mean in this case?

**Solution:**

First convert the cumulative frequency distribution into a continuous frequency distribution.

| Age (years)    | Frequency ( $f$ )   | Less than c.f. |
|----------------|---------------------|----------------|
| 0–10           | 2                   | 2              |
| 10–20          | $5 - 2 = 3$         | 5              |
| 20–30          | $9 - 5 = 4$         | 9              |
| 30–40          | $12 - 9 = 3$        | 12             |
| 40–50          | $14 - 12 = 2$       | 14             |
| 50–60          | $15 - 14 = 1$       | 15             |
| 60–70          | $15.5 - 15 = 0.5$   | 15.5           |
| 70 and over    | $15.6 - 15.5 = 0.1$ | 15.6           |
| $N = \sum f =$ |                     | 15.6           |

$$\frac{N}{2} = \frac{15.6}{2} = 7.8.$$

The cumulative frequency just greater than 7.8 is 9.  
Hence, the median class is

$$20 - 30.$$

Using the median formula,

$$\text{Median} = L + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

where

$$L = 20, \quad h = 10, \quad f = 4, \quad C = 5.$$

Therefore,

$$\begin{aligned} \text{Median} &= 20 + \frac{10}{4}(7.8 - 5) \\ &= 20 + \frac{5}{2} \times 2.8 \\ &= 20 + 7 \\ &= 27. \end{aligned}$$

Hence,

$$\boxed{\text{Median age} = 27 \text{ years.}}$$

**(ii) Why median is more suitable?**

The last class is an **open-end class** (70 years and over). Since the class mark of this class mid-value cannot be determined, the arithmetic mean cannot be computed accurately. Therefore, the median is a more suitable measure of central tendency for this distribution.

**Question-9**

The frequency distribution of the weights (in grams) of mangoes of a given variety is given below. Calculate the arithmetic mean and the median.

| Weight (g)     | 410–419 | 420–429 | 430–439 | 440–449 | 450–459 | 460–469 | 470–479 |
|----------------|---------|---------|---------|---------|---------|---------|---------|
| No. of Mangoes | 14      | 20      | 42      | 54      | 45      | 18      | 7       |

**Solution:**

Since the interpolation formula for median is based on continuous frequency distribution, we first convert the given inclusive class intervals into exclusive class intervals.

| Weight (g)   | Frequency (f)  | Mid-value (X) | $d = \frac{X - 444.5}{10}$ | fd              | Less than c.f. |
|--------------|----------------|---------------|----------------------------|-----------------|----------------|
| 409.5–419.5  | 14             | 414.5         | –3                         | –42             | 14             |
| 419.5–429.5  | 20             | 424.5         | –2                         | –40             | 34             |
| 429.5–439.5  | 42             | 434.5         | –1                         | –42             | 76             |
| 439.5–449.5  | 54             | 444.5         | 0                          | 0               | 130            |
| 449.5–459.5  | 45             | 454.5         | 1                          | 45              | 175            |
| 459.5–469.5  | 18             | 464.5         | 2                          | 36              | 193            |
| 469.5–479.5  | 7              | 474.5         | 3                          | 21              | 200            |
| <b>Total</b> | $\sum f = 200$ |               |                            | $\sum fd = -22$ |                |

**(i) Arithmetic Mean**

Using the assumed mean method,

$$\bar{X} = A + \frac{h \sum fd}{N},$$

where

$$A = 444.5, \quad h = 10, \quad N = 200.$$

Hence,

$$\begin{aligned}\bar{X} &= 444.5 + \frac{10(-22)}{200} \\ &= 444.5 - 1.1 \\ &= 443.4 \text{ grams.}\end{aligned}$$

Therefore,

$$\boxed{\bar{X} = 443.4 \text{ grams.}}$$

## (ii) Median

$$\frac{N}{2} = \frac{200}{2} = 100.$$

The cumulative frequency just greater than 100 is 130.

Hence, the median class is

$$439.5 - 449.5.$$

Using the median formula,

$$\text{Median} = L + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

where

$$L = 439.5, \quad h = 10, \quad f = 54, \quad C = 76.$$

Therefore,

$$\begin{aligned}\text{Median} &= 439.5 + \frac{10}{54}(100 - 76) \\ &= 439.5 + \frac{10 \times 24}{54} \\ &= 439.5 + 4.44 \\ &= 443.94 \text{ grams.}\end{aligned}$$

Hence,

$$\boxed{\text{Median} = 443.94 \text{ grams.}}$$

## Question-10

Find the missing frequency from the following distribution of daily sales of shops, given that the median sales of shops is Rs. 2400.

| Sales (in hundred Rs.) | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|------------------------|------|-------|-------|-------|-------|
| No. of shops           | 5    | 25    | ?     | 18    | 7     |

**Solution:**

Let the missing frequency be  $a$ . Then, the cumulative frequency table is

| Sales (hundred Rs.) | Frequency ( $f$ ) | Cumulative Frequency |
|---------------------|-------------------|----------------------|
| 0–10                | 5                 | 5                    |
| 10–20               | 25                | 30                   |
| 20–30               | $a$               | $30 + a$             |
| 30–40               | 18                | $48 + a$             |
| 40–50               | 7                 | $55 + a$             |
| $N = 55 + a$        |                   |                      |

Since the median sales is Rs. 2400 (=24 hundred), the median class is 20–30.

Using the median formula,

$$\text{Median} = L + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

where

$$L = 20, \quad h = 10, \quad f = a, \quad C = 30, \quad N = 55 + a.$$

Therefore,

$$24 = 20 + \frac{10}{a} \left( \frac{55 + a}{2} - 30 \right)$$

$$4 = \frac{10}{a} \left( \frac{a - 5}{2} \right)$$

$$4a = 5(a - 5)$$

$$4a = 5a - 25$$

$$a = 25.$$

Hence,

The missing frequency is 25.

**Question-11**

In the following frequency distribution of 100 families, the frequencies corresponding to expenditure groups 20–40 and 60–80 are missing. The median is known to be Rs. 50. Find the missing frequencies.

| Expenditure (Rs.) | 0–20 | 20–40 | 40–60 | 60–80 | 80–100 |
|-------------------|------|-------|-------|-------|--------|
| No. of families   | 14   | $f_1$ | 27    | $f_2$ | 15     |

**Solution:**

Let the missing frequencies be  $f_1$  and  $f_2$ .

Since

$$N = 100,$$

we have

$$14 + f_1 + 27 + f_2 + 15 = 100,$$

or

$$56 + f_1 + f_2 = 100.$$

Hence,

$$f_1 + f_2 = 44. \quad (1)$$

The cumulative frequency table is

| Expenditure (Rs.) | Frequency | Less than c.f.   |
|-------------------|-----------|------------------|
| 0–20              | 14        | 14               |
| 20–40             | $f_1$     | $14 + f_1$       |
| 40–60             | 27        | $41 + f_1$       |
| 60–80             | $f_2$     | $41 + f_1 + f_2$ |
| 80–100            | 15        | $56 + f_1 + f_2$ |
| $N = 100$         |           |                  |

Since the median is 50, the median class is

$$40 - 60.$$

Using the median formula,

$$50 = 40 + \frac{20}{27} (50 - (14 + f_1)).$$

Therefore,

$$10 = \frac{20}{27} (36 - f_1)$$

$$27 = 2(36 - f_1)$$

$$27 = 72 - 2f_1$$

$$2f_1 = 45$$

$$f_1 = 22.5.$$

Since frequency cannot be fractional,

$$f_1 \approx 23.$$

Using Equation (1),

$$f_2 = 44 - 23 = 21.$$

Hence,

$$\boxed{f_1 = 23, \quad f_2 = 21.}$$

### Question-12

A number of particular articles has been classified according to their weights. After drying for two weeks, the same articles have again been weighed and similarly classified.

It is known that the median weight in the first weighing was 20.83 oz., while in the second weighing it was 17.35 oz.

Some frequencies  $a$  and  $b$  in the first weighing and  $x$  and  $y$  in the second weighing are missing. It is known that

$$a = \frac{x}{3}, \quad b = \frac{y}{2}.$$

Find the missing frequencies.

| Class | Frequencies    |                 |
|-------|----------------|-----------------|
|       | First Weighing | Second Weighing |
| 0–5   | $a$            | $x$             |
| 5–10  | $b$            | $y$             |
| 10–15 | 11             | 40              |
| 15–20 | 52             | 50              |
| 20–25 | 75             | 30              |
| 25–30 | 22             | 28              |

### Solution:

Since

$$a = \frac{x}{3}, \quad b = \frac{y}{2},$$

we have

$$x = 3a, \quad y = 2b.$$

Hence,

$$N_1 = 160 + a + b,$$

and

$$N_2 = 148 + x + y = 148 + 3a + 2b.$$

| Class | First Weighing |                     | Second Weighing |                     |
|-------|----------------|---------------------|-----------------|---------------------|
|       | Frequency      | Less than c.f.      | Frequency       | Less than c.f.      |
| 0–5   | $a$            | $a$                 | $x$             | $x$                 |
| 5–10  | $b$            | $a + b$             | $y$             | $x + y$             |
| 10–15 | 11             | $a + b + 11$        | 40              | $x + y + 40$        |
| 15–20 | 52             | $a + b + 63$        | 50              | $x + y + 90$        |
| 20–25 | 75             | $a + b + 138$       | 30              | $x + y + 120$       |
| 25–30 | 22             | $a + b + 160 = N_1$ | 28              | $x + y + 148 = N_2$ |

Since the median in the first weighing is 20.83 oz.,  
the median class is

$$20 - 25.$$

Using the median formula,

$$\text{Median} = L + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

we obtain

$$20.83 = 20 + \frac{5}{75} \left[ \frac{160 + a + b}{2} - (63 + a + b) \right].$$

Therefore,

$$\begin{aligned} 15(20.83 - 20) &= \frac{160 + a + b}{2} - (63 + a + b) \\ 12.45 &= 17 - \frac{a + b}{2} \\ a + b &= 2(17 - 12.45) \\ &= 9.10 \approx 9. \end{aligned}$$

Since frequencies are integers,

$$a + b = 9. \tag{1}$$

Similarly, the median in the second weighing is 17.35 oz.,

whose median class is

$$15 - 20.$$

Hence,

$$17.35 = 15 + \frac{5}{50} \left[ \frac{148 + 3a + 2b}{2} - (40 + x + y) \right].$$

Since

$$x = 3a, \quad y = 2b,$$

we get

$$\begin{aligned} 10(2.35) &= 74 + \frac{3a + 2b}{2} - 40 - 3a - 2b \\ 23.5 &= 34 - \frac{3a + 2b}{2} \\ \frac{3a + 2b}{2} &= 34 - 23.5 \\ 3a + 2b &= 21. \end{aligned}$$

Therefore,

$$3a + 2b = 21. \quad (2)$$

Multiply Equation (1) by 3,

$$3a + 3b = 27. \quad (3)$$

Subtract Equation (2) from Equation (3),

$$b = 6.$$

Hence,

$$a = 9 - 6 = 3.$$

Finally,

$$x = 3a = 9,$$

and

$$y = 2b = 12.$$

Therefore,

|          |          |          |           |
|----------|----------|----------|-----------|
| $a = 3,$ | $b = 6,$ | $x = 9,$ | $y = 12.$ |
|----------|----------|----------|-----------|

### Question-13

For a group of 5000 workers, the hourly wages vary from Rs. 20 to Rs. 80. The wages of 4% of the workers are under Rs. 25 and those of 10% are under Rs. 30; 15% of the workers earn Rs. 60 and over, and 5% of them get Rs. 70 and over. The quartile wages are Rs. 40 and Rs. 54, and the sixth decile is Rs. 50. Put this information in the form of a frequency table.

**Solution:**

Given,

$$N = 5000.$$

1. Since

$$Q_1 = 40,$$
$$25\% = \frac{25}{100} \times 5000 = 1250,$$

therefore, **1250 workers earn less than Rs. 40.**

2. Since

$$D_6 = 50,$$
$$60\% = \frac{60}{100} \times 5000 = 3000,$$

therefore, **3000 workers earn less than Rs. 50.**

3. Since

$$Q_3 = 54,$$
$$75\% = \frac{75}{100} \times 5000 = 3750,$$

therefore, **3750 workers earn less than Rs. 54.**

Further,

(i)

$$4\% = \frac{4}{100} \times 5000 = 200,$$

therefore, **200 workers earn under Rs. 25.**

(ii)

$$10\% = \frac{10}{100} \times 5000 = 500,$$

therefore, **500 workers earn under Rs. 30.**

(iii)

$$15\% = \frac{15}{100} \times 5000 = 750,$$

therefore, **750 workers earn Rs. 60 and over.**

(iv)

$$5\% = \frac{5}{100} \times 5000 = 250,$$

therefore, **250 workers earn Rs. 70 and over.**

**Summary:**

| Hourly Wages (Rs.) | No. of Workers |
|--------------------|----------------|
| less than 40       | 1250           |
| less than 50       | 3000           |
| less than 54       | 3750           |
| less than 25       | 200            |
| less than 30       | 500            |
| 60 and over        | 750            |
| 70 and over        | 250            |

| Hourly Wages (Rs.) | No. of Workers |
|--------------------|----------------|
| less than 25       | 200            |
| less than 30       | 500            |
| less than 40       | 1250           |
| less than 50       | 3000           |
| less than 54       | 3750           |

| Hourly Wages (Rs.) | No. of Workers |
|--------------------|----------------|
| 60 and over        | 750            |
| 70 and over        | 250            |

| Hourly Wages (Rs.) | No. of Workers | Less than c.f. |
|--------------------|----------------|----------------|
| Under 25           | 200            | 200            |
| 25–30              | 300            | 500            |
| 30–40              | 750            | 1250           |
| 40–50              | 1750           | 3000           |
| 50–54              | 750            | 3750           |
| 54–60              | 500            | 4250           |

| Hourly Wages (Rs.) | No. of Workers | Greater than c.f. |
|--------------------|----------------|-------------------|
| 60–70              | 500            | 750               |
| 70 and over        | 250            | 250               |

Hence, the cumulative frequencies and class frequencies are obtained as follows.

| Hourly Wages (Rs.) | No. of Workers | Less than c.f. |
|--------------------|----------------|----------------|
| Under 25           | 200            | 200            |
| 25–30              | 300            | 500            |
| 30–40              | 750            | 1250           |
| 40–50              | 1750           | 3000           |
| 50–54              | 750            | 3750           |
| 54–60              | 500            | 4250           |
| 60–70              | 500            | 4750           |
| 70 and over        | 250            | 5000           |

Since the wages vary from Rs. 20 to Rs. 80, the first class may be taken as 20–25 and the last class as 70–80. Combining the unequal class intervals, the required frequency distribution with equal class width is obtained as follows.

| Hourly Wages (Rs.) | No. of Workers     |
|--------------------|--------------------|
| 20–30              | $200 + 300 = 500$  |
| 30–40              | 750                |
| 40–50              | 1750               |
| 50–60              | $750 + 500 = 1250$ |
| 60–70              | 500                |
| 70–80              | 250                |
| <b>Total</b>       | <b>5000</b>        |

Dr. Boina Anil Kumar

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Measures of Dispersion

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

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## Dispersion

### Introduction

Measures of central tendency such as Mean, Median, and Mode provide information about the central or typical value of a dataset. However, they do not indicate how far the observations are scattered around the central value.

**Dispersion** refers to the extent to which the observations in a dataset are scattered or spread around a central value.

### Definition

*Dispersion is the degree to which the observations of a dataset are scattered around a measure of central tendency.*

### Illustration

For example, consider the following two datasets:

$A :$  48, 49, 50, 51, 52

$B :$  20, 35, 50, 65, 80

For both data sets,

$$\bar{X} = 50.$$

However, the observations in Dataset B are much more widely spread than those in Dataset A. Therefore, Dataset B has greater dispersion.

Thus,

Dispersion = Measure of spread or variability of data

### Need for Measuring Dispersion

Measures of dispersion are useful because they:

- indicate the extent of variability in the data;
- show how closely the observations are concentrated around the central value;
- help in comparing the variability of two or more data sets;

- indicate the consistency or stability of observations;
- provide additional information along with measures of central tendency; and
- are widely used in statistical analysis and decision-making.

## Measures of Dispersion

The statistical measures used to determine the amount of variation or spread in a dataset are called **Measures of Dispersion**.

These measures can be classified into two broad categories:

### Absolute Measures of Dispersion

Absolute measures express the amount of dispersion in the same units as the original observations, except variance, which is expressed in squared units.

The important absolute measures are:

- Range
- Quartile Deviation
- Mean Deviation
- Variance
- Standard Deviation

### Relative Measures of Dispersion

Relative measures are generally expressed as ratios or percentages. They are useful for comparing the variability of two or more datasets.

The important relative measures are:

- Coefficient of Range
- Coefficient of Quartile Deviation
- Coefficient of Mean Deviation
- Coefficient of Variation

## Various Measures of Dispersion

### Range

The **Range** is the simplest measure of dispersion. Formula is

$$R = x_{\max} - x_{\min}$$

where

$x_{\max}$  = Largest observation      and       $x_{\min}$  = Smallest observation.

## Coefficient of Range

$$\text{Coefficient of Range} = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$$

### Example 1:

Calculate the range and the coefficient of range of A's monthly earnings for a year.

| Month | Monthly earnings<br>(In '00 Rs.) | Month | Monthly earnings<br>(In '00 Rs.) | Month | Monthly earnings<br>(In '00 Rs.) |
|-------|----------------------------------|-------|----------------------------------|-------|----------------------------------|
| 1     | 139                              | 5     | 157                              | 9     | 162                              |
| 2     | 150                              | 6     | 158                              | 10    | 162                              |
| 3     | 151                              | 7     | 160                              | 11    | 173                              |
| 4     | 151                              | 8     | 161                              | 12    | 175                              |

### Solution.

Largest earning  $X_{\max} = \text{Rs. } 17,500$ ;

Smallest earning  $X_{\min} = \text{Rs. } 13,900$ .

The coefficient of range is

Therefore,

$$\begin{aligned}\text{Range} &= x_{\max} - x_{\min} \\ &= 17,500 - 13,900 \\ &= \text{Rs. } 3,600.\end{aligned}$$

$$\begin{aligned}\text{Coefficient of range} &= \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}} \\ &= \frac{17,500 - 13,900}{17,500 + 13,900} \\ &= 0.115.\end{aligned}$$

### Example 2:

The following table gives the age distribution of a group of 50 individuals.

| Age (in years) | 16–20 | 21–25 | 26–30 | 31–36 |
|----------------|-------|-------|-------|-------|
| No. of persons | 10    | 15    | 17    | 8     |

Calculate range and the coefficient of range.

### Solution.

Since age is a continuous variable, we should first convert the given classes into continuous classes.

The first class becomes

$$15.5 - 20.5$$

and the last class becomes

$$30.5 - 35.5.$$

Therefore,

$$x_{\max} = \text{Largest value} = 35.5, \quad x_{\min} = \text{Smallest value} = 15.5.$$

Hence,

$$\begin{aligned}\text{Range} &= 35.5 - 15.5 \\ &= 20 \text{ years.}\end{aligned}$$

The coefficient of range is

$$\begin{aligned}\text{Coefficient of range} &= \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}} \\ &= \frac{35.5 - 15.5}{35.5 + 15.5} \\ &= \frac{20}{51} \\ &= 0.39.\end{aligned}$$

### Quartile Deviation

Quartile Deviation is based on the first quartile ( $Q_1$ ) and the third quartile ( $Q_3$ ).

The Interquartile Range is

$$IQR = Q_3 - Q_1$$

The Quartile Deviation is half of the interquartile range:

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

Quartile Deviation measures the spread of the middle 50% of the observations.

### Coefficient of Quartile Deviation

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

#### Example 1:

Find:

- (i) Inter-Quartile Range,
- (ii) Quartile Deviation,
- (iii) Coefficient of Quartile Deviation.

For the following distribution:

| Class Interval | 0–15 | 15–30 | 30–45 | 45–60 | 60–75 | 75–90 | 90–105 |
|----------------|------|-------|-------|-------|-------|-------|--------|
| Frequency      | 8    | 26    | 30    | 45    | 20    | 17    | 4      |

**Solution:**

The corresponding cumulative frequency table is:

| Class Interval | $f$       | Less than c.f. |
|----------------|-----------|----------------|
| 0–15           | 8         | 8              |
| 15–30          | 26        | 34             |
| 30–45          | 30        | 64             |
| 45–60          | 45        | 109            |
| 60–75          | 20        | 129            |
| 75–90          | 17        | 146            |
| 90–105         | 4         | 150            |
| <b>Total</b>   | $N = 150$ |                |

$$N = 150$$

For the first quartile,

$$\frac{N}{4} = \frac{150}{4} = 37.5$$

The c.f. just greater than 37.5 is 64. Therefore,  $Q_1$  lies in the class 30 – 45.

Using

$$Q_1 = L + \frac{h}{f} \left( \frac{N}{4} - C \right),$$

we get

$$\begin{aligned} Q_1 &= 30 + \frac{15}{30}(37.5 - 34) \\ &= 30 + \frac{15}{30}(3.5) \\ &= 30 + 1.75 \\ &= \boxed{31.75}. \end{aligned}$$

For the third quartile,

$$\frac{3N}{4} = \frac{3(150)}{4} = 112.5$$

The c.f. just greater than 112.5 is 129. Therefore,  $Q_3$  lies in the class 60 – 75.

Using

$$Q_3 = L + \frac{h}{f} \left( \frac{3N}{4} - C \right),$$

we get

$$\begin{aligned}
 Q_3 &= 60 + \frac{15}{20}(112.5 - 109) \\
 &= 60 + \frac{15}{20}(3.5) \\
 &= 60 + 2.625 \\
 &= \boxed{62.625}.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \text{Inter-Quartile Range} &= Q_3 - Q_1 \\
 &= 62.625 - 31.750 \\
 &= \boxed{30.875}.
 \end{aligned}$$

The Quartile Deviation is

$$\begin{aligned}
 Q.D. &= \frac{Q_3 - Q_1}{2} \\
 &= \frac{62.625 - 31.750}{2} \\
 &= \frac{30.875}{2} \\
 &= \boxed{15.44}.
 \end{aligned}$$

The Coefficient of Quartile Deviation is

$$\begin{aligned}
 \text{Coefficient of Q.D.} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\
 &= \frac{62.625 - 31.750}{62.625 + 31.750} \\
 &= \frac{30.875}{94.375} \\
 &\approx \boxed{0.33}.
 \end{aligned}$$

### Example 2:

Evaluate an appropriate measure of dispersion for the following data:

| Income<br>(in Rs.) | Less than<br>50 | 50–70 | 70–90 | 90–110 | 110–130 | 130–150 | 150 and<br>above |
|--------------------|-----------------|-------|-------|--------|---------|---------|------------------|
| No. of<br>persons  | 54              | 100   | 140   | 300    | 230     | 125     | 51               |

### Solution:

Since we are given the classes with open-end intervals, the only appropriate measure of dispersion is the **Quartile Deviation**.

Here,

$$N = 1000.$$

Therefore,

$$\frac{N}{4} = \frac{1000}{4} = 250$$

and

$$\frac{3N}{4} = \frac{3(1000)}{4} = 750.$$

The corresponding cumulative frequencies are:

| Income (in Rs.) | No. of persons | Less than c.f. |
|-----------------|----------------|----------------|
| Less than 50    | 54             | 54             |
| 50–70           | 100            | 154            |
| 70–90           | 140            | 294            |
| 90–110          | 300            | 594            |
| 110–130         | 230            | 824            |
| 130–150         | 125            | 949            |
| Above 150       | 51             | 1000           |

Since the c.f. just greater than 250 is 294,  $Q_1$  lies in the class 70 – –90.

Thus,

$$\begin{aligned}
 Q_1 &= l + \frac{h}{f} \left( \frac{N}{4} - C \right) \\
 &= 70 + \frac{20}{140} (250 - 154) \\
 &= 70 + \frac{96}{7} \\
 &= 70 + 13.714 \\
 &= \boxed{83.714}.
 \end{aligned}$$

Since the c.f. just greater than 750 is 824,  $Q_3$  lies in the class 110 – –130.

Thus,

$$\begin{aligned}
 Q_3 &= l + \frac{h}{f} \left( \frac{3N}{4} - C \right) \\
 &= 110 + \frac{20}{230} (750 - 594) \\
 &= 110 + \frac{2 \times 156}{23} \\
 &= 110 + 13.565 \\
 &= \boxed{123.565}.
 \end{aligned}$$

Therefore, the Quartile Deviation is

$$\begin{aligned} Q.D. &= \frac{Q_3 - Q_1}{2} \\ &= \frac{123.565 - 83.714}{2} \\ &= \frac{39.851}{2} \\ &= \boxed{19.925}. \end{aligned}$$

### Percentile Range

The **Percentile Range** is the difference between two selected percentiles. It is used to measure the spread of a specified portion of the observations.

$$\text{Percentile Range} = P_{100-k} - P_k,$$

#### Example:

The 10–90 percentile range is

$$P_{90} - P_{10}$$

Similarly,

$$\text{20–80 Percentile Range} = P_{80} - P_{20}$$

and

$$\text{25–75 Percentile Range} = P_{75} - P_{25}$$

### Coefficient of Percentile Range

The coefficient of percentile range is a relative measure of dispersion.

$$\text{Coefficient of Percentile Range} = \frac{P_{100-k} - P_k}{P_{100-k} + P_k}$$

For the 10–90 percentile range,

$$\text{Coefficient of Percentile Range} = \frac{P_{90} - P_{10}}{P_{90} + P_{10}}$$

#### Example

Suppose

$$P_{10} = 20, \quad P_{90} = 80.$$

Then,

$$\text{10–90 Percentile Range} = P_{90} - P_{10} = 80 - 20 = 60.$$

The coefficient is

$$\text{Coefficient of Percentile Range} = \frac{80 - 20}{80 + 20} = \frac{60}{100} = 0.6.$$

**Example 1:**

Find the 10–90 percentile range for the following distribution.

| Class Interval | Frequency | Cumulative Frequency |
|----------------|-----------|----------------------|
| 0–10           | 5         | 5                    |
| 10–20          | 10        | 15                   |
| 20–30          | 15        | 30                   |
| 30–40          | 25        | 55                   |
| 40–50          | 20        | 75                   |
| 50–60          | 15        | 90                   |
| 60–70          | 7         | 97                   |
| 70–80          | 3         | 100                  |

**Solution:**

$$N = 100$$

For  $P_{10}$ ,

$$\frac{10N}{100} = 10.$$

The 10th item lies in the class 10 – –20.

Here,

$$L = 10, \quad C = 5, \quad f = 10, \quad h = 10.$$

Therefore,

$$P_{10} = 10 + \left( \frac{10 - 5}{10} \right) 10 = 15.$$

For  $P_{90}$ ,

$$\frac{90N}{100} = 90.$$

The 90th item lies in the class 50 – –60.

Here,

$$L = 50, \quad C = 75, \quad f = 15, \quad h = 10.$$

Therefore,

$$P_{90} = 50 + \left( \frac{90 - 75}{15} \right) 10 = 60.$$

Hence,

$$10-90 \text{ Percentile Range} = P_{90} - P_{10}$$

$$= 60 - 15 = \boxed{45}.$$

**Example 2:**

Find the 20–80 percentile range from the following data.

| Marks | Frequency | C.F. |
|-------|-----------|------|
| 0–10  | 8         | 8    |
| 10–20 | 12        | 20   |
| 20–30 | 20        | 40   |
| 30–40 | 25        | 65   |
| 40–50 | 20        | 85   |
| 50–60 | 10        | 95   |
| 60–70 | 5         | 100  |

**Solution:**

$$N = 100.$$

For  $P_{20}$ ,

$$\frac{20N}{100} = 20.$$

The 20th item lies in the class 10 – –20.

$$P_{20} = 10 + \left( \frac{20 - 8}{12} \right) 10 = 20.$$

For  $P_{80}$ ,

$$\frac{80N}{100} = 80.$$

The 80th item lies in the class 40 – –50.

$$\begin{aligned} P_{80} &= 40 + \left( \frac{80 - 65}{20} \right) 10 \\ &= 40 + 7.5 = 47.5. \end{aligned}$$

Therefore,

$$20\text{--}80 \text{ Percentile Range} = P_{80} - P_{20}$$

$$= 47.5 - 20 = \boxed{27.5}.$$

**Example 3:**

Calculate the 25–75 percentile range for the following frequency distribution.

| Income (Rs.) | Frequency | C.F. |
|--------------|-----------|------|
| 100–200      | 10        | 10   |
| 200–300      | 15        | 25   |
| 300–400      | 25        | 50   |
| 400–500      | 30        | 80   |
| 500–600      | 15        | 95   |
| 600–700      | 5         | 100  |

**Solution:**

$$N = 100.$$

For  $P_{25}$ ,

$$\frac{25N}{100} = 25.$$

The 25th item lies in the class 200 – 300.

$$\begin{aligned} P_{25} &= 200 + \left( \frac{25 - 10}{15} \right) 100 \\ &= 200 + 100 = \boxed{300}. \end{aligned}$$

For  $P_{75}$ ,

$$\frac{75N}{100} = 75.$$

The 75th item lies in the class 400 – 500.

$$\begin{aligned} P_{75} &= 400 + \left( \frac{75 - 50}{30} \right) 100 \\ &= 400 + 83.33 = \boxed{483.33}. \end{aligned}$$

Therefore,

$$25\text{--}75 \text{ Percentile Range} = P_{75} - P_{25}$$

$$= 483.33 - 300 = \boxed{183.33}.$$

**Example 4:**

The following table gives the distribution of heights of 100 students. Find the 10–90 percentile range.

| Height (cm) | Frequency | C.F. |
|-------------|-----------|------|
| 140–150     | 6         | 6    |
| 150–160     | 14        | 20   |
| 160–170     | 25        | 45   |
| 170–180     | 30        | 75   |
| 180–190     | 18        | 93   |
| 190–200     | 7         | 100  |

**Solution:**

$$N = 100.$$

For  $P_{10}$ ,

$$\frac{10N}{100} = 10.$$

The 10th item lies in the class 150 – 160.

$$\begin{aligned} P_{10} &= 150 + \left( \frac{10 - 6}{14} \right) 10 \\ &= 150 + 2.857 = \boxed{152.86}. \end{aligned}$$

For  $P_{90}$ ,

$$\frac{90N}{100} = 90.$$

The 90th item lies in the class 180 – 190.

$$\begin{aligned} P_{90} &= 180 + \left( \frac{90 - 75}{18} \right) 10 \\ &= 180 + 8.333 = \boxed{188.33}. \end{aligned}$$

Therefore,

$$\begin{aligned} 10\text{--}90 \text{ Percentile Range} &= P_{90} - P_{10} \\ &= 188.33 - 152.86 = \boxed{35.47 \text{ cm}}. \end{aligned}$$

**Example 5:**

Find the 5–95 percentile range for the following distribution.

| Class Interval | Frequency | C.F. |
|----------------|-----------|------|
| 0–20           | 5         | 5    |
| 20–40          | 10        | 15   |
| 40–60          | 20        | 35   |
| 60–80          | 30        | 65   |
| 80–100         | 20        | 85   |
| 100–120        | 10        | 95   |
| 120–140        | 5         | 100  |

**Solution:**

$$N = 100.$$

For  $P_5$ ,

$$\frac{5N}{100} = 5.$$

The 5th item lies in the class 0 – 20.

$$P_5 = 0 + \left( \frac{5 - 0}{5} \right) 20 = \boxed{20}.$$

For  $P_{95}$ ,

$$\frac{95N}{100} = 95.$$

The 95th item lies in the class 100 – 120.

$$\begin{aligned} P_{95} &= 100 + \left( \frac{95 - 85}{10} \right) 20 \\ &= 100 + 20 = \boxed{120}. \end{aligned}$$

Therefore,

$$5\text{--}95 \text{ Percentile Range} = P_{95} - P_5$$

$$= 120 - 20 = \boxed{100}.$$

**Summary of Percentile Ranges**

| Percentile Range       | Formula           |
|------------------------|-------------------|
| 10–90 Percentile Range | $P_{90} - P_{10}$ |
| 20–80 Percentile Range | $P_{80} - P_{20}$ |
| 25–75 Percentile Range | $P_{75} - P_{25}$ |
| 5–95 Percentile Range  | $P_{95} - P_5$    |

$$\text{Percentile Range} = P_{\text{upper}} - P_{\text{lower}}$$

### Mean Deviation

The **Mean Deviation (M.D.)** is the arithmetic mean of the absolute deviations of the observations from a central value.

The deviations may be calculated from:

- (i) Mean,
- (ii) Median, or
- (iii) Mode.

For individual observations,

$$M.D. = \frac{\sum |X - A|}{n}$$

where  $A$  represents the chosen central value.

For a frequency distribution,

$$M.D. = \frac{\sum f|X - A|}{\sum f}$$

where

$f$  = frequency      and       $A$  = Mean, Median, or Mode.

Usually, the **Median** is preferred for calculating Mean Deviation because the sum of absolute deviations is minimum about the median.

### Coefficient of Mean Deviation

$$\text{Coefficient of M.D.} = \frac{M.D.}{A}$$

where  $A$  is the central value used for calculating Mean Deviation.

**Example :**

(a) Find the Mean Deviation from the Mean for the following data:

| Class Interval    | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 | 60 – 70 |
|-------------------|--------|---------|---------|---------|---------|---------|---------|
| Frequency ( $f$ ) | 8      | 12      | 10      | 8       | 3       | 2       | 7       |

(b) Also find the mean deviation about median.

(c) Compare the results obtained in (a) and (b).

**Solution:**

**Solution:**

**Calculation of mean deviation about the mean**

| Class Interval | Mid-value (X) | Frequency (f) | CF | fX               | X - $\bar{X}$ | f X - $\bar{X}$             |
|----------------|---------------|---------------|----|------------------|---------------|-----------------------------|
| 0 – 10         | 5             | 8             | 8  | 40               | 24            | 192                         |
| 10 – 20        | 15            | 12            | 20 | 180              | 14            | 168                         |
| 20 – 30        | 25            | 10            | 30 | 250              | 4             | 40                          |
| 30 – 40        | 35            | 8             | 38 | 280              | 6             | 48                          |
| 40 – 50        | 45            | 3             | 41 | 135              | 16            | 48                          |
| 50 – 60        | 55            | 2             | 43 | 110              | 26            | 52                          |
| 60 – 70        | 65            | 7             | 50 | 455              | 36            | 252                         |
| <b>Total</b>   |               | $N = 50$      |    | $\sum fX = 1450$ |               | $\sum f X - \bar{X}  = 800$ |

The total frequency is

$$N = \sum f = 50.$$

The arithmetic mean is

$$\bar{X} = \frac{1}{N} \sum fX = \frac{1450}{50} = 29.$$

The mean deviation about the mean is

$$\text{M.D. about mean} = \frac{1}{N} \sum f|X - \bar{X}| = \frac{800}{50} = 16.$$

Hence,

$$\boxed{\text{M.D. about mean} = 16}$$

**(b) Also find the mean deviation about median.**

We have

$$\frac{N}{2} = \frac{50}{2} = 25.$$

The cumulative frequency just greater than 25 is 30. Hence, the corresponding class 20 – 30 is the median class.

Thus,

$$l = 20, \quad h = 10, \quad f = 10, \quad C = 20.$$

Therefore,

$$Md = l + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

$$Md = 20 + \frac{10}{10}(25 - 20) = 25.$$

Now calculate the deviations from the median.

| Mid-value ( $X$ ) | Frequency ( $f$ ) | $ X - Md  =  X - 25 $ | $f X - Md $            |
|-------------------|-------------------|-----------------------|------------------------|
| 5                 | 8                 | 20                    | 160                    |
| 15                | 12                | 10                    | 120                    |
| 25                | 10                | 0                     | 0                      |
| 35                | 8                 | 10                    | 80                     |
| 45                | 3                 | 20                    | 60                     |
| 55                | 2                 | 30                    | 60                     |
| 65                | 7                 | 40                    | 280                    |
| <b>Total</b>      | $N = 50$          |                       | $\sum f X - Md  = 760$ |

Hence,

$$\text{M.D. about median} = \frac{1}{N} \sum f|X - Md| = \frac{760}{50} = 15.2.$$

Therefore,

$$\boxed{\text{M.D. about median} = 15.2}$$

**(c) Compare the results obtained in (a) and (b).**

From (a),

$$\text{M.D. about mean} = 16.$$

From (b),

$$\text{M.D. about median} = 15.2.$$

Therefore,

$$15.2 < 16.$$

Hence,

$$\boxed{\text{M.D. about median} < \text{M.D. about mean}}$$

Thus, we observe the following general result:

*“Mean deviation is least when taken about median.”*

### Example

Calculate mean deviation from the median for the following data:

|                 |     |    |    |    |    |    |    |    |
|-----------------|-----|----|----|----|----|----|----|----|
| Marks less than | 80  | 70 | 60 | 50 | 40 | 30 | 20 | 10 |
| No. of Students | 100 | 90 | 80 | 60 | 32 | 20 | 13 | 5  |

### Solution:

First of all, we shall convert the given cumulative frequency distribution into an ordinary frequency distribution.

| Marks        | c.f. | Frequency ( $f$ )  | Mid-value ( $X$ ) | $ X - Md $ | $f X - Md $                     |
|--------------|------|--------------------|-------------------|------------|---------------------------------|
| 0 – 10       | 5    | 5                  | 5                 | 41.43      | 207.15                          |
| 10 – 20      | 13   | $13 - 5 = 8$       | 15                | 31.43      | 251.44                          |
| 20 – 30      | 20   | $20 - 13 = 7$      | 25                | 21.43      | 150.01                          |
| 30 – 40      | 32   | $32 - 20 = 12$     | 35                | 11.43      | 137.16                          |
| 40 – 50      | 60   | $60 - 32 = 28$     | 45                | 1.43       | 40.04                           |
| 50 – 60      | 80   | $80 - 60 = 20$     | 55                | 8.57       | 171.40                          |
| 60 – 70      | 90   | $90 - 80 = 10$     | 65                | 18.57      | 185.70                          |
| 70 – 80      | 100  | $100 - 90 = 10$    | 75                | 28.57      | 285.70                          |
| <b>Total</b> |      | $N = \sum f = 100$ |                   |            | $\sum f X - Md  \approx 1428.6$ |

Here,

$$\frac{N}{2} = \frac{100}{2} = 50.$$

Since the cumulative frequency just greater than 50 is 60, the corresponding class 40 – 50 is the median class.

Using the median formula,

$$Md = l + \frac{h}{f} \left( \frac{N}{2} - C \right),$$

where

$$l = 40, \quad h = 10, \quad f = 28, \quad C = 32.$$

Therefore,

$$\begin{aligned} Md &= 40 + \frac{10}{28}(50 - 32) \\ &= 40 + \frac{180}{28} \\ &= 40 + 6.43 \\ &= 46.43. \end{aligned}$$

Thus,

$$Md = 46.43$$

The mean deviation about the median is

$$\begin{aligned} \text{M.D. about } Md &= \frac{1}{N} \sum f|X - Md| \\ &= \frac{1428.6}{100} \\ &= 14.286 \\ &\approx 14.29. \end{aligned}$$

Hence,

$$\text{M.D. about median} = 14.29$$

### Mean Deviation by Short-Cut Method

Since the median value comes out to be in fractions, we can solve the above question conveniently by the short-cut method, by taking the deviations from an arbitrary point

$$a = 45,$$

which lies in the median class.

| Marks ( $X$ ) | $f$            | $d = X - 45$ | $ d $ | $f d $             |
|---------------|----------------|--------------|-------|--------------------|
| 5             | 5              | -40          | 40    | 200                |
| 15            | 8              | -30          | 30    | 240                |
| 25            | 7              | -20          | 20    | 140                |
| 35            | 12             | -10          | 10    | 120                |
| 45            | 28             | 0            | 0     | 0                  |
| 55            | 20             | 10           | 10    | 200                |
| 65            | 10             | 20           | 20    | 200                |
| 75            | 10             | 30           | 30    | 300                |
| <b>Total</b>  | $\sum f = 100$ |              |       | $\sum f d  = 1400$ |

Using the short-cut formula,

$$\text{M.D. about } Md = \frac{1}{N} \left[ \sum f|d| + (Md - a) \left( \sum f'_B - \sum f'_A \right) \right].$$

Here,

$$\sum f'_B = 5 + 8 + 7 + 12 + 28 = 60,$$

and

$$\sum f'_A = N - \sum f'_B = 100 - 60 = 40.$$

Therefore,

$$\begin{aligned}
 \text{M.D. about } Md &= \frac{1}{100} [1400 + (46.43 - 45)(60 - 40)] \\
 &= \frac{1}{100} [1400 + 1.43(20)] \\
 &= \frac{1428.6}{100} \\
 &= 14.286 \\
 &\approx 14.29.
 \end{aligned}$$

Thus,

$$\boxed{\text{M.D. about median} = 14.29}$$

**Remark:** The values of mean deviation obtained by the direct method and the short-cut method are the same, since the arbitrary value  $a$  is taken in the median class.

### Example

Mean deviation may be calculated from the arithmetic mean, the median or the mode. Which of these three measures is the minimum?

Five towns  $A$ ,  $B$ ,  $C$ ,  $D$  and  $E$  lie in that order along a road. The distances in kilometres of the towns, measured from  $A$  are  $A = 0$ ,  $B = 5$ ,  $C = 9$ ,  $D = 16$ ,  $E = 20$ .

A new college is to be established at one of these places, and the number of students who would join the college are  $A = 39$ ,  $B = 911$ ,  $C = 46$ ,  $D = 193$ , and  $E = 716$ .

The criterion for choosing the location is that the total distance travelled by the students, as measured in student-kilometres, should be a minimum.

Thus, if the college is located at  $D$ , 46 students from  $C$  will travel in all  $46 \times 7 = 322$  student-kilometres.

**Where should the college be located? Justify your result.**

### Solution:

Mean deviation calculated from the median is the minimum.

Let  $X$  denote the distance (in kms.) as measured from the town  $A$ . Then the frequency distribution of the distances covered by the students in going to the college is as given in the following table.

| Town | Distance from town A ( $X$ ) | No. of students ( $f$ ) | Less than c.f. |
|------|------------------------------|-------------------------|----------------|
| $A$  | 0                            | 39                      | 39             |
| $B$  | 5                            | 911                     | 950            |
| $C$  | 9                            | 46                      | 996            |
| $D$  | 16                           | 193                     | 1189           |
| $E$  | 20                           | 716                     | $N = 1905$     |

We are given that the criterion for choosing the location of the college is that the total distance travelled by the students, as measured in student-kilometres, should be minimum.

Since mean deviation is minimum when calculated from median, the total distance travelled by the students will be minimum at the point  $X = \text{Median of the frequency distribution of } X$ .

Here,

$$\frac{N}{2} = \frac{1905}{2} = 952.5.$$

The c.f. just greater than 952.5 is 996.

Hence, the corresponding value of  $X = 9$  is the median.

Since  $X = 9$  corresponds to the town  $C$ , the college should be located at the town  $C$ .

The college should be located at town  $C$ .

## Variance

Variance is the average of the squared deviations of the observations from their arithmetic mean.

For individual observations,

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{N}$$

For a frequency distribution,

$$\sigma^2 = \frac{\sum f(X - \bar{x})^2}{\sum f}$$

where

$\bar{x}$  = Arithmetic Mean.

Variance is expressed in the **squared units** of the original observations.

**For example**, if the observations are measured in kilograms, variance is expressed in  $\text{kg}^2$ .

## Standard Deviation

The **Standard Deviation (S.D.)** is the positive square root of variance.

For individual observations,

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

For a frequency distribution,

$$\sigma = \sqrt{\frac{\sum f(X - \bar{x})^2}{\sum f}}$$

Standard Deviation is one of the most important and widely used measures of dispersion.

## Interpretation

- A small standard deviation indicates that the observations are closely concentrated around the mean.
- A large standard deviation indicates that the observations are widely scattered around the mean.
- If the standard deviation is zero, all observations are identical.

## Coefficient of Variation

The **Coefficient of Variation (C.V.)** is a relative measure of dispersion based on the Standard Deviation.

It is given by

$$C.V. = \frac{\sigma}{\bar{x}} \times 100$$

where

$\sigma$  = Standard Deviation

and

$\bar{x}$  = Arithmetic Mean.

The coefficient of variation is particularly useful for comparing the variability or consistency of two or more datasets.

## Interpretation

Smaller C.V.  $\Rightarrow$  Greater consistency

Larger C.V.  $\Rightarrow$  Less consistency

### Example 1:

Calculate the variance and standard deviation for the following observations:

2, 4, 6, 8, 10.

**Solution:**

$$\text{Variance} = \sigma^2 = 8$$

$$\text{Standard Deviation} = \sigma = 2.828$$

### Example 2:

Calculate the variance and standard deviation for the following frequency distribution:

|     |    |    |    |    |
|-----|----|----|----|----|
| $X$ | 10 | 20 | 30 | 40 |
| $f$ | 2  | 3  | 4  | 1  |

**Solution:**

| X | f | fX | $X - \bar{X}$ | $(X - \bar{X})^2$ | $f(X - \bar{X})^2$ |
|---|---|----|---------------|-------------------|--------------------|
|   |   |    |               |                   |                    |
|   |   |    |               |                   |                    |
|   |   |    |               |                   |                    |
|   |   |    |               |                   |                    |
|   |   |    |               |                   |                    |

$$\text{Variance} = \sigma^2 = 84$$

$$\text{Standard Deviation} = \sigma \approx 9.165$$

**Example 3:**

Calculate the variance and standard deviation for the following continuous frequency distribution:

| Class Interval | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 |
|----------------|--------|---------|---------|---------|
| Frequency (f)  | 5      | 8       | 4       | 3       |

**Solution:**

| Class Interval | f | X | fX | $X - \bar{X}$ | $(X - \bar{X})^2$ | $f(X - \bar{X})^2$ |
|----------------|---|---|----|---------------|-------------------|--------------------|
|                |   |   |    |               |                   |                    |
|                |   |   |    |               |                   |                    |
|                |   |   |    |               |                   |                    |
|                |   |   |    |               |                   |                    |
|                |   |   |    |               |                   |                    |

$$\text{Variance} = \sigma^2 = 98.75$$

$$\text{Standard Deviation} = \sigma \approx 9.937$$

**Example 4:**

Calculate the variance and standard deviation for the following less-than cumulative frequency distribution:

| Marks less than | Cumulative Frequency |
|-----------------|----------------------|
| 10              | 5                    |
| 20              | 13                   |
| 30              | 20                   |
| 40              | 32                   |
| 50              | 40                   |
| 60              | 50                   |

**Solution:**

First, convert the cumulative frequency distribution into an ordinary frequency distribution.

| Class Interval | f | CF | X | fX | $X - \bar{X}$ | $(X - \bar{X})^2$ | $f(X - \bar{X})^2$ |
|----------------|---|----|---|----|---------------|-------------------|--------------------|
|                |   |    |   |    |               |                   |                    |
|                |   |    |   |    |               |                   |                    |
|                |   |    |   |    |               |                   |                    |
|                |   |    |   |    |               |                   |                    |
|                |   |    |   |    |               |                   |                    |
|                |   |    |   |    |               |                   |                    |

$$\text{Variance} = 260$$

$$\text{Standard Deviation} \approx 16.125$$

**Example 5:**

The arithmetic mean and the standard deviation of a set of 9 items are 43 and 5 respectively. If an item of value 63 is added to the set, find the mean and standard deviation of 10 items.

**Solution:**

We are given

$$n = 9, \quad \bar{x} = 43, \quad \sigma = 5.$$

We know that

$$\bar{x} = \frac{\sum x}{n}$$

Therefore,

$$\sum x = n\bar{x} = 9(43) = 387. \quad (i)$$

Also,

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \quad \Rightarrow \quad \sum x^2 = n(\sigma^2 + \bar{x}^2)$$

Therefore,

$$\sum x^2 = n(\sigma^2 + \bar{x}^2) = 9(5^2 + 43^2) = 9(25 + 1849) = 9(1874) = 16866.$$

If a new item of value 63 is added, then the number of items becomes

$$n = 10.$$

The new sum is

$$\text{New } \sum x = 387 + 63 = 450.$$

Therefore, the new mean is

$$\bar{x}_{\text{new}} = \frac{450}{10} = \boxed{45}.$$

The new sum of squares is

$$\begin{aligned} \text{New } \sum x^2 &= 16866 + 63^2 \\ &= 16866 + 3969 = 20835. \end{aligned}$$

Hence, the new standard deviation is

$$\begin{aligned} \sigma_{\text{new}} &= \sqrt{\frac{\text{New } \sum x^2}{10} - (\text{New mean})^2} \\ &= \sqrt{\frac{20835}{10} - (45)^2} \\ &= \sqrt{2083.5 - 2025} \\ &= \sqrt{58.5} \\ &= \boxed{7.65}. \end{aligned}$$

Hence,

|                |                               |
|----------------|-------------------------------|
| New Mean = 45, | New Standard Deviation = 7.65 |
|----------------|-------------------------------|

### Example 6:

Initially there were 9 workers, all being paid a uniform wage. Later a 10th worker was added whose wage rate is Rs. 20 less than for the others. Compute

- (i) the effect on the mean wage.
- (ii) the standard deviation of wages for the group of 10 workers.

**Solution:**

Let the constant wage of each of the 9 workers be Rs.  $c$ .

The wage of the 10th worker is therefore

$$\text{Wage of 10th worker} = c - 20.$$

**(i) Effect on the mean wage**

The mean wage of the 9 workers is

$$\bar{x} = \frac{9c}{9} = c.$$

The mean wage of the 10 workers is

$$\begin{aligned}\bar{x} &= \frac{c + c + \cdots + c + (c - 20)}{10} \\ &= \frac{10c - 20}{10} \\ &= c - 2.\end{aligned}$$

Hence, with the addition of the 10th worker, the average wage has decreased by

$$\boxed{\text{Rs. } 2}.$$

**(ii) Standard deviation of wages**

| S.No.    | $x$      | $x - \bar{x} = x - (c - 2)$ | $(x - \bar{x})^2$ |
|----------|----------|-----------------------------|-------------------|
| 1        | $c$      | 2                           | 4                 |
| 2        | $c$      | 2                           | 4                 |
| $\vdots$ | $\vdots$ | $\vdots$                    | $\vdots$          |
| 9        | $c$      | 2                           | 4                 |
| 10       | $c - 20$ | -18                         | 324               |

Therefore,

$$\begin{aligned}\sigma^2 &= \frac{1}{10} \sum (x - \bar{x})^2 \\ &= \frac{1}{10} [(4 + 4 + \cdots + 4) + 324]\end{aligned}$$

Since there are nine values of 4,

$$\begin{aligned}\sigma^2 &= \frac{1}{10} (4 \times 9 + 324) \\ &= \frac{360}{10} = 36.\end{aligned}$$

Therefore,

$$\sigma = \sqrt{36} = \boxed{6}.$$

Hence,

$$\boxed{\text{Decrease in mean wage} = \text{Rs. } 2}$$

and

$$\boxed{\text{Standard deviation} = \text{Rs. } 6}.$$

Dr. Boina Anil Kumar

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Skewness and Kurtosis

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Skewness

Skewness is a measure of the degree of **asymmetry** in a distribution about its central value.

A distribution is said to be **symmetric** if the observations are equally distributed on both sides of the central value.

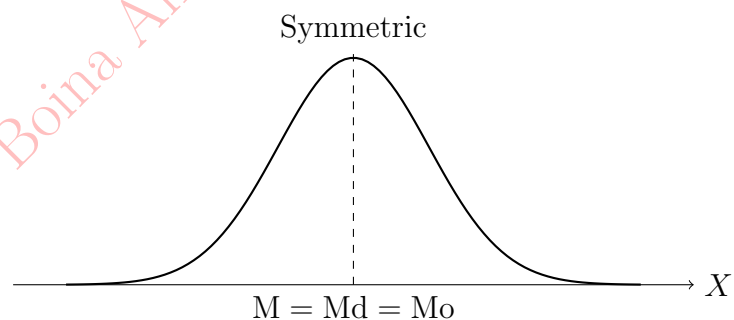
If the observations are not equally distributed on both sides, the distribution is said to be **skewed**.

There are three important types of distributions based on skewness.

### Symmetric Distribution

$$\text{Mean} = \text{Median} = \text{Mode}$$

$$\text{Skewness} = 0$$

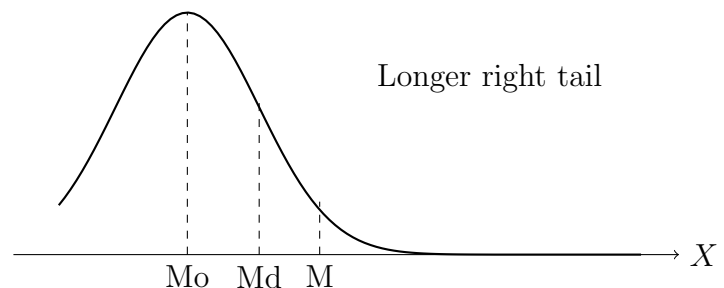


### Positively Skewed Distribution

$$\text{Mean} > \text{Median} > \text{Mode}$$

$$\text{Skewness} > 0$$

Long Tail towards right

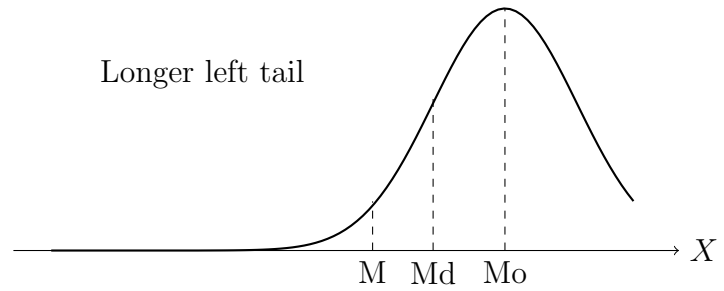


## Negatively Skewed Distribution

$$\text{Mean} < \text{Median} < \text{Mode}$$

$$\text{Skewness} < 0.$$

Long Tail towards left



## Measures of Skewness

The absolute measures of skewness are

- (i)  $Sk = \text{Mean} - \text{Median} = M - Md$
- (ii)  $Sk = \text{Mean} - \text{Mode} = M - Mo$
- (iii)  $Sk = (Q_3 - Md) - (Md - Q_1) = Q_3 + Q_1 - 2Md$

These are not much practically important because they cannot be used for comparative study of different distributions.

The commonly used relative measures of skewness are:

1. Karl Pearson's coefficient of skewness
2. Bowley's coefficient of skewness
3. Moment coefficient of skewness

## Karl Pearson's Coefficient of Skewness

Karl Pearson's coefficient of skewness is given by

$$Sk = \frac{\text{Mean} - \text{Mode}}{s.d} = \frac{\bar{X} - \text{Mode}}{\sigma}$$

When the mode is not clearly defined, the empirical relation

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

may be used.

Therefore,

$$Sk = \frac{3(\bar{X} - \text{Median})}{\sigma}$$

**Example 1:**

The mean, median and standard deviation of a distribution are 35, 32 and 6, respectively. Find Karl Pearson's coefficient of skewness.

**Solution:**

Given,

$$\bar{X} = 35, \quad \text{Median} = 32, \quad \sigma = 6$$

Using Karl Pearson's formula,

$$Sk = \frac{3(\bar{X} - \text{Median})}{\sigma} = \frac{3(35 - 32)}{6} = \boxed{1.5}$$

Since  $Sk > 0$ , the distribution is **positively skewed**.

**Example 2**

Calculate Karl Pearson's coefficient of skewness from the following data.

| Size ( $x$ )      | 1  | 2  | 3  | 4  | 5  | 6 | 7 |
|-------------------|----|----|----|----|----|---|---|
| Frequency ( $f$ ) | 10 | 18 | 30 | 25 | 12 | 3 | 2 |

**Solution.**

| Size ( $x$ ) | Frequency ( $f$ ) | $fx$            | $fx^2$             |
|--------------|-------------------|-----------------|--------------------|
| 1            | 10                | 10              | 10                 |
| 2            | 18                | 36              | 72                 |
| 3            | 30                | 90              | 270                |
| 4            | 25                | 100             | 400                |
| 5            | 12                | 60              | 300                |
| 6            | 3                 | 18              | 108                |
| 7            | 2                 | 14              | 98                 |
| <b>Total</b> | $N = 100$         | $\sum fx = 328$ | $\sum fx^2 = 1258$ |

Mean is

$$M = \frac{\sum fx}{N} = \frac{328}{100} = 3.28.$$

Standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} \\ &= \sqrt{\frac{1258}{100} - \left(\frac{328}{100}\right)^2} \\ &= \sqrt{12.58 - 10.7584} \\ &= \sqrt{1.8216} \\ &= 1.3497.\end{aligned}$$

Since the maximum frequency is 30, the corresponding value of  $x$ , namely 3, is the mode.

Thus,

$$Mo = 3.$$

Karl Pearson's coefficient of skewness is

$$Sk = \frac{M - Mo}{\sigma} = \frac{3.28 - 3.00}{1.3497} = 0.2075.$$

Hence, the distribution is **slightly positively skewed**.

### Example 3

*Calculate Pearson's measure of skewness on the basis of Mean, Mode and Standard Deviation.*

| Mid-value ( $X$ ) | 14.5 | 15.5 | 16.5 | 17.5 | 18.5 | 19.5 | 20.5 | 21.5 |
|-------------------|------|------|------|------|------|------|------|------|
| $f$               | 35   | 40   | 48   | 100  | 125  | 87   | 43   | 22   |

### Solution

Here, the values of  $X$  are the mid-values of the class intervals 14–15, 15–16, ..., and so on, of magnitude 1.

The corresponding continuous frequency distribution is

| Class Interval | Frequency ( $f$ ) |
|----------------|-------------------|
| 14–15          | 35                |
| 15–16          | 40                |
| 16–17          | 48                |
| 17–18          | 100               |
| 18–19          | 125               |
| 19–20          | 87                |
| 20–21          | 43                |
| 21–22          | 22                |

The maximum frequency is 125. Thus, the corresponding class 18 – 19 is the modal class.

$$\text{Mode} = 18.397 \approx 18.4$$

$$\bar{X} = 18.066 \approx 18.07$$

$$\sigma = 1.775$$

Karl Pearson's coefficient of skewness based on mean, mode and standard deviation is given by

$$Sk = \frac{M - M_o}{\sigma} = \frac{18.07 - 18.40}{1.775} = -0.186$$

Hence, the given distribution is slightly negatively skewed.

#### Example 4

Calculate Karl Pearson's coefficient of skewness from the following data.

| Hourly wages   | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 | 100-110 | 110-120 | 120-130 | 130-140 |
|----------------|-------|-------|-------|-------|-------|--------|---------|---------|---------|---------|
| No. of workers | 5     | 6     | 8     | 10    | 25    | 30     | 36      | 50      | 60      | 70      |

#### Solution

For a distribution for which Mean, Median and S.D. are known, Karl Pearson's coefficient of skewness may be calculated by

$$Sk = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

For the given distribution,

$$\text{Mean} = 110.93, \quad \text{Median} = 116, \quad \sigma = 22.3043.$$

Therefore,

$$Sk = \frac{3(110.93 - 116)}{22.3043} = -0.6819.$$

Hence, the distribution is **negatively skewed**.

#### Example 5

Find Karl Pearson's coefficient of skewness.

The following is the given “more than” cumulative frequency distribution:

| Wages more than (in Rs.) | 5   | 15 | 25 | 35 | 45 | 55 | 65 |
|--------------------------|-----|----|----|----|----|----|----|
| Number of earners        | 100 | 80 | 75 | 60 | 55 | 20 | 0  |

## Solution

We are given a “more than” cumulative frequency distribution. To compute Karl Pearson’s coefficient of skewness, which is based on mean, mode and standard deviation, we first convert it into a continuous frequency distribution (exclusive type) as given below.

| Wages<br>(in Rs.)  | No. of<br>earners ( $f$ ) | Less<br>than c.f. | Mid-value<br>( $X$ ) | $d = \frac{X - 30}{10}$ | $fd$           | $fd^2$            |
|--------------------|---------------------------|-------------------|----------------------|-------------------------|----------------|-------------------|
| 5–15               | $100 - 80 = 20$           | 20                | 10                   | –2                      | –40            | 80                |
| 15–25              | $80 - 75 = 5$             | 25                | 20                   | –1                      | –5             | 5                 |
| 25–35              | $75 - 60 = 15$            | 40                | 30                   | 0                       | 0              | 0                 |
| 35–45              | $60 - 55 = 5$             | 45                | 40                   | 1                       | 5              | 5                 |
| 45–55              | $55 - 20 = 35$            | 80                | 50                   | 2                       | 70             | 140               |
| 55–65              | $20 - 0 = 20$             | 100               | 60                   | 3                       | 60             | 180               |
| $\sum f = N = 100$ |                           |                   |                      |                         | $\sum fd = 90$ | $\sum fd^2 = 410$ |

Taking the assumed mean  $A = 30$  and class width  $h = 10$ ,

$$M = A + \frac{h \sum fd}{N} = 30 + \frac{10 \times 90}{100} = 30 + 9 = \boxed{\text{Rs. } 39.00}$$

The standard deviation is given by

$$\sigma = h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} = 10 \sqrt{\frac{410}{100} - \left(\frac{90}{100}\right)^2} = \boxed{18.138}$$

The modal class is

$$45 - 55$$

Therefore,

$$l = 45, \quad h = 10, \quad f_1 = 35, \quad f_0 = 5, \quad f_2 = 20.$$

Now,

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2} = 45 + \frac{10(35 - 5)}{2(35) - 5 - 20} = \boxed{51.6666}$$

Karl Pearson’s coefficient of skewness is given by

$$Sk = \frac{M - Mo}{\sigma} = \frac{39 - 51.6666}{18.138} \approx -0.69.$$

Since the coefficient of skewness is negative, the distribution is

moderately negatively skewed.

### Example 6

Consider the following distributions:

|                    | Distribution A | Distribution B |
|--------------------|----------------|----------------|
| Mean               | 100            | 90             |
| Median             | 90             | 80             |
| Standard Deviation | 10             | 10             |

Answer the following:

1. Distribution A has the same degree of variation as distribution B.
2. Both distributions have the same degree of skewness. True/False? Comment, giving reasons.

### Solution

(i) Coefficient of variation for distribution A is

$$C.V._A = 100 \times \frac{\sigma_A}{\bar{X}_A} = 100 \times \frac{10}{100} = 10.$$

Coefficient of variation for distribution B is

$$C.V._B = 100 \times \frac{\sigma_B}{\bar{X}_B} = 100 \times \frac{10}{90} = 11.11.$$

Since

$$C.V._B > C.V._A,$$

the distribution B is more variable than distribution A.

Hence, the statement that distribution A has the same degree of variation as distribution B is **wrong**.

(ii) Karl Pearson's coefficient of skewness for the distributions A and B is

$$Sk = \frac{3(M - Md)}{\sigma}.$$

For distribution A,

$$Sk(A) = \frac{3(100 - 90)}{10} = 3.$$

For distribution B,

$$Sk(B) = \frac{3(90 - 80)}{10} = 3.$$

Since

$$Sk(A) = Sk(B) = 3,$$

the statement that both distributions have the same degree of skewness is **true**.

**Example 7**

Pearson's coefficient of skewness for a distribution is 0.4 and its coefficient of variation is 30%. Its mode is 88. Find mean and median.

**Solution**

Given that

$$Mo = 88 \quad Sk = 0.4 \quad \text{C.V.} = 30$$

Therefore, we have

$$Sk = \frac{M - Mo}{\sigma} = 0.4 \quad \text{C.V.} = \frac{100\sigma}{M} = 30.$$

On simplification,

$$\frac{M - 88}{\sigma} = 0.4. \quad (1)$$

$$\sigma = \frac{30M}{100} = 0.3M. \quad (2)$$

From equations (1) and (2),

$$M - 88 = 0.4\sigma$$

and

$$M - 88 = 0.4(0.3M).$$

$$\Rightarrow M - 88 = 0.12M.$$

$$\Rightarrow M - 0.12M = 88,$$

$$\Rightarrow 0.88M = 88.$$

Hence,

$$\boxed{M = \frac{88}{0.88} = 100}.$$

Therefore,

$$\sigma = 0.3 \times 100 = 30.$$

Now,

$$3Md = Mo + 2M.$$

Substituting the values,

$$3Md = 88 + 2(100) = 288.$$

Hence,

$$\boxed{Md = \frac{288}{3} = 96}.$$

Therefore,

$$\boxed{\text{Mean} = 100, \quad \text{Median} = 96.}$$

### Example 8

Pearson's measure of skewness of a distribution is 0.5. Its median and mode are respectively 42 and 36. Find the Coefficient of Variation.

#### Solution

We are given:

$$\text{Median} = 42, \quad \text{Mode} = 36$$

Pearson's coefficient of skewness is:

$$Sk = \frac{\text{Mean} - \text{Mode}}{\sigma} = 0.5$$

To find the standard deviation ( $\sigma$ ), we shall first find the value of mean, by using the empirical relationship between mean, median and mode for a moderately asymmetrical distribution viz.,

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

Therefore,

$$\text{Mean} = \frac{3\text{Median} - \text{Mode}}{2}$$

$$\text{Mean} = \frac{3(42) - 36}{2} = \frac{126 - 36}{2} = \frac{90}{2} = 45$$

Substituting in the expression for Pearson's coefficient of skewness, we get:

$$\sigma = \frac{\text{Mean} - \text{Mode}}{0.5} = \frac{45 - 36}{0.5} = \frac{9}{0.5} = 18$$

The coefficient of variation is:

$$\text{C.V.} = \frac{100\sigma}{\text{Mean}} = \frac{100(18)}{45} = 40$$

Hence,

$$\boxed{\text{C.V.} = 40}$$

### Example 9

The sum of 50 observations is 500 and the sum of their squares is 6,000, and median is 12. Compute the coefficient of variation and the coefficient of skewness.

#### Solution

In the usual notations, we are given:

$$n = 50, \quad \sum x = 500, \quad \sum x^2 = 6000, \quad \text{Median} = 12$$

Mean:

$$\bar{x} = \frac{\sum x}{n} = \frac{500}{50} = 10$$

Variance:

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{6000}{50} - 100 = 120 - 100 = 20$$

Therefore,

$$\sigma = \sqrt{20} = 4.47$$

Coefficient of Variation:

$$\text{C.V.} = \frac{100\sigma}{\bar{x}} = \frac{100(4.47)}{10} = 44.7$$

Karl Pearson's coefficient of skewness is given by:

$$Sk = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(10 - 12)}{4.47} = \frac{-6}{4.47} = -1.34$$

Therefore,

$$\boxed{\text{C.V.} = 44.7} \quad \text{and} \quad \boxed{Sk = -1.34}$$

### Example 10

The following information was obtained from the records of a factory relating to wages:

|                 |           |
|-----------------|-----------|
| Arithmetic Mean | Rs. 56.80 |
| Median          | Rs. 59.50 |
| S.D.            | Rs. 12.40 |

Give as much information as you can about the distribution of wages.

### Solution

We can obtain the following information from the above data:

1. Since Median = Rs. 59.50, we conclude that 50% of the workers in the factory obtain wages above Rs. 59.50.

2.

$$\text{C.V.} = \frac{100\sigma}{M} = \frac{100 \times 12.40}{56.80} = 21.83$$

3. Karl Pearson's coefficient of skewness is given by:

$$Sk = \frac{3(M - Md)}{\sigma} = \frac{3(56.80 - 59.50)}{12.40} = -0.65$$

Hence, the distribution of wages is negatively skewed.

### Bowley's Coefficient of Skewness

Bowley's coefficient of skewness is based on quartiles and median.

$$\boxed{Sk_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1}}$$

The value of Bowley's coefficient of skewness lies between  $-1$  and  $+1$ .

### Example 1

For a distribution,

$$Q_1 = 20, \quad \text{Median} = 28, \quad Q_3 = 40.$$

Find Bowley's coefficient of skewness and comment on the distribution.

### Solution

Bowley's coefficient of skewness is

$$Sk_B = \frac{Q_3 + Q_1 - 2M}{Q_3 - Q_1} = \frac{40 + 20 - 2(28)}{40 - 20} = \boxed{0.20}$$

Since  $Sk_B > 0$ , the distribution is **positively skewed**.

### Example 2

For a distribution,

$$Q_1 = 15, \quad \text{Median} = 25, \quad Q_3 = 30.$$

Find Bowley's coefficient of skewness.

### Solution

$$Sk_B = \frac{Q_3 + Q_1 - 2M}{Q_3 - Q_1} = \frac{30 + 15 - 2(25)}{30 - 15} = \boxed{-0.333}$$

Since the coefficient is negative, the distribution is **negatively skewed**.

### Example 3

Calculate Bowley's coefficient of skewness of the following data:

| Weight (lbs) | No. of persons | Weight (lbs) | No. of persons | Weight (lbs) | No. of persons |
|--------------|----------------|--------------|----------------|--------------|----------------|
| Under 100    | 1              | 130–139      | 145            | 170–179      | 12             |
| 100–109      | 14             | 140–149      | 121            | 180–189      | 5              |
| 110–119      | 66             | 150–159      | 65             | 190–199      | 2              |
| 120–129      | 122            | 160–169      | 31             | 200 and over | 2              |

### Solution

Here we are given the frequency distribution with inclusive type classes. Since the formulae for median and quartiles are based on continuous frequency distribution with exclusive type classes without any gaps, we obtain the class boundaries which are given in the last column of the adjoining table.

| Weight (lbs) | No. of persons (f) | Less than c.f. | Class boundaries |
|--------------|--------------------|----------------|------------------|
| 0–99         | 1                  | 1              | 0 – –99.5        |
| 100–109      | 14                 | 15             | 99.5 – –109.5    |
| 110–119      | 66                 | 81             | 109.5 – –119.5   |
| 120–129      | 122                | 203            | 119.5 – –129.5   |
| 130–139      | 145                | 348            | 129.5 – –139.5   |
| 140–149      | 121                | 469            | 139.5 – –149.5   |
| 150–159      | 65                 | 534            | 149.5 – –159.5   |
| 160–169      | 31                 | 565            | 159.5 – –169.5   |
| 170–179      | 12                 | 577            | 169.5 – –179.5   |
| 180–189      | 5                  | 582            | 179.5 – –189.5   |
| 190–199      | 2                  | 584            | 189.5 – –199.5   |
| 200–209      | 2                  | $N = 586$      | 199.5 – –209.5   |

Here,

$$N = 586, \quad \frac{N}{4} = 146.5, \quad \frac{N}{2} = 293, \quad \frac{3N}{4} = 439.5.$$

The c.f. just greater than  $N/2$ , i.e., 293, is 348. Hence, the corresponding class 129.5 – –139.5 is the median class. Using the Median formula, we get

$$\begin{aligned} Md(Q_2) &= 129.5 + \frac{10}{145}(293 - 203) \\ &= 129.5 + \frac{10 \times 90}{145} = 129.50 + 6.21 = 135.71. \end{aligned}$$

The c.f. just greater than  $N/4$ , i.e., 146.5, is 203. Hence, the corresponding class 119.5 – –129.5 contains  $Q_1$ .

$$\begin{aligned} Q_1 &= 119.5 + \frac{10}{122}(146.5 - 81) \\ &= 119.5 + \frac{10 \times 65.5}{122} = 119.50 + 5.37 = 124.87. \end{aligned}$$

The c.f. just greater than  $3N/4$ , i.e., 439.5, is 469. Hence, the corresponding class 139.5 – –149.5 contains  $Q_3$ .

$$\begin{aligned} Q_3 &= 139.5 + \frac{10}{121}(439.5 - 348) \\ &= 139.5 + \frac{10 \times 91.5}{121} = 139.50 + 7.56 = 147.06. \end{aligned}$$

Bowley's coefficient of skewness is given by

$$Sk_{(\text{Bowley})} = \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1}.$$

Therefore,

$$Sk_{(\text{Bowley})} = \frac{147.06 + 124.87 - 2(135.71)}{147.06 - 124.87}$$

$$= \frac{271.93 - 271.42}{22.19} = \frac{0.51}{22.19} = 0.0298.$$

Hence,

$$Sk_{(\text{Bowley})} = 0.0298$$

#### Example 4

Calculate Bowley's coefficient of skewness from the following data:

|           |    |     |     |     |     |     |     |     |
|-----------|----|-----|-----|-----|-----|-----|-----|-----|
| Mid-value | 75 | 100 | 125 | 150 | 175 | 200 | 225 | 250 |
| Frequency | 35 | 40  | 48  | 100 | 125 | 80  | 50  | 22  |

#### Solution

Here we are given the mid-values of the class intervals and since we want to compute the coefficient of skewness based on  $Q_1$  and  $Q_3$ , we first convert the given distribution into continuous frequency distribution, the magnitude of the class interval being 25, as given in the adjoining table.

| Class boundaries | Mid-value | Frequency (f) | Less than c.f. |
|------------------|-----------|---------------|----------------|
| 62.5–87.5        | 75        | 35            | 35             |
| 87.5–112.5       | 100       | 40            | 75             |
| 112.5–137.5      | 125       | 48            | 123            |
| 137.5–162.5      | 150       | 100           | 223            |
| 162.5–187.5      | 175       | 125           | 348            |
| 187.5–212.5      | 200       | 80            | 428            |
| 212.5–237.5      | 225       | 50            | 478            |
| 237.5–262.5      | 250       | 22            | 500            |
|                  |           | N=500         |                |

Here

$$\frac{N}{4} = 125, \quad \frac{N}{2} = 250, \quad \frac{3N}{4} = 375.$$

Since c.f. greater than 125 is 223, the corresponding class 137.5 – 162.5 contains  $Q_1$ ; since c.f. greater than 250 is 348, the corresponding class 162.5 – 187.5 is the median class. Further, since c.f. greater than 375 is 428, the corresponding class 187.5 – 212.5 contains  $Q_3$ .

Hence,

$$Q_1 = 137.5 + \frac{25}{100}(125 - 123) = 137.5 + 0.5 = 138.0$$

$$Md = 162.5 + \frac{25}{125}(250 - 223) = 162.5 + 5.4 = 167.9$$

$$Q_3 = 187.5 + \frac{25}{80}(375 - 348) = 187.50 + 8.44 = 195.94$$

Bowley's coefficient of skewness is given by

$$\begin{aligned} Sk &= \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1} \\ &= \frac{195.94 + 138.0 - 2(167.9)}{195.94 - 138.0} \\ &= \frac{195.94 + 138.0 - 335.8}{57.94} = \frac{-1.86}{57.94} = -0.032. \end{aligned}$$

$$Sk = -0.032$$

### Example 5

Calculate the coefficient of skewness from the following data by using quartiles.

| Marks    | No. of students | Marks    | No. of students |
|----------|-----------------|----------|-----------------|
| Above 0  | 180             | Above 60 | 65              |
| Above 15 | 160             | Above 75 | 20              |
| Above 30 | 130             | Above 90 | 5               |
| Above 45 | 100             |          |                 |

### Solution

We are given “more than” cumulative frequency distribution. To compute quartiles, we first express it as a continuous frequency distribution without any gaps as given in the adjoining table.

| Marks    | Number of students (f) | Less than c.f. |
|----------|------------------------|----------------|
| 0–15     | 180-160=20             | 20             |
| 15–30    | 160-130=30             | 50             |
| 30–45    | 130-100=30             | 80             |
| 45–60    | 100-65=35              | 115            |
| 60–75    | 65-20=45               | 160            |
| 75–90    | 20-5=15                | 175            |
| above 90 | 5                      | 180            |
| Total    | $N = \sum f = 180$     |                |

$$\frac{N}{2} = \frac{180}{2} = 90, \quad \frac{N}{4} = \frac{180}{4} = 45, \quad \frac{3N}{4} = \frac{3(180)}{4} = 135.$$

The c.f. just greater than  $N/2$ , i.e., 90, is 115. Hence the corresponding class 45 – 60 is the median class.

Therefore,

$$\begin{aligned} Md &= l + \frac{h}{f} \left( \frac{N}{2} - C \right) \\ &= 45 + \frac{15}{35}(90 - 80) \\ &= 45 + \frac{15 \times 10}{35} = 45 + 4.29 = 49.29. \end{aligned}$$

The c.f. just greater than  $N/4$ , i.e., 45, is 50. Hence the corresponding class 15 – 30 contains  $Q_1$ .

$$\begin{aligned} Q_1 &= l + \frac{h}{f} \left( \frac{N}{4} - C \right) \\ &= 15 + \frac{15}{30}(45 - 20) \\ &= 15 + \frac{15 \times 25}{30} = 27.50. \end{aligned}$$

The c.f. just greater than  $3N/4$ , i.e., 135, is 160. Hence the corresponding class 60 – 75 contains  $Q_3$ .

$$\begin{aligned} Q_3 &= l + \frac{h}{f} \left( \frac{3N}{4} - C \right) \\ &= 60 + \frac{15}{45}(135 - 115) \\ &= 60 + \frac{15}{45}(20) = 60 + \frac{20}{3} = 66.67. \end{aligned}$$

Hence, Bowley's coefficient of skewness based on quartiles is

$$\begin{aligned} Sk(\text{Bowley}) &= \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1} \\ &= \frac{66.67 + 27.50 - 2(49.29)}{66.67 - 27.50} \\ &= \frac{94.17 - 98.58}{39.17} = \frac{-4.41}{39.17} = -0.1126. \end{aligned}$$

$$\boxed{Sk(\text{Bowley}) = -0.1126}$$

**Example 6**

If the first quartile is 142 and the semi-interquartile range is 18, find the median assuming the distribution to be symmetrical.

**Solution**

We are given

$$Q_1 = 142$$

and

$$\frac{Q_3 - Q_1}{2} = 18.$$

Therefore,

$$Q_3 = Q_1 + 2(18)$$

$$= 142 + 36 = 178.$$

If the distribution is symmetrical, then

$$Sk(\text{Bowley}) = \frac{(Q_3 - Md) - (Md - Q_1)}{(Q_3 - Md) + (Md - Q_1)} = 0.$$

Therefore,

$$Q_3 - Md = Md - Q_1.$$

Hence,

$$\begin{aligned} Md &= \frac{1}{2}(Q_1 + Q_3) \\ &= \frac{142 + 178}{2} = 160. \end{aligned}$$

Thus,

$$\boxed{Md = 160}$$

**Example 7**

In a frequency distribution the coefficient of skewness based on quartiles is 0.6. If the sum of the upper and lower quartiles is 100 and median is 38, find the value of upper and lower quartiles.

### Solution

We are given

$$Sk = \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1} = 0.6. \quad (*)$$

Also,

$$Q_3 + Q_1 = 100$$

and

$$\text{Median} = 38. \quad (**)$$

Substituting from (\*\*) in (\*), we get

$$\begin{aligned} \frac{100 - 2 \times 38}{Q_3 - Q_1} &= 0.6 \\ \Rightarrow Q_3 - Q_1 &= \frac{100 - 76}{0.6} = \frac{24}{0.6} = 40. \end{aligned}$$

Thus, we have

$$Q_3 + Q_1 = 100$$

and

$$Q_3 - Q_1 = 40.$$

Adding and subtracting, we get respectively

$$2Q_3 = 140$$

$$\Rightarrow Q_3 = \frac{140}{2} = 70$$

and

$$2Q_1 = 60$$

$$\Rightarrow Q_1 = \frac{60}{2} = 30.$$

Hence, the values of the upper and lower quartiles are 70 and 30 respectively.

$$\boxed{Q_3 = 70, \quad Q_1 = 30}$$

### Example 8

For a distribution the distance of the median from the first quartile is five times the distance of the third quartile from the median. Calculate a measure of skewness for this distribution. Give a rough sketch of the distribution.

### Solution

We are given

$$Md - Q_1 = 5(Q_3 - Md). \quad (*)$$

Bowley's coefficient of skewness is given by

$$Sk(\text{Bowley}) = \frac{(Q_3 - Md) - (Md - Q_1)}{(Q_3 - Md) + (Md - Q_1)}.$$

Using (\*), we get

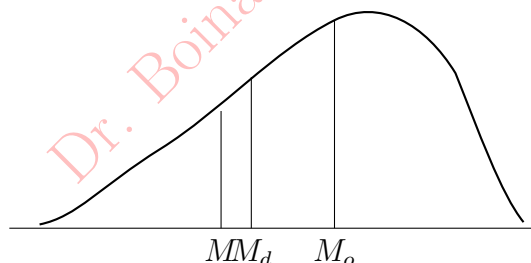
$$\begin{aligned} Sk(\text{Bowley}) &= \frac{(Q_3 - Md) - 5(Q_3 - Md)}{(Q_3 - Md) + 5(Q_3 - Md)} \\ &= \frac{-4(Q_3 - Md)}{6(Q_3 - Md)} \\ &= -\frac{2}{3} = -0.67. \end{aligned}$$

Thus, the measure of skewness is

$$Sk(\text{Bowley}) = -0.67$$

Further, since Bowley's coefficient lies between  $-1$  and  $+1$ , the distribution is fairly negatively skewed.

A rough sketch of the distribution is shown below.



Negatively Skewed Distribution

### Example 9

From the information given below, calculate Karl Pearson's coefficient of skewness and also quartile coefficient of skewness:

| Measure            | Place A | Place B |
|--------------------|---------|---------|
| Mean               | 150     | 140     |
| Median             | 142     | 155     |
| Standard deviation | 30      | 55      |
| Third quartile     | 195     | 260     |
| First quartile     | 62      | 80      |

**Solution**

**Place A:**

$$Sk \text{ (Karl Pearson)} = \frac{3(M - Md)}{\sigma} = \frac{3(150 - 142)}{30} = \frac{8}{10} = 0.8$$

$$\begin{aligned} Sk \text{ (Bowley)} &= \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1} = \frac{195 + 62 - 2 \times 142}{195 - 62} \\ &= \frac{257 - 284}{133} = -\frac{27}{133} = -0.203 \end{aligned}$$

**Place B:**

$$Sk \text{ (Karl Pearson)} = \frac{3(M - Md)}{\sigma} = \frac{3(140 - 155)}{55} = \frac{3 \times (-15)}{55} = -\frac{9}{11} = -0.82$$

$$\begin{aligned} Sk \text{ (Bowley)} &= \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1} = \frac{260 + 80 - 2 \times 155}{260 - 80} \\ &= \frac{340 - 310}{180} = \frac{1}{6} = 0.167 \end{aligned}$$

## Moment Coefficient of Skewness

The moment coefficient of skewness is

$$\gamma_1 = \frac{\mu_3}{\sigma^3}$$

where

$$\mu_3 = \frac{\sum(X - \bar{X})^3}{N}, \quad \sigma = \sqrt{\mu_2}, \quad \mu_2 = \frac{\sum(X - \bar{X})^2}{N}.$$

Thus,

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

$\mu_3 \rightarrow$  Third central Moment       $\mu_2 \rightarrow$  Second central Moment       $\sigma \rightarrow$  Standard Deviation

### Example 1

Find the coefficient of skewness for the data

2, 3, 4, 5, 6.

by using central moments

**Solution:**

The mean is

$$\bar{X} = \frac{2 + 3 + 4 + 5 + 6}{5} = 4.$$

The deviations from the mean are

$$X - \bar{X} = -2, -1, 0, 1, 2.$$

Therefore,

$$\mu_2 = \frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{5} = \frac{4 + 1 + 0 + 1 + 4}{5} = 2.$$

Similarly,

$$\mu_3 = \frac{(-2)^3 + (-1)^3 + 0^3 + 1^3 + 2^3}{5} = \frac{-8 - 1 + 0 + 1 + 8}{5} = 0.$$

Hence,

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{0}{2^{3/2}}$$

$$\boxed{\gamma_1 = 0}$$

Therefore, the distribution is **symmetric**.

**Kelly's Measure of Skewness**

**Kelly's measure of skewness** is a positional measure of skewness based on deciles or percentiles.

The coefficient of Kelly's skewness is given by

$$Sk_K = \frac{D_9 + D_1 - 2D_5}{D_9 - D_1} = \frac{D_9 + D_1 - 2Md}{D_9 - D_1}$$

or

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

**Example 1**

The values of the first, fifth and ninth deciles of a distribution are 20, 35 and 60, respectively. Find Kelly's coefficient of skewness and interpret the result.

**Solution**

Given,

$$D_1 = 20, \quad D_5 = 35, \quad D_9 = 60.$$

Kelly's coefficient of skewness is

$$Sk_K = \frac{D_9 + D_1 - 2D_5}{D_9 - D_1} = \frac{60 + 20 - 2(35)}{60 - 20} = \boxed{0.25}$$

Since  $Sk_K > 0$ , the distribution is

positively skewed.

### Example 2:

For a distribution,

$$P_{10} = 15, \quad P_{50} = 30, \quad P_{90} = 55.$$

Find Kelly's coefficient of skewness.

**Solution:**

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} = \frac{55 + 15 - 2(30)}{55 - 15} = \boxed{0.25}$$

Therefore, the distribution is

positively skewed.

### Example 3

The first, fifth and ninth deciles of a distribution are 20, 40 and 60, respectively. Find Kelly's coefficient of skewness.

**Solution:**

Given,

$$D_1 = 20, \quad D_5 = 40, \quad D_9 = 60.$$

Therefore,

$$Sk_K = \frac{60 + 20 - 2(40)}{60 - 20} = \boxed{0}.$$

Hence, the distribution is

symmetric.

### Example 4

The values of  $D_1$ ,  $D_5$  and  $D_9$  are 10, 30 and 45, respectively. Find Kelly's coefficient of skewness and interpret the result.

**Solution**

Given,

$$D_1 = 10, \quad D_5 = 30, \quad D_9 = 45.$$

Using Kelly's formula,

$$\begin{aligned} Sk_K &= \frac{D_9 + D_1 - 2D_5}{D_9 - D_1} \\ &= \frac{45 + 10 - 2(30)}{45 - 10} \\ &= \frac{55 - 60}{35} \\ &= -\frac{5}{35} \end{aligned}$$

$$Sk_K = -0.143$$

Since  $Sk_K < 0$ , the distribution is

negatively skewed.

**Example 5**

Calculate Kelly's coefficient of skewness for the following observations:

$$10, 12, 15, 18, 20, 24, 28, 35, 42, 50.$$

**Solution**

The observations are already arranged in ascending order.

The formula for the  $p$ th percentile in an individual series is

$$P_p = \left( \frac{p(N+1)}{100} \right)^{\text{th}} \text{ item.}$$

Here,

$$N = 10.$$

For  $P_{10}$ ,

$$P_{10} = \left( \frac{10(10+1)}{100} \right)^{\text{th}} \text{ item} = 1.1^{\text{th}} \text{ item.}$$

Therefore,

$$P_{10} = 10 + 0.1(12 - 10) = 10.2.$$

For  $P_{50}$ ,

$$P_{50} = \left( \frac{50(11)}{100} \right)^{\text{th}} \text{ item} = 5.5^{\text{th}} \text{ item.}$$

Therefore,

$$P_{50} = 20 + 0.5(24 - 20) = 22.$$

For  $P_{90}$ ,

$$P_{90} = \left( \frac{90(11)}{100} \right)^{\text{th}} \text{ item} = 9.9^{\text{th}} \text{ item.}$$

Therefore,

$$P_{90} = 42 + 0.9(50 - 42) = 49.2.$$

Hence,

$$\begin{aligned} Sk_K &= \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} \\ &= \frac{49.2 + 10.2 - 2(22)}{49.2 - 10.2} \\ &= \frac{59.4 - 44}{39} \\ &= \frac{15.4}{39} \end{aligned}$$

$$Sk_K \approx 0.395$$

Since  $Sk_K > 0$ , the distribution is

positively skewed.

### Example 6

Calculate Kelly's coefficient of skewness for the following distribution.

| Value ( $X$ ) | Frequency ( $f$ ) | Cumulative Frequency |
|---------------|-------------------|----------------------|
| 10            | 5                 | 5                    |
| 20            | 8                 | 13                   |
| 30            | 12                | 25                   |
| 40            | 15                | 40                   |
| 50            | 10                | 50                   |
| 60            | 6                 | 56                   |
| 70            | 4                 | 60                   |

Total frequency is

$$N = 60.$$

For a discrete frequency distribution, the position of the  $p$ th percentile is

$$\frac{pN}{100}.$$

Thus,

$$P_{10} = \frac{10(60)}{100} = 6^{\text{th}} \text{ item.}$$

From the cumulative frequency column,

$$P_{10} = 20.$$

Similarly,

$$P_{50} = \frac{50(60)}{100} = 30^{\text{th}} \text{ item.}$$

Therefore,

$$P_{50} = 40.$$

Also,

$$P_{90} = \frac{90(60)}{100} = 54^{\text{th}} \text{ item.}$$

Therefore,

$$P_{90} = 60.$$

Hence,

$$\begin{aligned} Sk_K &= \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} \\ &= \frac{60 + 20 - 2(40)}{60 - 20} \\ &= \frac{80 - 80}{40} \end{aligned}$$

$$\boxed{Sk_K = 0}.$$

Therefore, the distribution is

$$\boxed{\text{symmetric.}}$$

**Example 7**

Calculate Kelly's coefficient of skewness for the following continuous frequency distribution.

| Class Interval | Frequency ( $f$ ) | Cumulative Frequency |
|----------------|-------------------|----------------------|
| 0–10           | 5                 | 5                    |
| 10–20          | 10                | 15                   |
| 20–30          | 20                | 35                   |
| 30–40          | 30                | 65                   |
| 40–50          | 20                | 85                   |
| 50–60          | 10                | 95                   |
| 60–70          | 5                 | 100                  |

Total frequency is

$$N = 100.$$

**Calculation of  $P_{10}$ :**

$$\frac{10N}{100} = 10.$$

The 10th item lies in the class 10 – –20.

Thus,

$$L = 10, \quad C = 5, \quad f = 10, \quad h = 10.$$

Therefore,

$$\begin{aligned} P_{10} &= 10 + \left( \frac{10 - 5}{10} \right) (10) \\ &= 10 + 5 \end{aligned}$$

$$\boxed{P_{10} = 15}.$$

**Calculation of  $P_{50}$ :**

$$\frac{50N}{100} = 50.$$

The 50th item lies in the class 30 – –40.

Thus,

$$L = 30, \quad C = 35, \quad f = 30, \quad h = 10.$$

Therefore,

$$P_{50} = 30 + \left( \frac{50 - 35}{30} \right) (10)$$

$$= 30 + 5$$

$$\boxed{P_{50} = 35}.$$

**Calculation of  $P_{90}$ :**

$$\frac{90N}{100} = 90.$$

The 90th item lies in the class 50 – 60.

Thus,

$$L = 50, \quad C = 85, \quad f = 10, \quad h = 10.$$

Therefore,

$$P_{90} = 50 + \left( \frac{90 - 85}{10} \right) (10)$$

$$= 50 + 5$$

$$\boxed{P_{90} = 55}.$$

Hence,

$$Sk_K = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

$$= \frac{55 + 15 - 2(35)}{55 - 15}$$

$$= \frac{70 - 70}{40}$$

$$\boxed{Sk_K = 0}.$$

Therefore, the distribution is

$$\boxed{\text{symmetric.}}$$

### **Merits of Kelly's Measure of Skewness**

1. It is based on positional measures such as deciles and percentiles.
2. It is not affected significantly by the exact values of the observations.
3. It is useful when extreme observations are present.
4. It can be used for distributions where the mode is not clearly defined.
5. It is particularly useful for open-ended distributions.

### **Limitations of Kelly's Measure of Skewness**

1. It does not use all the observations in the distribution.
2. The calculation of percentiles or deciles may be lengthy for large frequency distributions.
3. It is less commonly used than Karl Pearson's measure of skewness.
4. The value of the coefficient depends on the positional measures used.

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# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Skewness and Kurtosis

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

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## Moments

Moments are numerical measures used to describe the important characteristics of a frequency distribution. They are useful in studying the **central tendency, dispersion, skewness, and kurtosis** of a distribution.

There are two important types of moments:

1. Moments about the mean (Central Moments)
2. Moments about an arbitrary point  $A$  (Raw Moments)

### Moments about the Mean

The moments calculated by taking deviations of the observations from their arithmetic mean are called **moments about the mean** or **central moments**.

Let the observations be

$$x_1, x_2, \dots, x_N$$

and let their arithmetic mean be  $\bar{x}$ .

The deviation of an observation  $x$  from the mean is

$$d = x - \bar{x}.$$

The  $r$ th moment about the mean is defined as

$$\mu_r = \frac{1}{N} \sum (x - \bar{x})^r$$

For a frequency distribution,

$$\mu_r = \frac{\sum f(x - \bar{x})^r}{\sum f}$$

where

$$N = \sum f.$$

### Zeroth Central Moment

$$\mu_0 = 1$$

### First Central Moment

$$\mu_1 = \frac{\sum f(x - \bar{x})}{N} = 0.$$

### Second Central Moment

$$\mu_2 = \frac{\sum f(x - \bar{x})^2}{N}$$

The second central moment is the **variance**.

Hence,

$$\sigma^2 = \mu_2 \quad \text{and} \quad \sigma = \sqrt{\mu_2}.$$

### Third Central Moment

$$\mu_3 = \frac{\sum f(x - \bar{x})^3}{N}$$

The third central moment is used to study the **skewness** of a distribution.

### Fourth Central Moment

$$\mu_4 = \frac{\sum f(x - \bar{x})^4}{N}$$

The fourth central moment is used to study the **kurtosis** of a distribution.

### Moments about an Arbitrary Point $A$

Moments calculated by taking deviations of the observations from an arbitrary constant  $A$  are called **moments about the arbitrary point  $A$** .

The  $r$ th moment about the arbitrary point  $A$  is

$$\mu'_r(A) = \frac{1}{N} \sum (x - A)^r$$

For a frequency distribution,

$$\mu'_r(A) = \frac{\sum f(x - A)^r}{\sum f}$$

where

$$N = \sum f.$$

### Zeroth Moment

$$\mu'_0(A) = 1$$

### First Moment

$$\mu'_1(A) = \frac{\sum f(x - A)}{N}$$

### Second Moment

$$\mu'_2(A) = \frac{\sum f(x - A)^2}{N}$$

### Third Moment

$$\mu'_3(A) = \frac{\sum f(x - A)^3}{N}$$

### Fourth Moment

$$\mu'_4(A) = \frac{\sum f(x - A)^4}{N}$$

### Relation between Moments about the Mean and Moments about an Arbitrary Point

Let

$$\mu_r = r\text{th central moment}$$

and

$$\mu'_r = r\text{th moment about the arbitrary point } A.$$

Then the relations between the moments are:

$$\mu'_1 = \bar{x} - A \quad \text{and} \quad \bar{x} = \mu'_1 + A$$

$$\mu_2 = \mu'_2 - \mu_1'^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4$$

### Moment coefficient of Skewness

The coefficient of skewness is

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}.$$

The moment coefficient of skewness is

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}.$$

That is,

$$\gamma_1 = \sqrt{\beta_1} \quad \text{or} \quad \beta_1 = \gamma_1^2.$$

### Examples

#### Example 1:

Calculate the first four moments about the mean for the data

$$2, 4, 6, 8, 10.$$

**Solution:**

The arithmetic mean is

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6.$$

Construct the following table:

| $x$          | $x - \bar{x}$ | $(x - \bar{x})^2$ | $(x - \bar{x})^3$ | $(x - \bar{x})^4$ |
|--------------|---------------|-------------------|-------------------|-------------------|
| 2            | -4            | 16                | -64               | 256               |
| 4            | -2            | 4                 | -8                | 16                |
| 6            | 0             | 0                 | 0                 | 0                 |
| 8            | 2             | 4                 | 8                 | 16                |
| 10           | 4             | 16                | 64                | 256               |
| <b>Total</b> | 0             | 40                | 0                 | 544               |

Therefore,

$$\mu_1 = \frac{\sum(x - \bar{x})}{N} = \frac{0}{5} = 0.$$

$$\mu_2 = \frac{\sum(x - \bar{x})^2}{N} = \frac{40}{5} = 8.$$

$$\mu_3 = \frac{\sum(x - \bar{x})^3}{N} = \frac{0}{5} = 0.$$

$$\mu_4 = \frac{\sum(x - \bar{x})^4}{N} = \frac{544}{5} = 108.8.$$

Hence, the first four central moments are

|              |              |              |                 |
|--------------|--------------|--------------|-----------------|
| $\mu_1 = 0,$ | $\mu_2 = 8,$ | $\mu_3 = 0,$ | $\mu_4 = 108.8$ |
|--------------|--------------|--------------|-----------------|

**Example 2:**

Find the first four moments about the arbitrary point  $A = 4$  for the data

$$2, 4, 6, 8, 10.$$

**Solution:**

Here,

$$A = 4.$$

| $x$          | $x - 4$ | $(x - 4)^2$ | $(x - 4)^3$ | $(x - 4)^4$ |
|--------------|---------|-------------|-------------|-------------|
| 2            | -2      | 4           | -8          | 16          |
| 4            | 0       | 0           | 0           | 0           |
| 6            | 2       | 4           | 8           | 16          |
| 8            | 4       | 16          | 64          | 256         |
| 10           | 6       | 36          | 216         | 1296        |
| <b>Total</b> | 10      | 60          | 280         | 1584        |

Thus,

$$\mu'_1(A) = \frac{10}{5} = 2.$$

$$\mu'_2(A) = \frac{60}{5} = 12.$$

$$\mu'_3(A) = \frac{280}{5} = 56.$$

$$\mu'_4(A) = \frac{1584}{5} = 316.8.$$

Hence,

$$\mu'_1 = 2, \quad \mu'_2 = 12, \quad \mu'_3 = 56, \quad \mu'_4 = 316.8$$

### Example 3:

The first four moments about the point  $A = 2$  are

$$\mu'_1 = 3, \quad \mu'_2 = 11, \quad \mu'_3 = 45, \quad \mu'_4 = 233.$$

Find the first four moments about the mean.

### Solution:

We know the relations are

- $\bar{x} = \mu'_1 + A$
- $\mu_2 = \mu'_2 - \mu_1'^2$
- $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$
- $\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4$

Then

$$\bar{x} = \mu'_1 + A = 3 + 2 = 5$$

and the central moments are

$$\mu_1 = 0, \quad \mu_2 = 2, \quad \mu_3 = 0, \quad \mu_4 = 44.$$

#### Example 4

Calculate the first four central moments about the mean for the following frequency distribution.

|     |   |   |    |    |    |    |   |   |
|-----|---|---|----|----|----|----|---|---|
| $x$ | 0 | 1 | 2  | 3  | 4  | 5  | 6 | 7 |
| $f$ | 2 | 5 | 10 | 16 | 20 | 15 | 8 | 4 |

#### Solution

The mean is

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{304}{80} = 3.8.$$

The direct calculation table is:

| $x$    | $f$ | $d = x - 3.8$ | $d^2$ | $d^3$   | $d^4$    | $fd$  | $fd^2$ | $fd^3$   | $fd^4$   |
|--------|-----|---------------|-------|---------|----------|-------|--------|----------|----------|
| 0      | 2   | -3.8          | 14.44 | -54.872 | 208.5136 | -7.6  | 28.88  | -109.744 | 417.0272 |
| 1      | 5   | -2.8          | 7.84  | -21.952 | 61.4656  | -14.0 | 39.20  | -109.760 | 307.3280 |
| 2      | 10  | -1.8          | 3.24  | -5.832  | 10.4976  | -18.0 | 32.40  | -58.320  | 104.9760 |
| 3      | 16  | -0.8          | 0.64  | -0.512  | 0.4096   | -12.8 | 10.24  | -8.192   | 6.5536   |
| 4      | 20  | 0.2           | 0.04  | 0.008   | 0.0016   | 4.0   | 0.80   | 0.160    | 0.0320   |
| 5      | 15  | 1.2           | 1.44  | 1.728   | 2.0736   | 18.0  | 21.60  | 25.920   | 31.1040  |
| 6      | 8   | 2.2           | 4.84  | 10.648  | 23.4256  | 17.6  | 38.72  | 85.184   | 187.4048 |
| 7      | 4   | 3.2           | 10.24 | 32.768  | 104.8576 | 12.8  | 40.96  | 131.072  | 419.4304 |
| $\sum$ | 80  |               |       |         |          | 0     | 212.8  | -43.68   | 1473.856 |

The central moments are given by

$$\mu_r = \frac{\sum f(x - \bar{x})^r}{N}.$$

Therefore,

$$\mu_1 = \frac{\sum f(x - \bar{x})}{N} = \frac{0}{80} = \boxed{0}.$$

The second central moment is

$$\begin{aligned}\mu_2 &= \frac{\sum f(x - \bar{x})^2}{N} \\ &= \frac{212.80}{80} \\ &= \boxed{2.66}.\end{aligned}$$

The third central moment is

$$\begin{aligned}\mu_3 &= \frac{\sum f(x - \bar{x})^3}{N} \\ &= \frac{-43.68}{80} \\ &= \boxed{-0.546}.\end{aligned}$$

The fourth central moment is

$$\begin{aligned}\mu_4 &= \frac{\sum f(x - \bar{x})^4}{N} \\ &= \frac{1473.856}{80} \\ &= \boxed{18.4232}.\end{aligned}$$

Hence, the first four central moments about the mean are

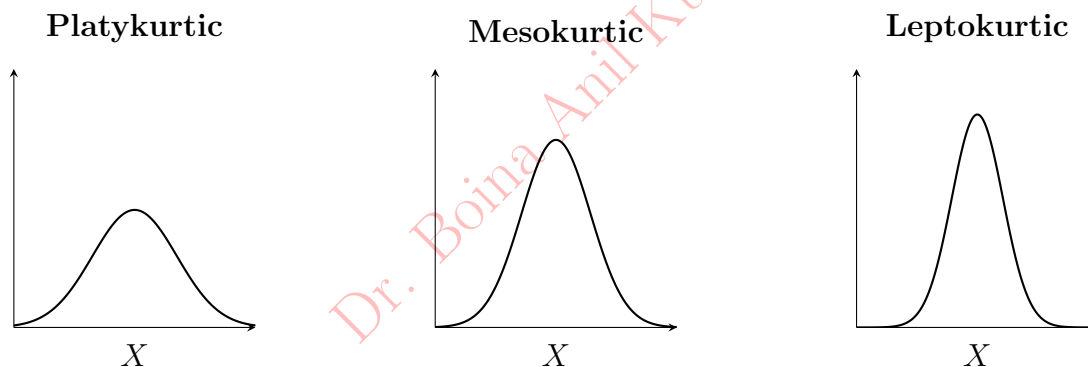
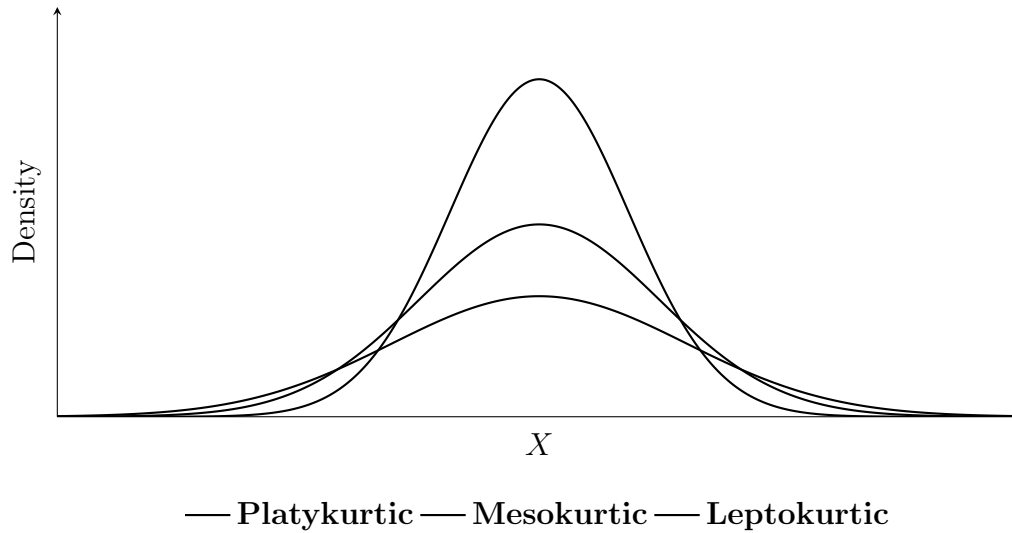
$$\boxed{\mu_1 = 0, \quad \mu_2 = 2.66, \quad \mu_3 = -0.5335, \quad \mu_4 = 18.4232}$$

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## Kurtosis

Kurtosis is a measure of the **degree of peakedness or flatness** of a frequency distribution.

It describes the shape of a distribution in comparison with the normal distribution.



## Measures of Kurtosis

As a measure of Kurtosis, Karl Pearson gave the following coefficients to classify the distributions.

The coefficient of kurtosis is

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

where

$$\mu_2 = \frac{\sum(X - \bar{X})^2}{N} \quad \text{and} \quad \mu_4 = \frac{\sum(X - \bar{X})^4}{N}.$$

The excess kurtosis is defined as

$$\gamma_2 = \beta_2 - 3 \quad \text{or} \quad \gamma_2 = \frac{\mu_4}{\mu_2^2} - 3$$

## Types of Kurtosis

The distribution is of type

1. Mesokurtic, if

$$\boxed{\beta_2 = 3} \quad \text{and} \quad \boxed{\gamma_2 = 0}.$$

2. Leptokurtic, if

$$\boxed{\beta_2 > 3} \quad \text{and} \quad \boxed{\gamma_2 > 0}.$$

3. Platykurtic, if

$$\boxed{\beta_2 < 3} \quad \text{and} \quad \boxed{\gamma_2 < 0}.$$

### Example 1

Consider the following frequency distribution

|     |   |   |    |    |    |    |   |   |   |
|-----|---|---|----|----|----|----|---|---|---|
| $x$ | 1 | 2 | 3  | 4  | 5  | 6  | 7 | 8 | 9 |
| $f$ | 1 | 6 | 13 | 25 | 30 | 22 | 9 | 5 | 2 |

- Find the first four central moments about  $A=5$
- Using the results in (a), find the first four central moments on mean.
- Comment on the nature of the distribution based on Skewness and Kurtosis

### Solution

| $x$ | $f$      | $x - 5$        | $f(x - 5)$      | $f(x - 5)^2$      | $f(x - 5)^3$      | $f(x - 5)^4$      |
|-----|----------|----------------|-----------------|-------------------|-------------------|-------------------|
| 1   | 1        | -4             | -4              | 16                | -64               | 256               |
| 2   | 6        | -3             | -18             | 54                | -162              | 486               |
| 3   | 13       | -2             | -26             | 52                | -104              | 208               |
| 4   | 25       | -1             | -25             | 25                | -25               | 25                |
| 5   | 30       | 0              | 0               | 0                 | 0                 | 0                 |
| 6   | 22       | 1              | 22              | 22                | 22                | 22                |
| 7   | 9        | 2              | 18              | 36                | 72                | 144               |
| 8   | 5        | 3              | 15              | 45                | 135               | 405               |
| 9   | 2        | 4              | 8               | 32                | 128               | 512               |
|     | $\sum f$ | $\sum (x - 5)$ | $\sum f(x - 5)$ | $\sum f(x - 5)^2$ | $\sum f(x - 5)^3$ | $\sum f(x - 5)^4$ |
|     | =113     | =0             | =-10            | =282              | =2                | =2058             |

## Solution

### Moments About the Point $x = 5$

$$\mu'_1 = \frac{\sum f(x-5)}{N} = \frac{-10}{113} = -0.0885,$$

$$\mu'_2 = \frac{\sum f(x-5)^2}{N} = \frac{282}{113} = 2.4956,$$

$$\mu'_3 = \frac{\sum f(x-5)^3}{N} = \frac{2}{113} = 0.0177,$$

and

$$\mu'_4 = \frac{\sum f(x-5)^4}{N} = \frac{2058}{113} = 18.2124.$$

The mean is

$$\bar{X} = A + \mu'_1 = 5 + (-0.0885) = 4.9115.$$

### Moments About Mean

The first central moment is

$$\mu_1 = 0.$$

The second central moment is

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

The third central moment is

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = 0.6789.$$

The fourth central moment is

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 = 18.3357.$$

### Moment Coefficients of Skewness

The coefficient  $\beta_1$  is

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.0299.$$

The coefficient  $\gamma_1$  is

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{\mu_3}{\mu_2\sqrt{\mu_2}} = 0.173.$$

## Kurtosis

The coefficient of kurtosis is

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \boxed{2.9626}.$$

The excess coefficient of kurtosis is

$$\gamma_2 = \beta_2 - 3 = \boxed{-0.0373}.$$

## Interpretation

Since

$$\gamma_1 = 0.173,$$

the given distribution is **very slightly positively skewed**.

Since

$$\beta_2 = 2.9626 \approx 3,$$

the distribution is more or less normal or **mesokurtic**.

However, since

$$\gamma_2 = -0.0373 < 0,$$

the frequency curve is **slightly platykurtic** and is slightly flatter than the normal curve.

## Example 2

Calculate the first four central moments about the mean for the following frequency distribution. Also, comment on the nature of the distribution.

|     |   |   |    |    |    |    |   |   |
|-----|---|---|----|----|----|----|---|---|
| $x$ | 0 | 1 | 2  | 3  | 4  | 5  | 6 | 7 |
| $f$ | 2 | 5 | 10 | 16 | 20 | 15 | 8 | 4 |

## Solution

The mean is

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{304}{80} = 3.8.$$

The direct calculation table is:

| $x$      | $f$ | $d = x - 3.8$ | $d^2$ | $d^3$   | $d^4$    | $fd$  | $fd^2$ | $fd^3$   | $fd^4$   |
|----------|-----|---------------|-------|---------|----------|-------|--------|----------|----------|
| 0        | 2   | -3.8          | 14.44 | -54.872 | 208.5136 | -7.6  | 28.88  | -109.744 | 417.0272 |
| 1        | 5   | -2.8          | 7.84  | -21.952 | 61.4656  | -14.0 | 39.20  | -109.760 | 307.3280 |
| 2        | 10  | -1.8          | 3.24  | -5.832  | 10.4976  | -18.0 | 32.40  | -58.320  | 104.9760 |
| 3        | 16  | -0.8          | 0.64  | -0.512  | 0.4096   | -12.8 | 10.24  | -8.192   | 6.5536   |
| 4        | 20  | 0.2           | 0.04  | 0.008   | 0.0016   | 4.0   | 0.80   | 0.160    | 0.0320   |
| 5        | 15  | 1.2           | 1.44  | 1.728   | 2.0736   | 18.0  | 21.60  | 25.920   | 31.1040  |
| 6        | 8   | 2.2           | 4.84  | 10.648  | 23.4256  | 17.6  | 38.72  | 85.184   | 187.4048 |
| 7        | 4   | 3.2           | 10.24 | 32.768  | 104.8576 | 12.8  | 40.96  | 131.072  | 419.4304 |
| $\Sigma$ | 80  |               |       |         |          | 0     | 212.8  | -43.68   | 1473.856 |

Hence, the first four central moments about the mean are

$$\mu_1 = 0, \quad \mu_2 = 2.66, \quad \mu_3 = -0.546, \quad \mu_4 = 18.4232$$

### Moment Coefficients of Skewness

The coefficient  $\beta_1$  is

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(-0.546)^2}{2.66^3} = \boxed{0.015839}.$$

The coefficient  $\gamma_1$  is

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{(-0.546)}{2.66^{3/2}} = \boxed{-0.12585}.$$

### Kurtosis

The coefficient of kurtosis is

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{18.4232}{2.66^2} = \boxed{2.6037}.$$

The excess coefficient of kurtosis is

$$\gamma_2 = \beta_2 - 3 = \boxed{-0.3963}.$$

### Interpretation

Since

$$\gamma_1 = -0.12585,$$

the given distribution is **very slightly negatively skewed**.

Since

$$\beta_2 = 2.602 \quad \text{and} \quad \gamma_2 = -0.3963 < 0,$$

the distribution is **platykurtic**.

### Example 3

From the following data, calculate moments about assumed mean 25 and convert them into central moments.

| Class        | Mid-value ( $X$ ) | $f$               | $d = \frac{x - 25}{10}$ | $fd$           | $fd^2$          | $fd^3$           |
|--------------|-------------------|-------------------|-------------------------|----------------|-----------------|------------------|
| 0 – 10       | 5                 | 1                 | –2                      | –2             | 4               | –8               |
| 10 – 20      | 15                | 3                 | –1                      | –3             | 3               | –3               |
| 20 – 30      | 25                | 4                 | 0                       | 0              | 0               | 0                |
| 30 – 40      | 35                | 2                 | 1                       | 2              | 2               | 2                |
| <b>Total</b> |                   | $N = \sum f = 10$ |                         | $\sum fd = -3$ | $\sum fd^2 = 9$ | $\sum fd^3 = -9$ |

Also,

$$\sum fd^4 = 21.$$

### Moments About the Point $x = 25$

Here,

$$d = \frac{x - 25}{10}, \quad h = 10, \quad N = \sum f = 10.$$

The first moment about the point  $x = 25$  is

$$\begin{aligned}\mu'_1 &= h \frac{\sum fd}{N} \\ &= \frac{10(-3)}{10} \\ &= \boxed{-3}.\end{aligned}$$

The second moment is

$$\begin{aligned}\mu'_2 &= h^2 \frac{\sum fd^2}{N} \\ &= \frac{10^2(9)}{10} \\ &= \boxed{90}.\end{aligned}$$

The third moment is

$$\begin{aligned}\mu'_3 &= h^3 \frac{\sum fd^3}{N} \\ &= \frac{10^3(-9)}{10} \\ &= \boxed{-900}.\end{aligned}$$

The fourth moment is

$$\begin{aligned}\mu'_4 &= h^4 \frac{\sum fd^4}{N} \\ &= \frac{10^4(21)}{10} \\ &= \boxed{21000}.\end{aligned}$$

### Central Moments (Moments About Mean)

The first central moment is

$$\boxed{\mu_1 = 0}.$$

The second central moment is

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= 90 - (-3)^2 \\ &= 90 - 9 \\ &= \boxed{81}.\end{aligned}$$

The third central moment is

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= -900 - 3(90)(-3) + 2(-3)^3 \\ &= -900 + 810 - 54 \\ &= \boxed{-144}.\end{aligned}$$

The fourth central moment is

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 21000 - 4(-900)(-3) + 6(90)(-3)^2 - 3(-3)^4 \\ &= 21000 - 10800 + 4860 - 243 \\ &= \boxed{14817}.\end{aligned}$$

### Example 4

The first three moments of a distribution about the value 67 of the variable are 0.45, 8.73 and 8.91. Calculate the second and third central moments, and the moment coefficient of skewness. Indicate the nature of the distribution.

### Solution

In the usual notations, we are given

$$A = 67, \quad \mu'_1 = 0.45, \quad \mu'_2 = 8.73, \quad \mu'_3 = 8.91.$$

The second and third central moments are given by

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= 8.73 - (0.45)^2 \\ &= 8.73 - 0.2025 \\ &= \boxed{8.5275}.\end{aligned}$$

Also,

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= 8.91 - 3(8.73)(0.45) + 2(0.45)^3 \\ &= 8.91 - 11.7855 + 0.18225 \\ &= \boxed{-2.6933}.\end{aligned}$$

Karl Pearson's moment coefficient of skewness is

$$\begin{aligned}
 \gamma_1 &= \frac{\mu_3}{\mu_2^{3/2}} \\
 &= \frac{\mu_3}{\mu_2 \sqrt{\mu_2}} \\
 &= \frac{-2.6933}{8.5275 \times 2.9202} \\
 &= \frac{-2.6933}{24.9020} \\
 &= \boxed{-0.1082}.
 \end{aligned}$$

Since  $\gamma_1$  is negative, the given distribution is **negatively skewed**. In other words, the frequency curve has a longer tail towards the left.

Also,

$$\begin{aligned}
 \beta_1 &= \frac{\mu_3^2}{\mu_2^3} \\
 &= (-0.1082)^2 \\
 &= \boxed{0.0117}.
 \end{aligned}$$

Since  $\gamma_1$  and  $\beta_1$  are approximately zero, the given distribution is approximately symmetrical.

### Example 5:

If the first four central moments of a distribution are

$$\mu_1 = 0, \quad \mu_2 = 16, \quad \mu_3 = 32, \quad \mu_4 = 768,$$

find the coefficients of skewness and kurtosis.

### Solution:

The coefficient of skewness is

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \implies \beta_1 = \frac{(32)^2}{(16)^3} = \frac{1024}{4096} = 0.25.$$

Therefore,

$$\boxed{\beta_1 = 0.25}.$$

The coefficient of kurtosis is

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \implies \beta_2 = \frac{768}{(16)^2} = \frac{768}{256} = 3.$$

Therefore,

$$\boxed{\beta_2 = 3}.$$

Since

$$\beta_2 = 3,$$

the distribution is

$$\boxed{\text{Mesokurtic}}.$$

### Example 6

The first four moments of a distribution about the origin are 1, 4, 10 and 46, respectively. Obtain the various characteristics of the distribution on the basis of the information.

### Solution

We are given the first four moments about the origin.

$$A = 0, \quad \mu'_1 = 1, \quad \mu'_2 = 4, \quad \mu'_3 = 10, \quad \mu'_4 = 46.$$

The measure of central tendency is given by

$$\text{Mean } (\bar{x}) = \text{First moment about origin} = \mu'_1 = 1.$$

The measure of dispersion is given by

$$\text{Variance } (\sigma^2) = \mu_2 = \mu'_2 - (\mu'_1)^2 = 4 - 1 = 3.$$

Hence,

$$\text{s.d. } (\sigma) = \sqrt{3} = 1.732.$$

The third central moment is

$$\begin{aligned} \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= 10 - 3(4)(1) + 2(1)^3 \\ &= 10 - 12 + 2 \\ &= 0. \end{aligned}$$

Karl Pearson's moment coefficient of skewness is given by

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{0}{3^{3/2}} = 0.$$

Therefore,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.$$

Since  $\gamma_1 = 0$ , the given distribution is symmetrical, i.e.,

$$\text{Mean} = \text{Median} = \text{Mode}.$$

Moreover, the quartiles are equidistant from the median, i.e.,

$$Q_3 - \text{Median} = \text{Median} - Q_1.$$

The fourth central moment is

$$\begin{aligned} \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 46 - 4(10)(1) + 6(4)(1)^2 - 3(1)^4 \\ &= 46 - 40 + 24 - 3 \\ &= 27. \end{aligned}$$

Hence, Karl Pearson's measure of kurtosis is

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{27}{3^2} = 3.$$

Also,

$$\gamma_2 = \beta_2 - 3 = 3 - 3 = 0.$$

Since  $\beta_2 = 3$ , the given distribution is **Normal (Meso-kurtic)**.

Thus, the various characteristics of the distribution are

$$\boxed{\text{Mean} = 1}$$

$$\boxed{\text{Variance} = 3, \quad \text{s.d.} = \sqrt{3} = 1.732}$$

$$\boxed{\beta_1 = 0, \quad \beta_2 = 3}$$

and the distribution is

$$\boxed{\text{Symmetrical and Meso-kurtic (Normal).}}$$

### Example 7

The following measures were computed from a problem in Statistics:

$$\text{Mode} = 83, \quad \text{Arithmetic Mean} = 74, \quad \text{Median} = 77.$$

Sketch a curve indicating the general shape of the distribution and comment. Is it possible for this distribution to have a positive third central moment? Explain.

### Solution

We are given:

$$\text{Mode} = 83, \quad \text{A.M.} = 74, \quad \text{Median} = 77.$$

Karl Pearson's coefficient of skewness is given by

$$S_k = \frac{M - M_o}{\sigma} = \frac{3(M - M_d)}{\sigma}.$$

Since

$$M < M_d \quad \text{and} \quad M < M_o,$$

the distribution is negatively skewed, i.e.,

$$S_k < 0.$$

Hence, the given distribution is negatively skewed.

A rough sketch of the frequency curve is shown below:

Negatively skewed distribution

$$M = 74, \quad M_d = 77, \quad M_o = 83$$

The moment coefficient of skewness is given by

$$\gamma_1 = \frac{\mu_3}{\frac{\mu_2^{3/2}}{\sigma^3}} = \frac{\mu_3}{\sigma^3}, \quad \sigma > 0.$$

Since the given distribution is negatively skewed,

$$\gamma_1 < 0 \quad \Rightarrow \quad \mu_3 < 0.$$

Hence, it is **not possible** for the distribution to have a positive third central moment.

### Example 8

If  $\beta_1 = +1$ ,  $\beta_2 = 4$  and variance = 9, find the values of  $\mu_3$  and  $\mu_4$  and comment on the nature of the distribution.

### Solution

We are given

$$\beta_1 = +1, \quad \beta_2 = 4$$

and

$$\text{Variance} = \mu_2 = 9.$$

Now,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 1.$$

Therefore,

$$\frac{\mu_3^2}{9^3} = 1,$$

so that

$$\mu_3^2 = 9 \times 9 \times 9 = (3 \times 3 \times 3)^2 = (27)^2.$$

Hence,

$$\mu_3 = \pm 27.$$

Also,

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 4.$$

Therefore,

$$\mu_4 = 4 \times 9^2 = 4 \times 9 \times 9 = 324.$$

Thus,

$$\boxed{\mu_3 = \pm 27, \quad \mu_4 = 324.}$$

**Nature of the Distribution.** Since

$$\beta_1 \neq 0$$

but

$$\beta_1 = 1,$$

the distribution is **moderately skewed**. Further, since  $\mu_3 = \pm 27$  can be positive or negative, we cannot tell the direction of the skewness.

Also,

$$\beta_2 = 4 > 3.$$

Hence, the given distribution is **lepto-kurtic**, i.e., more peaked than the normal curve.

### Example 9

For a meso-kurtic distribution, the first moment about 7 is 23 and the second moment about origin is 1000. Find the coefficient of variation and the fourth moment about the mean.

### Solution

Since the distribution is given to be meso-kurtic, we have

$$\beta_2 = 3.$$

Therefore,

$$\frac{\mu_4}{\mu_2^2} = 3,$$

and hence

$$\mu_4 = 3\mu_2^2.$$

First moment about 7 is 23, i.e.,

$$\mu'_1 \text{ (about 7)} = 23 \quad (\text{Given}).$$

Therefore,

$$\text{Mean} = 7 + \mu'_1 = 7 + 23 = 30.$$

But the mean is the first moment about origin.

Therefore,

$$\mu'_1 \text{ (about origin)} = 30.$$

**Moments About Origin** We have

$$\mu'_1 = \text{Mean} = 30, \quad \mu'_2 = 1000 \quad (\text{Given}).$$

The second central moment is

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= 1000 - 30^2 \\ &= 1000 - 900 \\ &= 100.\end{aligned}$$

Therefore,

$$\text{Variance } (\sigma^2) = 100$$

and

$$\text{s.d. } (\sigma) = 10.$$

The coefficient of variation is

$$\begin{aligned}\text{C.V.} &= \frac{100 \times \text{s.d.}}{\text{Mean}} \\ &= \frac{100 \times 10}{30} \\ &= 33.33.\end{aligned}$$

Thus,

$$\boxed{\text{C.V.} = 33.33\%}.$$

Substituting the value of  $\mu_2 = 100$  in

$$\mu_4 = 3\mu_2^2,$$

we get

$$\mu_4 = 3 \times 100^2 = 30,000.$$

Hence,

$$\boxed{\mu_4 = 30,000}.$$

### Example 10

(a) You are given the following information:

| Class    | $f$ | Class    | $f$ | Class    | $f$ |
|----------|-----|----------|-----|----------|-----|
| 40 – –41 | 1   | 45 – –46 | 20  | 50 – –51 | 19  |
| 41 – –42 | 2   | 46 – –47 | 38  | 51 – –52 | 14  |
| 42 – –43 | 4   | 47 – –48 | 52  | 52 – –53 | 6   |
| 43 – –44 | 7   | 48 – –49 | 40  | 53 – –54 | 4   |
| 44 – –45 | 12  | 49 – –50 | 29  | 54 – –55 | 2   |

Calculate the first four moments about 47.5. Convert these into moments about the mean and calculate  $\beta_1$  and  $\beta_2$ .

(b) Also apply Sheppard's corrections to moments.

### Solution

For the calculation of moments, take

$$A = 47.5, \quad d = X - 47.5.$$

Since the class interval is 1, we have  $h = 1$ .

The calculation table is:

| Class   | $X$  | $f$       | $d = X - 47.5$ | $fd$ | $fd^2$ | $fd^3$ | $fd^4$ |
|---------|------|-----------|----------------|------|--------|--------|--------|
| 40 – 41 | 40.5 | 1         | –7             | –7   | 49     | –343   | 2391   |
| 41 – 42 | 41.5 | 2         | –6             | –12  | 72     | –432   | 2592   |
| 42 – 43 | 42.5 | 4         | –5             | –20  | 100    | –500   | 2500   |
| 43 – 44 | 43.5 | 7         | –4             | –28  | 112    | –448   | 1792   |
| 44 – 45 | 44.5 | 12        | –3             | –36  | 108    | –324   | 972    |
| 45 – 46 | 45.5 | 20        | –2             | –40  | 80     | –160   | 320    |
| 46 – 47 | 46.5 | 38        | –1             | –38  | 38     | –38    | 38     |
| 47 – 48 | 47.5 | 52        | 0              | 0    | 0      | 0      | 0      |
| 48 – 49 | 48.5 | 40        | 1              | 40   | 40     | 40     | 40     |
| 49 – 50 | 49.5 | 29        | 2              | 58   | 116    | 232    | 464    |
| 50 – 51 | 50.5 | 19        | 3              | 57   | 171    | 513    | 1539   |
| 51 – 52 | 51.5 | 14        | 4              | 56   | 224    | 896    | 3584   |
| 52 – 53 | 52.5 | 6         | 5              | 30   | 150    | 750    | 3750   |
| 53 – 54 | 53.5 | 4         | 6              | 24   | 144    | 864    | 5184   |
| 54 – 55 | 54.5 | 2         | 7              | 14   | 98     | 986    | 4802   |
| Total   |      | $N = 250$ |                | 98   | 1502   | 1736   | 29968  |

**First four moments about  $A = 47.5$ .** Since  $h = 1$ , we have

$$\mu'_1 = \frac{1}{N} \sum fd = \frac{98}{250} = 0.392,$$

$$\mu'_2 = \frac{1}{N} \sum fd^2 = \frac{1502}{250} = 6.008,$$

$$\mu'_3 = \frac{1}{N} \sum fd^3 = \frac{1736}{250} = 6.944,$$

and

$$\mu'_4 = \frac{1}{N} \sum fd^4 = \frac{29968}{250} = 119.872.$$

**First four moments about the mean.** The mean is

$$\bar{x} = 47.5 + \mu'_1 = 47.5 + 0.392 = 47.892.$$

The first central moment is

$$\mu_1 = 0.$$

The second central moment is

$$\begin{aligned}\mu_2 &= \mu'_2 - (\mu'_1)^2 \\ &= 6.008 - (0.392)^2 \\ &= 6.008 - 0.1537 \\ &= 5.854.\end{aligned}$$

The third central moment is

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ &= 6.944 - 3(6.008)(0.392) + 2(0.392)^3 \\ &= 6.944 - 7.0654 + 0.1204 \\ &\approx -0.001 \approx 0.\end{aligned}$$

The fourth central moment is

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 119.872 - 4(6.944)(0.392) + 6(6.008)(0.392)^2 - 3(0.392)^4 \\ &= 119.872 - 10.888 + 5.539 - 0.071 \\ &= 114.452.\end{aligned}$$

Hence,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \approx 0 \quad (\because \mu_3 = 0),$$

and

$$\begin{aligned}\beta_2 &= \frac{\mu_4}{\mu_2^2} \\ &= \frac{114.452}{(5.854)^2} \\ &= \frac{114.452}{34.269} \\ &= 3.34.\end{aligned}$$

**(b) Sheppard's Corrections** Using Sheppard's correction, we have

$$\mu_2(\text{Corrected}) = \mu_2 - \frac{h^2}{12}.$$

Since  $h = 1$ ,

$$\begin{aligned}
\mu_2(\text{Corrected}) &= 5.854 - \frac{1}{12} \\
&= 5.854 - 0.083 \\
&= 5.771.
\end{aligned}$$

For the third moment,

$$\mu_3(\text{Corrected}) \approx \mu_3 \approx 0.$$

For the fourth moment,

$$\mu_4(\text{Corrected}) = \mu_4 - \frac{1}{2}h^2\mu_2 + \frac{7}{240}h^4.$$

Since  $h = 1$ ,

$$\begin{aligned}
\mu_4(\text{Corrected}) &= 114.452 - \frac{1}{2}(5.854) + \frac{7}{240} \\
&= 114.452 - 2.927 + 0.029 \\
&= 111.554.
\end{aligned}$$

Thus,

|                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| $\mu_2(\text{Corrected}) = 5.771, \quad \mu_3(\text{Corrected}) \approx 0, \quad \mu_4(\text{Corrected}) = 111.554.$ |
|----------------------------------------------------------------------------------------------------------------------|

### Example 11

The first four moments of a distribution about the value 4 of the variable are  $-1.5$ ,  $17$ ,  $-30$  and  $108$ . Find the moments about mean,  $\beta_1$ , and  $\beta_2$ .

Find also the moments about

- (i) the origin, and
- (ii) the point  $x = 2$ .

### Solution

In the usual notations, we are given

$$A = 4$$

and

$$\mu'_1 = -1.5, \quad \mu'_2 = 17, \quad \mu'_3 = -30, \quad \mu'_4 = 108.$$

### Moments about Mean:

$$\mu_2 = \mu'_2 - \mu_1'^2 = 17 - (-1.5)^2 = 17 - 2.25 = 14.75$$

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 \\ &= -30 - 3(17)(-1.5) + 2(-1.5)^3 \\ &= -30 + 76.5 - 6.75 \\ &= 39.75\end{aligned}$$

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4 \\ &= 108 - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4 \\ &= 108 - 180 + 229.5 - 15.1875 \\ &= 142.3125\end{aligned}$$

Hence,

$$\begin{aligned}\beta_1 &= \frac{\mu_3^2}{\mu_2^3} = \frac{(39.75)^2}{(14.75)^3} = \frac{1580.06}{3209.05} = 0.4924 \\ \beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{142.3125}{(14.75)^2} = \frac{142.3125}{217.5625} = 0.7541\end{aligned}$$

Also,

$$\text{Mean } (\bar{x}) = A + \mu'_1 = 4 + (-1.5) = 2.5$$

### Moments about Origin

We have

$$\bar{x} = 2.5, \quad \mu_2 = 14.75, \quad \mu_3 = 39.75, \quad \mu_4 = 142.31 \text{ (approx.)}$$

We know

$$\bar{x} = A + \mu'_1,$$

where  $\mu'_1$  is the first moment about the point  $x = A$ .

Taking  $A = 0$ , we get the first moment about origin as

$$\mu'_1 = \text{mean} = 2.5.$$

Using (7-35), we get

$$\mu'_2 = \mu_2 + \mu_1'^2 = 14.75 + (2.5)^2 = 14.75 + 6.25 = 21$$

$$\begin{aligned}\mu'_3 &= \mu_3 + 3\mu_2\mu'_1 + \mu_1'^3 \\ &= 39.75 + 3(14.75)(2.5) + (2.5)^3 \\ &= 39.75 + 110.625 + 15.625 \\ &= 166\end{aligned}$$

$$\begin{aligned}
\mu'_4 &= \mu_4 + 4\mu_3\mu'_1 + 6\mu_2\mu_1'^2 + \mu_1'^4 \\
&= 142.3125 + 4(39.75)(2.5) + 6(14.75)(2.5)^2 + (2.5)^4 \\
&= 142.3125 + 397.5 + 553.125 + 39.0625 \\
&= 1132
\end{aligned}$$

### Moments about the Point $x = 2$

We have

$$\bar{x} = A + \mu'_1.$$

Taking  $A = 2$ , the first moment about the point  $x = 2$  is

$$\mu'_1 = \bar{x} - 2 = 2.5 - 2 = 0.5$$

Using (7-35), we get

$$\mu'_2 = \mu_2 + \mu_1'^2 = 14.75 + 0.25 = 15$$

$$\begin{aligned}
\mu'_3 &= \mu_3 + 3\mu_2\mu'_1 + \mu_1'^3 \\
&= 39.75 + 3(14.75)(0.5) + (0.5)^3 \\
&= 39.75 + 22.125 + 0.125 \\
&= 62
\end{aligned}$$

$$\begin{aligned}
\mu'_4 &= \mu_4 + 4\mu_3\mu'_1 + 6\mu_2\mu_1'^2 + \mu_1'^4 \\
&= 142.3125 + 4(39.75)(0.5) + 6(14.75)(0.5)^2 + (0.5)^4 \\
&= 142.3125 + 79.5 + 22.125 + 0.0625 \\
&= 244
\end{aligned}$$

### Example 12

Examine whether the following results of a piece of computation for obtaining second central moment are consistent or not;

$$N = 120, \quad \sum fX = -125, \quad \sum fX^2 = 128.$$

**Solution.** We have

$$\begin{aligned}
\mu_2 &= \frac{\sum fX^2}{N} - \left( \frac{\sum fX}{N} \right)^2 \\
&= \frac{128}{120} - \left( \frac{-125}{120} \right)^2 \\
&= 1.0670 - 1.0816 \\
&= -0.0146
\end{aligned}$$

which is impossible, since variance cannot be negative.  
Hence, the given data are inconsistent.

# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Correlation and Regression

B. Tech 3<sup>rd</sup> semester. 2025-2029 Batch

Prepared by: Dr. Boina Anil Kumar

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## Introduction

In many practical situations, two or more variables are related to each other. For example, the height and weight of a person, rainfall and agricultural production, temperature and electricity consumption, and study time and examination marks may show some degree of relationship.

**Correlation** is a statistical technique used to measure the **degree and direction of relationship between two variables**.

For example, if the height of a person increases, his or her weight generally tends to increase. Thus, height and weight are said to be positively correlated.

## Definition

Correlation is a statistical measure that describes the **strength and direction of association between two variables**.

If two variables are denoted by  $X$  and  $Y$ , correlation measures the extent to which changes in  $X$  are associated with changes in  $Y$ .

## Examples of Correlation

- Height and weight of individuals.
- Income and expenditure of families.
- Temperature and electricity consumption.
- Rainfall and crop production.
- Study hours and examination marks.
- Speed of a vehicle and time required to cover a fixed distance.

## Types of Correlation

Correlation can be classified in several ways.

## 1. Positive Correlation

Two variables are positively correlated when an increase in one variable is generally associated with an increase in the other variable, and a decrease in one is associated with a decrease in the other.

$$X \uparrow \implies Y \uparrow$$

### Examples:

- Height and weight.
- Income and expenditure.
- Study time and marks.

## 2. Negative Correlation

Two variables are negatively correlated when an increase in one variable is generally associated with a decrease in the other variable.

$$X \uparrow \implies Y \downarrow$$

### Examples:

- Speed and time taken to cover a fixed distance.
- Price and demand, under suitable market conditions.

## 3. Zero Correlation

When there is no systematic relationship between two variables, their correlation is said to be zero.

$$r = 0$$

## 4. Perfect Positive Correlation

If all the observations lie exactly on a straight line with positive slope, the correlation is perfectly positive.

$$r = +1$$

## 5. Perfect Negative Correlation

If all the observations lie exactly on a straight line with negative slope, the correlation is perfectly negative.

$$r = -1$$

## 6. Linear Correlation

When the relationship between two variables can be represented approximately by a straight line, the correlation is called **linear correlation**.

## 7. Non-linear Correlation

When the relationship between two variables cannot be adequately represented by a straight line, the correlation is called **non-linear** or **curvilinear correlation**.

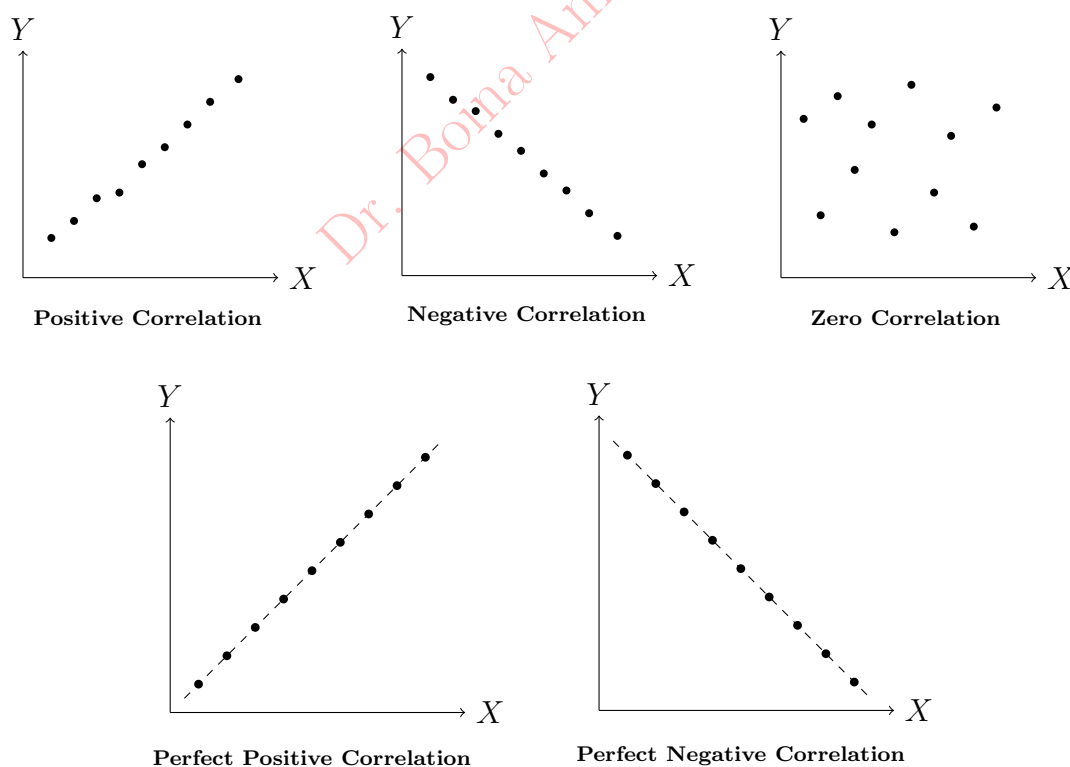
| Type             | Nature of Relationship        | Typical Value |
|------------------|-------------------------------|---------------|
| Perfect Positive | Exact increasing relationship | $r = +1$      |
| Positive         | Increasing relationship       | $0 < r < 1$   |
| Zero             | No linear relationship        | $r = 0$       |
| Negative         | Decreasing relationship       | $-1 < r < 0$  |
| Perfect Negative | Exact decreasing relationship | $r = -1$      |

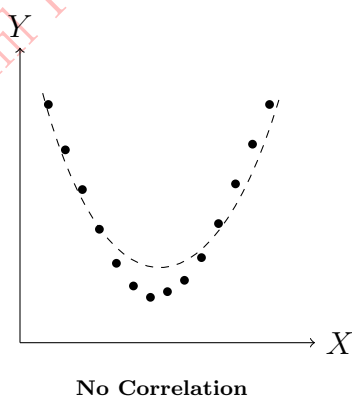
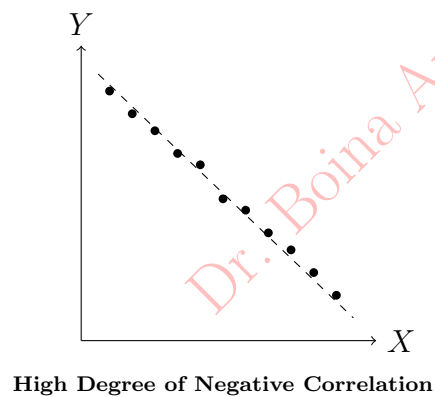
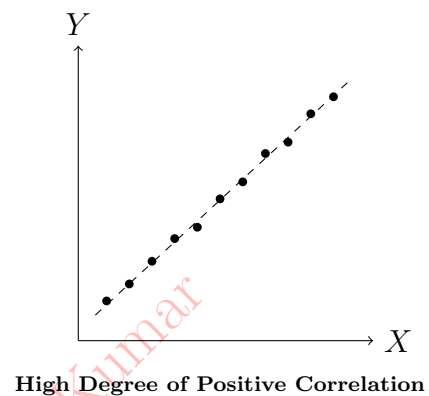
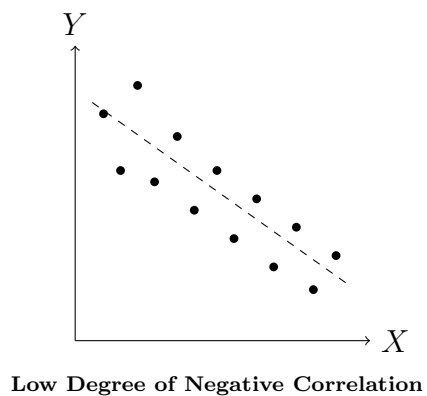
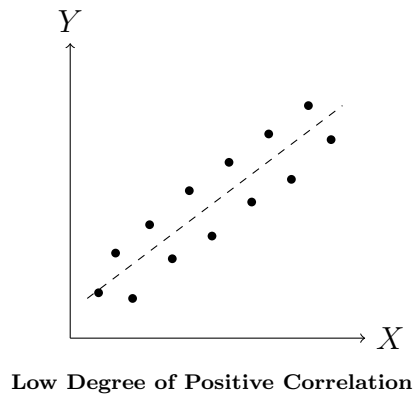
### Scatter Diagram

A **scatter diagram**, also called a **scatter plot**, is a graphical method of studying the relationship between two variables.

The values of one variable are represented on the horizontal axis and the values of the other variable are represented on the vertical axis. Each pair  $(x, y)$  is represented by a point on the graph.

### Schematic Representation of Scatter Diagrams





## Interpretation of Scatter Diagrams

The general patterns are:

- Points rising from left to right indicate **positive correlation**.
- Points falling from left to right indicate **negative correlation**.
- Points scattered randomly indicate **little or no correlation**.
- Points closely concentrated around a straight line indicate **strong correlation**.
- Points widely scattered around a straight line indicate **weak correlation**.

**Example:**

The following data give the marks obtained by 8 students in Mathematics ( $X$ ) and Statistics ( $Y$ ):

|     |    |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|----|
| $X$ | 40 | 45 | 50 | 55 | 60 | 65 | 70 | 75 |
| $Y$ | 42 | 48 | 51 | 58 | 61 | 67 | 70 | 78 |

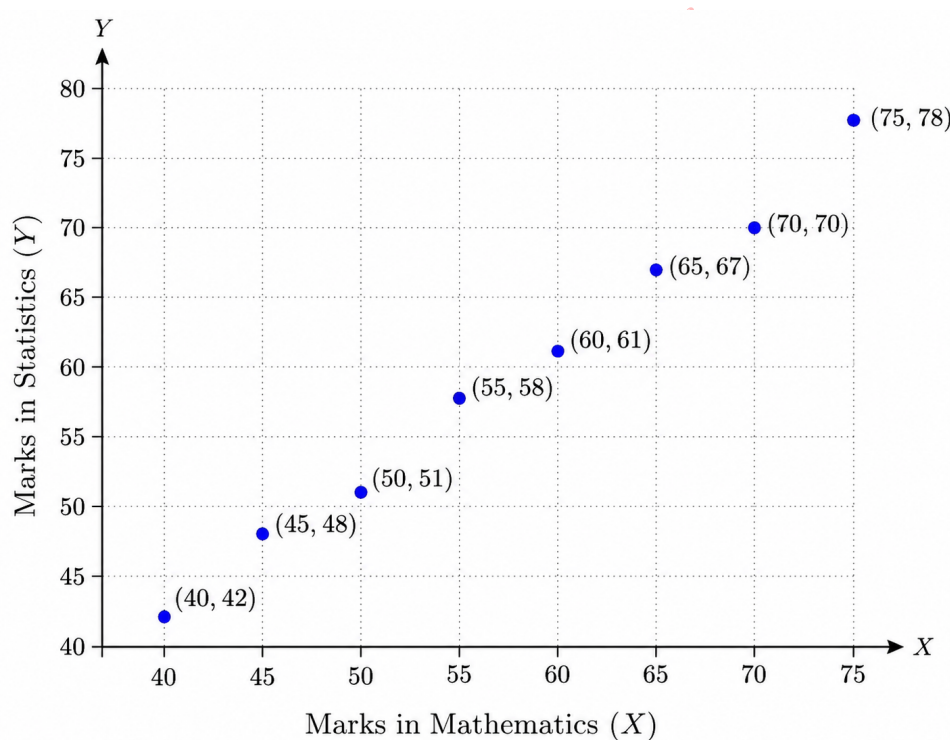
1. Draw a scatter diagram for the given data.
2. Comment on the type and degree of correlation between Mathematics and Statistics marks.

**Answer:**

The pairs of given observations are

$(40, 42)$ ,  $(45, 48)$ ,  $(50, 51)$ ,  $(55, 58)$ ,  $(60, 61)$ ,  $(65, 67)$ ,  $(70, 70)$ ,  $(75, 78)$

The scatter diagram of given observations is



The points in the scatter diagram are concentrated around an upward-sloping straight line. Therefore, as the marks in Mathematics increase, the marks in Statistics also tend to increase.

Hence, the data indicate a

Positive Correlation

and, since the points lie fairly close to a straight line, the correlation is

High Degree of Positive Correlation.

### Conclusion:

There is a high degree of positive correlation between Mathematics marks and Statistics marks.

### Example

Plot a Scatter Plot for the following heights and weights of 10 students.

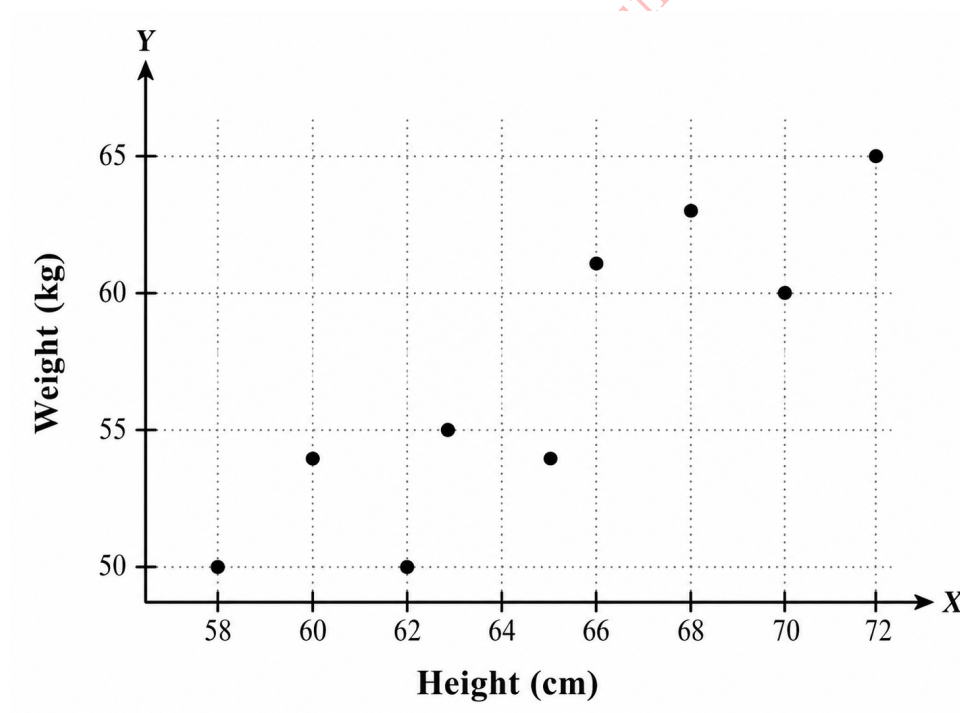
| Height | 62 | 72 | 68 | 58 | 65 | 70 | 66 | 63 | 60 | 72 |
|--------|----|----|----|----|----|----|----|----|----|----|
| Weight | 50 | 65 | 63 | 50 | 54 | 60 | 61 | 55 | 54 | 65 |

### Answer:

The corresponding pairs of observations  $(X, Y)$  are

$(62, 50)$ ,  $(72, 65)$ ,  $(68, 63)$ ,  $(58, 50)$ ,  $(65, 54)$ ,  
 $(70, 60)$ ,  $(66, 61)$ ,  $(63, 55)$ ,  $(60, 54)$ ,  $(72, 65)$ .

Taking **Height** on the horizontal axis and **Weight** on the vertical axis, the scatter diagram is obtained as follows.



From the scatter diagram, the points tend to move upward from left to right. Hence, as the height of a student increases, the weight also tends to increase.

Therefore, there is a

|                                     |
|-------------------------------------|
| High Degree of Positive Correlation |
|-------------------------------------|

between height and weight.

### **Advantages of Scatter Diagram**

- It is simple and easy to understand.
- It gives a visual impression of the relationship.
- It indicates the direction of correlation.
- It gives a rough idea about the strength of correlation.
- It helps identify unusual observations or outliers.

### **Limitations**

- It does not provide an exact numerical measure of correlation.
- Interpretation may be subjective.
- It may not clearly identify weak relationships.

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# APPLIED STATISTICS

( for AIML Students)

Unit-2 (Descriptive Statistics)

Topic: Correlation and Regression

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## Karl Pearson's Coefficient of Correlation

The **Karl Pearson's coefficient of correlation** is one of the most widely used numerical measures of linear correlation between two variables.

It is denoted by  $r$ .

$$r = \frac{Cov(x, y)}{\sigma_x \sigma_y}.$$

$$Cov(x, y) = \frac{1}{N} \sum (X - \bar{X})(Y - \bar{Y}); \quad \sigma_x = \sqrt{\frac{1}{N} \sum (X - \bar{X})^2}; \quad \sigma_y = \sqrt{\frac{1}{N} \sum (Y - \bar{Y})^2};$$

**Formulas:**

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{[N \sum X^2 - (\sum X)^2][N \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{N \sum xy - (\sum x)(\sum y)}{\sqrt{[N \sum x^2 - (\sum x)^2][N \sum y^2 - (\sum y)^2]}}$$

Here,

$$x = X - A, \quad y = Y - B, \quad \text{where } A \text{ and } B \text{ are assumed means}$$

## Properties of Karl Pearson's Coefficient

1. The value of  $r$  lies between  $-1$  and  $+1$ .
2.  $r = +1$  indicates perfect positive correlation.
3.  $r = -1$  indicates perfect negative correlation.
4.  $r = 0$  indicates no linear correlation.
5.  $0 < r < 1$  indicates positive correlation.
6.  $-1 < r < 0$  indicates negative correlation.
7.  $r$  is independent of change of origin and scale.

8. Correlation is a symmetric measure:

$$r_{XY} = r_{YX}.$$

9. Pearson's coefficient measures the degree of **linear** relationship.

**Example 1:**

Calculate Karl Pearson's coefficient of correlation between expenditure on advertising and sales from the data given below.

|                                        |    |    |    |    |    |    |    |    |    |    |
|----------------------------------------|----|----|----|----|----|----|----|----|----|----|
| <b>Advertising expenses ('000 Rs.)</b> | 39 | 65 | 62 | 90 | 82 | 75 | 25 | 98 | 36 | 78 |
| <b>Sales (lakh Rs.)</b>                | 47 | 53 | 58 | 86 | 62 | 68 | 60 | 91 | 51 | 84 |

**Solution.**

Let the advertising expenses (in '000 Rs.) be denoted by the variable  $x$  and the sales (in lakh Rs.) be denoted by the variable  $y$ .

| $x$                 | $y$                 | $x - \bar{x}$<br>$= x - 65$ | $y - \bar{y}$<br>$= y - 66$ | $(x - \bar{x})^2$<br>$= (x - 65)^2$ | $(y - \bar{y})^2$<br>$= (y - 66)^2$ | $(x - \bar{x})(y - \bar{y})$<br>$= (x - 65)(y - 66)$ |
|---------------------|---------------------|-----------------------------|-----------------------------|-------------------------------------|-------------------------------------|------------------------------------------------------|
| 39                  | 47                  | -26                         | -19                         | 676                                 | 361                                 | 494                                                  |
| 65                  | 53                  | 0                           | -13                         | 0                                   | 169                                 | 0                                                    |
| 62                  | 58                  | -3                          | -8                          | 9                                   | 64                                  | 24                                                   |
| 90                  | 86                  | 25                          | 20                          | 625                                 | 400                                 | 500                                                  |
| 82                  | 62                  | 17                          | -4                          | 289                                 | 16                                  | -68                                                  |
| 75                  | 68                  | 10                          | 2                           | 100                                 | 4                                   | 20                                                   |
| 25                  | 60                  | -40                         | -6                          | 1600                                | 36                                  | 240                                                  |
| 98                  | 91                  | 33                          | 25                          | 1089                                | 625                                 | 825                                                  |
| 36                  | 51                  | -29                         | -15                         | 841                                 | 225                                 | 435                                                  |
| 78                  | 84                  | 13                          | 18                          | 169                                 | 324                                 | 234                                                  |
| $\sum x$<br>$= 650$ | $\sum y$<br>$= 660$ | $\sum (x - 65)$<br>$= 0$    | $\sum (y - 66)$<br>$= 0$    | $\sum (x - 65)^2$<br>$= 5398$       | $\sum (y - 66)^2$<br>$= 2224$       | $\sum (x - 65)(y - 66)$<br>$= 2704$                  |

$$\bar{x} = \frac{\sum x}{n} = \frac{650}{10} = 65,$$

$$\bar{y} = \frac{\sum y}{n} = \frac{660}{10} = 66.$$

Karl Pearson's coefficient of correlation is

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} = \frac{2704}{\sqrt{5398 \times 2224}} = 0.7804.$$

$$r = 0.7804$$

Here, there is a fairly high degree of positive correlation between expenditure on advertisement and sales. We may, therefore conclude that in general, sales have increased with an increase in the advertisement expenses.

### Example 2:

Calculate Karl Pearson's coefficient of correlation for the following data:

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $X$ | 10 | 20 | 30 | 40 | 50 |
| $Y$ | 15 | 25 | 35 | 45 | 55 |

**Solution:**

| $X$            | $Y$            | $X^2$             | $Y^2$             | $XY$             |
|----------------|----------------|-------------------|-------------------|------------------|
| 10             | 15             | 100               | 225               | 150              |
| 20             | 25             | 400               | 625               | 500              |
| 30             | 35             | 900               | 1225              | 1050             |
| 40             | 45             | 1600              | 2025              | 1800             |
| 50             | 55             | 2500              | 3025              | 2750             |
| $\sum X = 150$ | $\sum Y = 175$ | $\sum X^2 = 5500$ | $\sum Y^2 = 7125$ | $\sum XY = 6250$ |

$$N = 5$$

Using Karl Pearson's formula,

$$\begin{aligned}
 r &= \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{[N \sum X^2 - (\sum X)^2][N \sum Y^2 - (\sum Y)^2]}} \\
 r &= \frac{5(6250) - (150)(175)}{\sqrt{[5(5500) - 150^2][5(7125) - 175^2]}} \\
 r &= \frac{31250 - 26250}{\sqrt{(27500 - 22500)(35625 - 30625)}} \\
 r &= \frac{5000}{\sqrt{5000 \times 5000}} = \frac{5000}{5000} = 1
 \end{aligned}$$

Hence,

$$r = +1$$

Therefore, there is a **perfect positive correlation** between  $X$  and  $Y$ .

### Example 3

The marks obtained by six students in Mathematics and Statistics are given below. Calculate Karl Pearson's coefficient of correlation.

|                     |    |    |    |    |    |    |
|---------------------|----|----|----|----|----|----|
| Mathematics ( $X$ ) | 40 | 45 | 50 | 55 | 60 | 65 |
| Statistics ( $Y$ )  | 42 | 48 | 46 | 55 | 58 | 62 |

#### Solution:

Construct the following table.

| $X$            | $Y$            | $X^2$              | $Y^2$              | $XY$              |
|----------------|----------------|--------------------|--------------------|-------------------|
| 40             | 42             | 1600               | 1764               | 1680              |
| 45             | 48             | 2025               | 2304               | 2160              |
| 50             | 46             | 2500               | 2116               | 2300              |
| 55             | 55             | 3025               | 3025               | 3025              |
| 60             | 58             | 3600               | 3364               | 3480              |
| 65             | 62             | 4225               | 3844               | 4030              |
| $\sum X = 315$ | $\sum Y = 311$ | $\sum X^2 = 16975$ | $\sum Y^2 = 16417$ | $\sum XY = 16675$ |

Here,

$$N = 6.$$

Therefore,

$$r = \frac{6(16675) - (315)(311)}{\sqrt{[6(16975) - 315^2][6(16417) - 311^2]}}.$$

Hence,

$$r \approx 0.976$$

Thus, Mathematics marks and Statistics marks have a **very strong positive linear correlation**.

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## Spearman's Rank Correlation Coefficient

Spearman's Rank Correlation Coefficient is a statistical measure used to determine the **degree and direction of association between two variables based on their ranks**. It is particularly useful when the data are expressed in terms of ranks or when the relationship between the variables is not necessarily linear.

The coefficient is denoted by  $\rho$  (rho).

### Formula

When there are no tied ranks, Spearman's rank correlation coefficient is given by

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where

- $n$  = number of observations,
- $d$  = difference between the two ranks for each observation.

## Ranks are given and No Tie between Ranks

### Example 1:

The ranks obtained by 8 students in Mathematics and Physics are given below. Find Spearman's rank correlation coefficient.

| Student              | A | B | C | D | E | F | G | H |
|----------------------|---|---|---|---|---|---|---|---|
| Mathematics Rank (X) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Physics Rank (Y)     | 2 | 1 | 4 | 3 | 5 | 7 | 8 | 6 |

**Solution:**

| Student    | X | Y | $d = X - Y$ | $d^2$ |
|------------|---|---|-------------|-------|
| A          | 1 | 2 | -1          | 1     |
| B          | 2 | 1 | 1           | 1     |
| C          | 3 | 4 | -1          | 1     |
| D          | 4 | 3 | 1           | 1     |
| E          | 5 | 5 | 0           | 0     |
| F          | 6 | 7 | -1          | 1     |
| G          | 7 | 8 | -1          | 1     |
| H          | 8 | 6 | 2           | 4     |
| $\sum d^2$ |   |   |             | = 10  |

Here,

$$n = 8, \quad \sum d^2 = 10.$$

Therefore,

$$\rho = 1 - \frac{6(10)}{8(8^2 - 1)} = 1 - \frac{60}{504} \approx 0.881$$

Hence,

$$\boxed{\rho \approx 0.881}$$

Therefore, there is a **strong positive rank correlation** between the students' performance in Mathematics and Physics.

## Raw data is given and No Tie between their Ranks

### Example 2:

The marks obtained by 8 students in Mathematics and Statistics are given below. Calculate Spearman's Rank Correlation Coefficient.

| Student     | A  | B  | C  | D  | E  | F  | G  | H  |
|-------------|----|----|----|----|----|----|----|----|
| Mathematics | 78 | 65 | 90 | 55 | 72 | 84 | 60 | 95 |
| Statistics  | 82 | 70 | 88 | 60 | 75 | 80 | 65 | 92 |

**Solution:**

The formula for Spearman's Rank Correlation Coefficient is

$$\boxed{\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}}$$

The scores of Mathematics in descending order is

$$95, 90, 84, 78, 72, 65, 60, 55$$

The ranks assigned respectively as

1, 2, 3, 4, 5, 6, 7, 8

The scores of Statistics in descending order is

92, 88, 82, 80, 75, 70, 65, 60

The ranks assigned respectively as

1, 2, 3, 4, 5, 6, 7, 8

The required calculations are shown below.

| Mathematics |       |         | Statistics |         |       |
|-------------|-------|---------|------------|---------|-------|
| Student     | Score | Rank(X) | score      | Rank(Y) | $d^2$ |
| A           | 78    | 4       | 82         | 3       | 1     |
| B           | 65    | 6       | 70         | 6       | 0     |
| C           | 90    | 2       | 88         | 2       | 0     |
| D           | 55    | 8       | 60         | 8       | 0     |
| E           | 72    | 5       | 75         | 5       | 0     |
| F           | 84    | 3       | 80         | 4       | 1     |
| G           | 60    | 7       | 65         | 7       | 0     |
| H           | 95    | 1       | 92         | 1       | 0     |
| $\sum d^2$  |       |         |            |         | 2     |

Here,

$$n = 8, \quad \sum d^2 = 2.$$

Therefore,

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(2)}{8(8^2 - 1)} = 0.97619.$$

Hence,

$$\boxed{\rho \approx 0.976}$$

### Interpretation:

Since  $\rho$  is very close to +1, there is a **very strong positive rank correlation** between the marks obtained in Mathematics and Statistics.

Very strong positive correlation

## Spearman's Rank Correlation Coefficient with Tied Ranks

### Standard Formula with Tie Correction:

When ties are present, Spearman's rank correlation can also be calculated using the tie-correction formula

$$\rho = 1 - \frac{6}{n(n^2 - 1)} \left[ \sum d^2 + \frac{1}{12} \sum m(m^2 - 1) \right]$$

Here,  $\frac{1}{12} \sum m(m^2 - 1)$  is correction factor added to  $\sum d^2$  for each tied item and  $m$  is the number of times the item repeated.

### Example 3:

The marks obtained by 8 students in Mathematics and Statistics are given below.

| Student     | A  | B  | C  | D  | E  | F  | G  | H  |
|-------------|----|----|----|----|----|----|----|----|
| Mathematics | 85 | 75 | 90 | 75 | 60 | 68 | 55 | 55 |
| Statistics  | 78 | 82 | 90 | 70 | 65 | 75 | 60 | 60 |

Calculate Spearman's Rank Correlation Coefficient.

### Solution:

The descending order of Mathematics Score is

90, 85, 75, 75, 68, 60, 55, 55

- 90 is ranked 1.
- 85 is ranked 2.
- 75 occurs twice and occupies ranks 3 and 4. Therefore, each is assigned

$$\frac{3 + 4}{2} = 3.5.$$

- 68 is ranked 5.
- 60 is ranked 6.
- 55 occurs twice and occupies ranks 7 and 8. Therefore, each is assigned

$$\frac{7 + 8}{2} = 7.5.$$

Since there are tied observations, the tied observations are assigned their **average ranks**.

The descending order of Statistics Score is

90, 82, 78, 75, 70, 65, 60, 60

- 90 is ranked 1.
- 82 is ranked 2.
- 78 is ranked 3.
- 75 is ranked 4.
- 70 is ranked 5.
- 65 is ranked 6.
- 60 occurs twice and occupies ranks 7 and 8. Therefore, each is assigned

$$\frac{7+8}{2} = 7.5.$$

| Mathematics |       |         | Statistics |         |            |
|-------------|-------|---------|------------|---------|------------|
| Student     | Score | Rank(X) | score      | Rank(Y) | $d^2$      |
| A           | 85    | 2       | 78         | 3       | 1          |
| B           | 75    | 3.5     | 82         | 2       | 2.25       |
| C           | 90    | 1       | 90         | 1       | 0          |
| D           | 75    | 3.5     | 70         | 5       | 2.25       |
| E           | 60    | 6       | 65         | 6       | 0          |
| F           | 68    | 5       | 75         | 4       | 1          |
| G           | 55    | 7.5     | 60         | 7.5     | 0          |
| H           | 55    | 7.5     | 60         | 7.5     | 0          |
| $\sum d^2$  |       |         |            |         | <b>6.5</b> |

### Calculation of Correction Factor

| Repeated value |       |           | Correction Factor         |
|----------------|-------|-----------|---------------------------|
| Series         | Value | Frequency | $\frac{m(m^2-1)}{12}$     |
| X-series       | 75    | 2         | $\frac{2(4-1)}{12} = 0.5$ |
| X-series       | 55    | 2         | $\frac{2(4-1)}{12} = 0.5$ |
| Y-series       | 60    | 2         | $\frac{2(4-1)}{12} = 0.5$ |

$$\frac{1}{12} \sum m(m^2 - 1) = \mathbf{1.5}$$

$$\rho = 1 - \frac{6}{n(n^2 - 1)} \left[ \sum d^2 + \frac{1}{12} \sum m(m^2 - 1) \right] = 1 - \frac{6}{8(8^2 - 1)} [6.5 + 1.5] = 0.9047$$

### Interpretation:

Since  $\rho$  is very close to +1, there is a **very strong positive rank correlation** between the marks obtained in Mathematics and Statistics.

# APPLIED STATISTICS

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## Regression

**Regression Analysis** is a statistical technique used to study the relationship between a dependent variable and one or more independent variables.

The main purpose of regression analysis is to:

1. determine the mathematical relationship between variables,
2. estimate the value of one variable from the known value of another variable, and
3. make predictions or forecasts.

## Regression Variables

Suppose two variables  $X$  and  $Y$  are related.

- $X$  is called the **independent variable** or **predictor variable**.
- $Y$  is called the **dependent variable** or **response variable**.

For example, if marks obtained depend on the number of hours studied, then

$$X = \text{Hours studied}, \quad Y = \text{Marks obtained}.$$

Thus, regression attempts to express  $Y$  in terms of  $X$ .

## Regression Lines

For two variables  $X$  and  $Y$ , there are generally two regression lines:

1. Regression line of  $Y$  on  $X$ :  $\boxed{Y = a + bX}$

2. Regression line of  $X$  on  $Y$ :  $\boxed{X = c + dY}$

Here,

$a$  = intercept,  $b$  = slope or regression coefficient.

## Regression and Correlation

Correlation measures the **degree and direction of association** between two variables, whereas regression provides a **mathematical equation for estimation and prediction**.

| Basis     | Correlation                                                | Regression                                                      |
|-----------|------------------------------------------------------------|-----------------------------------------------------------------|
| Purpose   | Measures strength and direction of relationship            | Used for estimation and prediction                              |
| Variables | No distinction between dependent and independent variables | Distinction is made between dependent and independent variables |
| Result    | Gives a correlation coefficient $r$                        | Gives regression equation(s)                                    |
| Range     | $-1 \leq r \leq 1$                                         | Regression coefficients need not lie between $-1$ and $1$       |

## Method of Least Squares

The **method of least squares** is one of the most important methods for fitting a mathematical equation to observed data.

Suppose the observed values are

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n).$$

We want to fit a straight line

$$\hat{Y} = a + bX$$

to these observations.

For each observation, the difference between the observed value and the fitted value is called the **residual** or **error**.

Thus,

$$e_i = y_i - \hat{y}_i.$$

The least-squares method chooses  $a$  and  $b$  such that

$$\sum e_i^2$$

is minimum.

Therefore,

$$\sum (y_i - a - bx_i)^2 = \text{minimum.}$$

This is the fundamental principle of the method of least squares.

**Note:** In simple terms,

- (i) the objective of the **Method of Least Squares** is to determine the constants  $a$  and  $b$  such that the regression line of  $Y$  on  $X$ ,

$$\boxed{Y = a + bX},$$

provides the **best fit to the given data by minimizing the sum of the squares of the errors**, and

- (ii) for the regression line of  $X$  on  $Y$ , the objective is to determine the constants  $c$  and  $d$  such that

$$\boxed{X = c + dY},$$

provides the **best fit to the given data by minimizing the sum of the squares of the errors**.

### Derivation of the Normal Equations

Consider the straight-line equation

$$\hat{Y} = a + bX.$$

The sum of squared errors is

$$S = \sum (y - a - bx)^2.$$

For  $S$  to be minimum,

$$\frac{\partial S}{\partial a} = 0$$

and

$$\frac{\partial S}{\partial b} = 0.$$

Differentiating with respect to  $a$ ,

$$\frac{\partial S}{\partial a} = -2 \sum (y - a - bx) = 0.$$

Therefore,

$$\sum y - na - b \sum x = 0.$$

Hence,

$$\boxed{\sum y = na + b \sum x}.$$

Differentiating with respect to  $b$ ,

$$\frac{\partial S}{\partial b} = -2 \sum x(y - a - bx) = 0.$$

Therefore,

$$\sum xy - a \sum x - b \sum x^2 = 0.$$

Hence,

$$\boxed{\sum xy = a \sum x + b \sum x^2}.$$

These two equations are called the **normal equations**.

## Normal Equations

For the straight line

$$Y = a + bX,$$

the normal equations are

$$\sum Y = na + b \sum X$$

and

$$\sum XY = a \sum X + b \sum X^2.$$

Solving these two simultaneous equations gives the values of  $a$  and  $b$ .

The required regression line is then

$$\hat{Y} = a + bX.$$

## Alternative Form Using Means

Since

$$\bar{X} = \frac{\sum X}{n}, \quad \bar{Y} = \frac{\sum Y}{n},$$

the first normal equation gives

$$\bar{Y} = a + b\bar{X}.$$

Therefore,

$$a = \bar{Y} - b\bar{X}.$$

The slope is

$$b = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}.$$

Hence, the regression equation can be written as

$$\hat{Y} - \bar{Y} = b(X - \bar{X}).$$

Thus,

$$\hat{Y} = \bar{Y} + b(X - \bar{X}).$$

## Shortcut Formula for the Slope

The slope can also be calculated using

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}.$$

Therefore,

$$a = \bar{Y} - b\bar{X}.$$

## Procedure for Fitting a Straight Line

To fit the regression line

$$\hat{Y} = a + bX$$

by the method of least squares:

1. Prepare columns for  $X$ ,  $Y$ ,  $X^2$ , and  $XY$ .
2. Calculate  $\sum X$ ,  $\sum Y$ ,  $\sum X^2$ ,  $\sum XY$ .
3. Use the normal equations  $\sum Y = na + b\sum X$  and  $\sum XY = a\sum X + b\sum X^2$ .
4. Solve for  $a$  and  $b$ .
5. Write the fitted regression equation  $\hat{Y} = a + bX$ .
6. Use the equation to estimate  $Y$  for a given value of  $X$ .

## Examples

### Example 1:

Fit a straight line

$$Y = a + bX$$

to the following data and estimate  $Y$  when  $X = 6$ .

|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $X$ | 1 | 2 | 3 | 4 | 5 |
| $Y$ | 2 | 4 | 5 | 4 | 6 |

### Solution:

The required table is

| $X$           | $Y$           | $X^2$           | $XY$           |
|---------------|---------------|-----------------|----------------|
| 1             | 2             | 1               | 2              |
| 2             | 4             | 4               | 8              |
| 3             | 5             | 9               | 15             |
| 4             | 4             | 16              | 16             |
| 5             | 6             | 25              | 30             |
| $\sum X = 15$ | $\sum Y = 21$ | $\sum X^2 = 55$ | $\sum XY = 71$ |

The normal equations are

$$\sum Y = na + b\sum X \quad \Rightarrow \quad 5a + 15b = 21$$

and

$$\sum XY = a\sum X + b\sum X^2 \quad \Rightarrow \quad 15a + 55b = 71$$

On solving, we have

$$\boxed{a = 1.8} \quad \boxed{b = 0.8}.$$

Hence the fitted regression line is

$$\boxed{\hat{Y} = 1.8 + 0.8X}.$$

For  $X = 6$ ,

$$\hat{Y} = 1.8 + 0.8(6) = 6.6$$

$$\boxed{\hat{Y} = 6.6}.$$

### Example 2:

Fit the regression line of  $Y$  on  $X$  for the following data:

|     |   |   |   |   |    |
|-----|---|---|---|---|----|
| $X$ | 2 | 4 | 6 | 8 | 10 |
| $Y$ | 3 | 5 | 7 | 8 | 10 |

Also estimate  $Y$  when  $X = 12$ .

### Solution:

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6 \quad \text{and} \quad \bar{Y} = \frac{3 + 5 + 7 + 8 + 10}{5} = \frac{33}{5} = 6.6.$$

The required table is

| $X$ | $Y$ | $X - \bar{X}$ | $Y - \bar{Y}$ | $(X - \bar{X})(Y - \bar{Y})$               | $(X - \bar{X})^2$               |
|-----|-----|---------------|---------------|--------------------------------------------|---------------------------------|
| 2   | 3   | -4            | -3.6          | 14.4                                       | 16                              |
| 4   | 5   | -2            | -1.6          | 3.2                                        | 4                               |
| 6   | 7   | 0             | 0.4           | 0                                          | 0                               |
| 8   | 8   | 2             | 1.4           | 2.8                                        | 4                               |
| 10  | 10  | 4             | 3.4           | 13.6                                       | 16                              |
|     |     |               |               | $\sum(X - \bar{X})(Y - \bar{Y})$<br>$= 34$ | $\sum(X - \bar{X})^2$<br>$= 40$ |

Therefore,

$$b = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2} = \frac{34}{40} = 0.85.$$

The regression equation is

$$\hat{Y} - \bar{Y} = b(X - \bar{X}).$$

Hence,

$$\hat{Y} - 6.6 = 0.85(X - 6) \quad \Rightarrow \quad \boxed{\hat{Y} = 1.5 + 0.85X}$$

For  $X = 12$ ,

$$\hat{Y} = 1.5 + 0.85(12) \quad \Rightarrow \quad \boxed{\hat{Y} = 11.7}.$$

**Example 3:**

The following data give the number of hours studied  $X$  and marks obtained  $Y$  by five students.

|                       |    |    |    |    |    |
|-----------------------|----|----|----|----|----|
| Hours studied ( $X$ ) | 1  | 2  | 3  | 4  | 5  |
| Marks ( $Y$ )         | 35 | 40 | 50 | 55 | 65 |

Find the regression line of  $Y$  on  $X$  and estimate the marks of a student who studies for 6 hours.

**Solution:**

The required table is

| $X$ | $Y$ | $X^2$ | $XY$ |
|-----|-----|-------|------|
| 1   | 35  | 1     | 35   |
| 2   | 40  | 4     | 80   |
| 3   | 50  | 9     | 150  |
| 4   | 55  | 16    | 220  |
| 5   | 65  | 25    | 325  |
| 15  | 245 | 55    | 810  |

Here,  $n = 5$  and

$$\sum X = 15, \quad \sum Y = 245, \quad \sum X^2 = 55, \quad \sum XY = 810.$$

Now, the regression coefficient

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{5(810) - (15)(245)}{5(55) - 15^2} = 7.5.$$

Thus,

$$\boxed{b = 7.5}.$$

Now,

$$\bar{X} = \frac{15}{5} = 3 \quad \text{and} \quad \bar{Y} = \frac{245}{5} = 49.$$

So,

$$a = \bar{Y} - b\bar{X} = 49 - (7.5)(3) = 26.5.$$

Hence the regression equation is

$$\boxed{\hat{Y} = 26.5 + 7.5X}.$$

For  $X = 6$ ,

$$\hat{Y} = 26.5 + 7.5(6) = 26.5 + 45 = 71.5$$

$$\implies \boxed{\hat{Y} = 71.5}.$$

Thus, the estimated marks are approximately

$$\boxed{72}.$$